BOOK OF ABSTRACTS

Leakage - Inaugural Conference of stsing e.V.
March 19-22, 2023
TU Dresden

Unless indicated otherwise all panels will be accessible on site in the designated rooms and on Zoom (link distributed separately for registered participants).

Sofern nicht anders angegeben, werden alle Panels vor Ort in den angezeigten Räumen und über Zoom zugänglich sein (Link wird separat an die registrierten Teilnehmer verteilt).

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Panel #23 Technopolitics and Counter-Leaks
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Panel #26 Leaking (Into) Homes: The Integrational Work Within Everyday Smart Homes
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Panel #29 Leaks, Cracks, Spills, Overflows – Waste, Pollution and Toxic Substances in Motion
Panel #30 State Power and Regimes of Control and Containment
Panel #31 Leakage and Post-Colonial Critique
#32 Off-Site Walk Wireless Walk and Talk – Tracing Mobile Communication Signals Through Dresden
Panel #33 Biopolitics of Marginalization
Panel #34 Workshop: Editors Meet Authors and Reviewers: Re-figuring STS Publications Together
Panel #35 (Un)durchlässigkeiten kompostieren in/durch/mit Ruiniertem des Anthropozäns
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#40 Workshop Memories of future histories - Situating “Best practices” and Speculating About “Good Relations”
Panel #41 The Black Triangle – Sensing Borderland Coal Ecologies
Panel #42 Art, Poetics, and Technoscience of Microscopic Vision

Friday, March 22

Panel #43 Planetary Data
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Panel #50 (Techno)Biopolitics in Healthcare and Welfare Institutions
Panel #51 Up to Date in STS: Early Career Reading Group
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Panel #53 Technoscientific Worldmaking, Environments and Futurity
Panel #54 Queer-Feminist Transformations (German)
Panel #55 States of Emergency and Technological Sensing
Panel #56 Leaking Ethnographies - A Conversation Café on Methodography

Panel #1 Feminist STS Meets Artistic Research

Stream: Ecological Leakage and the More-Than-Human
ABS/E11
Chair: Ramona Haegele

Xenolith

Caitlin Berrigan (Academy of Fine Arts Vienna, Austria)
remote

I want to start with this feeling: an out-of-place feeling. A contradiction. A contamination. To me this is familiar: the feeling of being contaminated. An alienation, an othering, and also when the contaminant is the self. I would like to propose sharing excerpts from a new assemblage of video works and poetic holograms that invite being-with the inhuman: experimental documentaries that trace encounters with difference, contamination, and queer metabolisms through the material metaphor of the xenolith—a geological term for alien stone. Much of my research centers around a geothermal power plant in Iceland where excess carbon dioxide gas is capped, liquefied, and injected into subterranean bedrock. Here it mineralizes as calcite within the porous cavities of basalt stone—forming a synthetic xenolith. It is a material of queer subjectivity: a transformed body whose holes are filled with foreign matter. The fossil fuel industry first developed the technique to maximize fuel extraction from emptied reservoirs. It is now a burgeoning industry with billions of euros of investment to cap and bank greenhouse gases from existing polluters. Although geoengineering is proposed to combat global warming, it leaves extractive infrastructures intact and creates new supply chains and economies for contaminants. Synthetic xenoliths as carbon capture and sequestration participate in greenwashing, but they also present as troubled metaphors for adapting to life in a contaminated world. Imagine xenoliths as what Lauren Berlant calls a glitch: “an interruption within a transition, a troubled transmission. A glitch is also the revelation of an infrastructural failure.” I want to follow how minerals are transformed and mobilized by different conceptual and political agendas in the climate crisis, and to inscribe geology as media—sculptural, visual, digital, textual. If minerals have agencies, what are their modes of communication, and what are their narrations?

Leaky bodies, Permeable Flows. On the Pervasive in Hybrid Art

Regine Rapp (Art Laboratory Berlin / Münster Academy of Art, Germany)
In person

Permeability, leaky systems and places as well as permeable thinking and acting are essential approaches, thematic areas as well as parameters for possible instructions for action in numerous artistic projects in current Hybrid Art. Essential is the direct engagement with fluid elements that surround and permeate us (water, cells, menstrual blood). Essential is a way of thinking that relies on networked, permeable and leaky bodies and investigates and aesthetically explores forms of dissolution of boundaries. In my lecture I want to present selected artistic examples. Numerous projects of the new artistic research namely design worlds beyond binary models and break through oppositions like culture/nature, inside/outside, human/non-human or female/male. Approaches such as porosity and permeability are rethought from the perspective of a post-anthropocentric era and intertwined with theories from posthumanism and New Materialism (especially hydrofeminism/ Astrida Neimanis). Finally, we can also perceive interesting structural moments of a permeability - namely in the context of transdisciplinary work in the STEAM* field: when artistic and scientific practices intersect in the context of an Art Science workshop and thus dense collective knowledge flows together.

Floating Transmissions

Julia Nordholz (Independent Artist, Germany)
In Person

"FLOATING TRANSMISSIONS" - A Radio Sound Art Festival is presented as a project that addresses the themes of leakage and sensory exploration within the context of artistic spatial practice. This project, curated and funded by the Behörde für Kultur und Medien Hamburg's Art in Public Spaces programme, is an innovative fusion of artistic exploration and urban river spaces developed by Julia Nordholz and her team to connect and showcase 70 international artists. In early September 2023, the MS Stubnitz, a converted ship, served as a floating radio station, stage, exhibition space and laboratory. In an experimental and transdisciplinary approach, the 'river space' was broadcasted via FM and other protocols. Participating artists presented their work, resulting in a live radio programme with diverse artistic contributions. This programme transformed urban river spaces into immersive environments, inviting audiences to engage with auditory media in new ways.
A sensory experience: "FLOATING TRANSMISSIONS" aimed to engage the senses through sound, providing audiences with a unique auditory and spatial experience as they explored urban river spaces. The river space was made accessible through aesthetic experiences, primarily through sound. Immersive environments: The project included multimedia sound sculptures, immersive performances on the Elbe and Bille rivers, and sound installations that explored the city's soundscapes. Transdisciplinary Collaboration: With the participation of 70 international artists, the project fostered collaboration between different artistic disciplines, encouraging a mix of perspectives. Radio broadcast: The formats created during the project were broadcast on several radio stations, reaching a global audience and expanding the boundaries of urban art engagement. Floating Transmissions embodies the concept of leakage as it challenges boundaries and perceptions of urban spaces. It offers a generative perspective on the interplay between technology, art and urban environments, reflecting themes of environmental exposure, socio-technical entanglement and the transformative potential of leakage. Through the project's programme, artists and participants explored the material unconscious, submerged histories and enabling conditions of urban river spaces.

Panel #2 Leaky Diseases: Negotiating Medical Instabilities and Symptomatic Overlap

Stream: Bodies and Governance
ABS/E04
Chair: Aurora A. Sauter

Aurora A. Sauter (Johannes Gutenberg-Universität Mainz, Germany)
Paula Muhr (University of Zurich, Switzerland and Technische Universität Berlin, Germany)
Lisa Wiedemann (Helmut Schmidt University, Germany)

Medical diagnoses classify ailing bodies and connect them to treatment plans and health care insurance accountings. Their instabilities and ambivalences remain hidden (Bowker and Star 2008) while they categorize bodily signs and symptoms, attempting to impose order onto unruly, ‘leaky bodies’ (Shildrick 1997). In short, diagnoses set boundaries. However, not every disease is medically clearly categorizable - some oppose the closure and leak through existing medical boundaries. In our panel, we draw attention to leaky diseases with ‘fuzzy boundaries’ (Dumit 2006). We will apply the traditional STS perspective of ‘doing disease’ (e.g., Mol 2002) to the phenomenon of leakiness in the context of medical standardization and classification politics, ‘chronic care infrastructures’ (Langstrup 2013) and bodily experiences. Leaky diseases such as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), functional neurological disorder (FND), fibromyalgia and Post-COVID-syndrome disrupt medical standardizations and procedures. Each of these diseases is characterized by heterogeneous bodily symptoms, many of which are regarded medically unspecific and difficult to measure ‘objectively’ (e.g., pain, fatigue, brain fog). Moreover, symptoms of these diseases overlap with those of other diseases and generalizable bodily experiences, making their differential diagnosis and acknowledgement challenging. Such diagnostic leakage is surrounded by a broader culture of

contestation since only clear and delimitable medical signs materialize illness in physical bodies. Hence, leaky diseases fall off the grid of routine medical examinations, often being psychologized or dismissed as pretense. In our panel, we bring together leaky diseases with overlapping and still divergent medical histories and forms of socio-technical embeddedness. The questions we will discuss are: How do illnesses leak through diagnostic boundaries, and what are the consequences for clinical practice and patient experience? What can we learn about knowledge production and health care if we concentrate on leakages? What kinds of ethics and politics are required if we conceptualize some diseases as leaky?

Paradoxical Leakage: Diagnostic Drift and Dual Diagnosis

Kristin Barker (The University of New Mexico, USA) & Chiara Moretti (University of Bologna, Italy)
Remote

Long COVID (LC) is one of several labels used to describe a puzzling and often debilitating array of ongoing symptoms in the wake of COVID-19 disease (Thaweethai et al. 2023). Although first articulated through online-patient forums (Callard and Perego 2021), the task of creating formal diagnostic criteria for post-COVID illnesses falls on medical authorities (Pfaff, E. et al. 2022). In previous research, I demonstrated how the official diagnostic unfolding of LC hinged on a series of conceptual maneuvers, all of which reflected and produced medical ignorance vis-à-vis this newly emerging disorder (Barker et al. 2022). In this presentation, I address how these conceptual maneuvers, and the diagnostic boundaries they established, have given way to dynamics of diagnostic leakage. As particularly useful ways of foregrounding this leakage, I present the two paradoxes: diagnostic drift away from Fibromyalgia Syndrome (FMS) toward LC, and the dual diagnosis of FMS-LC. Both cases demonstrate the cultural potency of diagnoses as medical knowledge and meaning making frameworks, despite (or perhaps because of) the inherent leakage they imbue.

Leakiness and the Mobilization of Vulnerability

Lisa Diedrich (Stony Brook University, USA)
Remote

The experience of illness is multiple, and tracking that multiplicity demonstrates that even as diagnostic categories, treatments, and illness narratives seek to contain and reduce an illness experience, the experience always also overspills the diagnostic container and gets messy. In this presentation, I discuss leakiness as a key aspect of how illness is done diagnostically (in the clinic) and politically (in both online and in-person public spaces). Drawing on the example of the diagnostically leaky illness categories ME/CFS and Long Covid, I explore how those deemed outside the limits of the political—the sick and bedbound—have, nonetheless, appeared in public in a deliberate mobilization of vulnerability as a form of illness politics.

The Power of Porosity: Symptom Overlap and Contested Illness Coalition Building

Abigail A. Dumes (University of Michigan, USA)
Remote

In previous work (Dumes 2020), I have described contested illness as “any condition whose biological basis is disputed (7). These conditions tend to present with more symptoms (subjective markers of disease) than they do signs (objective markers of disease), and the symptoms themselves are often “non- specific.” As a result, the embodied and epistemic boundaries between contested illnesses such as myalgic encephalomyelitis (ME/CFS) and chronic Lyme disease tend to be “leaky.” Prior to the emergence of Long COVID in 2020, this leakiness was often a point of friction and defensiveness among patients and patient advocates because symptom overlap was

perceived to undermine the specificity of the biological basis of each condition. Since 2020, however, symptom overlap among and between ME/CFS, Long COVID, and chronic Lyme disease has become a site of mobilization toward broader collective recognition. This paper explores the power of porosity in contested illness coalition building and the conditions of possibility for the recent transformation in how symptom overlap is perceived among contested illness patient advocates.

Panel #3 Data Collection and Smart Interfaces

Stream: Technology and Data
ABS/08
Chair: Laura Kocksch

Smart Clothing in a Circular Economy: Anticipating Futures of Surveillance and Digital Divide through Data Leakage

Camillo Sanchez, Felix Anand Epp (Aalto University, Finland)
remote

Anticipatory practices materialize and reframe uncertain futures. The present work investigates the merging points of future trajectories normally observed in isolation. Our case examines the two future imaginaries of circular economy and smart garments by staging a future through prototyping and scenario development, i.e., critical uncertainties. The physical prototype we developed, is a reusable smart shirt for tracking personal and environmental data, and self-reflection. Participants experience the prototype in their own environment for two weeks and are confronted with a fictional data leakage. The scenario development identified two critical uncertainties: surveillance and digital divide. We look at these uncertainties along four exclusive scenarios, including state and corporate surveillance, as well as social divide along wealth and technology access. The anticipatory work done in the form of scenario development and an in-the-wild study serves to understand the disparity of participants' current privacy beliefs and data practices, their imaginaries of sustainable futures, and their concerns for data privacy and eventually identify the implications of the collision of future trajectories beyond a techno-economic perspective, i.e., environmental, social, ethical, and political. We examine people's privacy practices and concerns through contamination and contextual integrity. These concepts support us in disclosing the anticipatory assumptions in narratives of a sustainable future. These assumptions are contested through a semi-structured interview, which presents the prototype in the different critical uncertainties scenarios to inspire critical reflection on how past and present experience shapes future imaginaries.

Data Integration and Analysis Platforms: Desilosation and Socio-Technical Complexity

Simon Egbert (Bielefeld University, Germany)
In Person

In contemporary society that is (nearly) completely digitized, many institutions are faced with the problem of making use of the ever-increasing amount of data that can be found in an ever-increasing number of places. Organizations and companies are increasingly faced with the challenge of making a seemingly unmanageable mass of data processable in order to derive meaningful and actionable information from it. In this context, digital platforms play a prominent role not only as a major creator of this flood of data, but also as the currently most prominently discussed tools with which this flood of data is to be made manageable. It is above all data integration and analysis platforms, such as those currently most frequently used by the US software company Palantir Technologies, that have established themselves as a promising data research and analysis technology. Before this backdrop, in my paper I will introduce data integration and analysis platforms

as special subtype of digital platforms, discussing their major commonalities and differences to the more common internet platforms (Amazon, Face-book, Uber etc.), highlighting the generative role they should play: by connecting third parties (their clients) with data, as well as data (banks) among each other, they aim to enhance interoperability. That is, their main goal is desilosation, which means to transgress data boundaries, to break up data silos – data sources unconnected to each other and, hence, only accessible separately without the possibility of association. And even if there are many points of criticism associated with these platforms – one keyword: surveillance –, the criticism of automation that is otherwise often expressed with regard to algorithmic decision-making systems is ineffective here. Because these platforms are characterized by a special socio-technical interaction, in the context of which algorithms and analysts together generate new knowledge. This makes it particularly difficult, as I will show, to analyze what exactly is being done with these platforms and what risks arise from this.

Situating “Smart Grids” – Posthuman Perspectives and Alternative Visions of Energy Intelligence

Jakob Roschka (University of Kassel, Germany)
In person

This contribution delves into the historical context of intelligent power grids, commonly referred to as "Smart Grids." Through an examination of this infrastructural discourse, it becomes evident how an industrial and energy policy program, now touted as a prerequisite for climate-neutral energy supply, has its roots in the liberalization, infrastructure investment backlog, and the securitization of infrastructures. "Smart Grids" initially emerged as a response to demands primarily driven by security and liberalization discourses in the 1990s. The promise of enabling climate-neutral power supply through flexible load balancing and the resultant integration of renewable energy sources was only appended as an additional rationale to this program in the 2000s. With Smart Grids, power grids and electricity markets are electrified, interlinked with other sectors, rendered flexible, digitized, and liberalized. This vision portrays the posthuman "Internet of Things" or "Internet of Energy," where individual consumption, generation, and storage facilities are intelligently interconnected. These facilities collectively assume responsibility for network stability and other system services while also harmonizing supply and demand structures in electricity markets. This concept is underpinned by a cybernetic logic of expansion, marked by its perpetual incompleteness, continuously incorporating new practices and contextual associations. The logic of saving energy or not utilizing it at all stands diametrically opposed to this principle, almost appearing irreconcilable with the expansion and growth logic of the Smart Grid. Recent events, such as Putin's military aggression against Ukraine, underscore the vulnerability and fragility of energy infrastructures, which are now increasingly targeted for resolution through smartification. The historical and discursive contextualization of the rationale behind Smart Grids enables us to elucidate the contingency of their design and grasp potential alternative forms of energy intelligence, such as passive energy and latent energy, among others. This contribution presents findings from a discourse and pattern analysis of smart electrification from 1990 to 2015, along with initial theses from the mapping of social worlds (A. Strauss/A. Clarke) in energy system design based on interviews with experts in the field.

Panel #4 Best Practice Working Group: Code of Conduct

Stream: Stsing
ABS/105
Chair: Milena Bister

Code of Conduct Drafting Initiative
In person

stsing e.V.'s 2023 General Assembly approved an amendment to the association's statutes introducing an ombudsgroup as the fifth organ of the Association. The ombudsgroup is mandated to address issues of accessibility, anti-discrimination, good cooperation, and good conduct. Statuses aside, the precise area of activity for the stsing e.V. ombudsgroup was/is to be outlined in detail. This process has been started in the context of a year-long collective endeavor–the bottom-up drafting of stsing's code of conduct (or #CoCo). This interactive conversation seeks to reflect the key insights of the Code of Conduct Drafting Initiative (CoCoDI), which has prepared the first version of the living code that will be voted on by members at the March 2024 General Assembly. Most notable example of a collective insight was the realization that the notion of power abuse (Machtmissbrauch) was missing from the 4S Ethics and Code of Conduct Policy that was guiding our work as a 'template'. We approach this as a leak because 'power' must have appeared as an issue in the process that produced that policy. We approach it also as links worth tracing and conversing about as matters of working out our academic, organizational, and workplace politics in STS, which are at risk of leakage in the day-to-day academic practice. In deciding to attend to these leaky matters, we developed three encompassing directions for coding conduct in stsing: A) good labor relations; B) good practices of collaboration; and C) shared accountability in resolving conflict. We believe that each notion engages power relations/power differentials through a distinct sensitivity toward their pervasive effects and intersectional vulnerabilities. In their co-articulation they have propelled our drafting initiative forward. We look forward to sharing insights and actively learning from the participants' different approaches to code conduct and support good practices in and as STS.

Panel #5 Law and Leakage

Stream: Bodies and Governance
ABS/114
Chair: Ozan Altinok

Terra Saturata? Thinking International Law through Leakage

Olivier Barsalou (Université du Québec à Montréal, Canada), remote
Michael Picard (The University of Edinburgh, UK), remote

Historically, the discipline of international law justified sovereign claims over land through the doctrine of terra nullius. Terra nullius was a 19th Century legal fiction engineered to dispossess "non-civilized" peoples from their lands and activities and to justify the unlimited exploitation and transformation of nature. Terra nullius rests on an epistemology and an imaginary of the void, of the lack, of the absence, or the promise of a return to an imagined primordial wilderness.
The application of this doctrine has paradoxically led to the framing of a mirror doctrine of terra saturata, which defines an ecology saturated with pollution, waste, and toxicity in need of legal redress. Whereas the legal fiction of terra nullius created a void to be filled, terra saturata rather posits leakage to be prevented or contained by bestowing "rights of nature" to a polluted earth. However, because leakage overflows the sink capacity of the Earth and escapes the governance models designed to regulate it, international law is now engulfed and consumed by failed attempts

to account for the spillover effect of waste.
Our presentation will start by delineating the classical concept of “environmental leakage” under international law and showing its limitations in the current crisis. Second, our paper will explore emerging legal theories of leakage, following scholarships on wet ontology and volume geography. Leakage implies thinking international law and global governance with and through overflows and spillovers beyond the flat telluric/territorial/terrestrial ontologies and the logic of stasis, control, points and lines of classical international law. Terra saturata identifies three forms of saturation at work in the current global legal order: a saturation of state boundaries caused by rising sea levels, a saturation of sink capacities exemplified by ever-rising levels of carbon dioxide in the atmosphere, and a legal saturation which is seen as both the symptom and the cure of the first two forms of saturation. Saturation compels us to rethink the foundation and the role of international law in addressing not only the overflow of waste, pollution, and toxicity produced by the human species, but also how international law mediates the relations between the human species and the earth systems, terrestrial and aquatic.

Rights of Nature as a Leakage in Law? The Case of Mar Menor

Nils Richterich (Goethe University Frankfurt, Germany)
In person

In view of multiple developing crises, with the loss of biodiversity being at the forefront, Rights of Nature are being established in a growing number of countries (Ecuador, Bolivia, New Zealand). With this legal technology, 'nature' is to be granted a legal personality that makes it possible to sue for 'nature' in order to be 'better' (re)presented in the political sphere. The Spanish Mar Menor saltwater lagoon was the first ecosystem in Europe that received rights of nature. In my presentation, I will use ethnographic field notes about this space to show how local Spanish activists are aiming for much more than a legal change in 'nature'. In doing so, I will argue that these movements seek 1) legal, 2) ontological-epistemological and 3) infrastructural changes in relation to their respective 'body of water'. To stay with the image of leakage: These changes cannot be understood as an absolute rupture. Rather, the activists' efforts reveal porous zones in an anthropogenic law, in the local exploitative agricultural and tourist industry and in a dualistic nature-culture image.

Technoscientific Normativity and the Environment in the Anthropocene

Bertram Turner (Max Planck Institute for Social Anthropology, Germany)
In Person

In an STS inspired anthropology of law, the legal effectiveness of technoscientific interventions is taken into account in the conceptualization of onto-legal registers that assemble in plural legal configurations. Such further development of the concept of legal pluralism allows for addressing legal challenges of the anthropocene. The entanglements of law with technology and scientific inventories of knowledge are exemplified with respect to the world legal order’s failure to protect the environment, regulate nature conservation and care for multispecies lifeworlds by legal means. How are envisioned socio-technical futures shaped by, and respond to, changes in their nomoscapes, how do they transform the law and are transformed by it?
Such coproduction of technoscientific and legal world making is substantiated with reference to the transformation of the argan tree ecosystem in Southwest Morocco from ‘being a forest area’ into a techno scientifically ‘optimized’ plantation via lawmaking, bioprospecting, techno-normative tree standardization and molecular tree enhancement. The argan region has been designated a UNESCO biosphere reserve on account of its unique forest ecosystem while it also constitutes a ‘Garden of Europe’, i.e. the site of high-standard agrobusiness cash-crop production and has seen progressive technologization of extractive practices. At the same time, the argan zone continues to be a site of multispecies encounter and more-than-human normativity. Collaborative resistance to technoscientific optimization of the argan woodlands and the legal consequences of this connects humans, trees, mycorrhizal fungi and many others. The arganosphere thus constitutes an extended

community of plural law and legal practice. The normative power of technoscientific practices and artifacts, of matter and materiality, joins with the plural legal configurations on the ground, creating socio-techno-legal assemblages.
This research draws attention to the ways in which the analysis of such techno-legal assemblages contributes to the decolonization of the nomosphere by challenging the notion of a unifying legal universe.

Panel #6 Extractivism: Contested Energy/Futures

Stream: Energy Infrastructures & Contamination
ABS/213
Chair: Desirée Hetzel

Containing Carbon: Remediating Oil Sands Bitumen

Hannah Tollefson (McGill University, Canada)
remote

From oil spills to atmospheric carbon emissions, the threat of leakage is central to the politics of oil extraction, circulation, and combustion. Bitumen is an extremely viscous hydrocarbon abundant in Treaty 8 territory, the Athabasca region of western Canada, where it is both a celebrated source of resource wealth and a demonized source of pollution, toxicity, and greenhouse gas emissions. Faced with political resistance from Indigenous-led environmental movements to the movement of liquid oil in pipelines and the prospect of a net-zero emissions future, various actors in the Canadian oil sands have been developing and marketing oil sands bitumen for non-combustion uses and devising formats for transporting these materials in solid-phase. This paper approaches these efforts as attempts to rehabilitate oil sands bitumen materially and discursively under conditions of environmental, political, and economic duress.
Such attempts at remediating oil sands bitumen rely on both material and representational strategies of containment. Liquid oil's leaky properties are contained by resource formats like solid-phase pucks and pellets. Logistical infrastructures promise to hold these commodities safely to market. Non-combustion uses for bitumen like asphalt for roads are referred to as "contained oil spills," promising to "lock-in" carbon and forgo emissions. Representational and marketing strategies, from infographics, technical drawings, animations, and process diagrams, seek to convey clean and controlled processes aligning bitumen's commodification with the green capitalist promise of "decarbonizing heavy oil." Analyzing both the material, aesthetic, and ideological practices of containment performed by solid-phase and non-combustion bitumen products and their associated media and infrastructure, this paper investigates how leakage and containment inform the politics of this contested resource and its uncertain future.

Dark Ecologies of Decarbonization: European Lithium and the Risk of Leakage

Robert Stock (Humboldt-Universität zu Berlin, Germany)
In Person

Lithium is one of the central elements for designing future energy systems. Against the backdrop of a hoped-for end to fossil fuels, lithium, copper, and other materials are proving to be key components of a so-called sustainable energy future. But as sites in South America and China where such metals are mined show, the production of these and other raw materials is situated in the context of extractivist colonial economies. These violent practices impact landscapes, ecologies, and their non-/human inhabitants in extreme ways. Yet it is not only the destruction of territories by mining that is significant. It is also the effluents and wastewaters from these industrial processes that endanger adjacent areas and populations living there. In this presentation, I aim to show how

such risks of exposure (Alaimo 2016) to regulated, yet potentially damaging liquid leaks are negotiated focusing on the planned mine in Covas do Barroso, Portugal. First, I intend to map the actors, their knowledge, and their motivations in the face of mine planning and the associated risks. In particular, I am interested in the ways in which the risk of wastewater leakage and consequent potential pollution of the land are articulated and visualized. By drawing on a variety of materials (published documents, journal articles, audiovisual material), I aim to chart the ways in which natural ecologies, cultural landscape and industrial threats are imagined. I contextualize these processes through theoretical approaches such as transcorporality (Alaimo) and invalid/ating landscapes (Voyles 2017) that highlight the permeability of mattering flows circulating between contaminated territories, air, water and non-/human organisms. In the face of a continuous ecological crisis, environmental injustices and intensifying global conflicts, the example from Portugal raises important questions of how the dark ecologies of a fossil-free future and so-called green mining strategies are imagined from and situated within a (marginal) European perspective.

Panel #7 Architectures and the More-Than-Human

Stream: Ecological Leakage and the More-Than-Human
ABS/214
Chair: Indrawan Prabaharyaka

Xeno_Monads of Control Architectures

Sebastian Gatz (Konstfack / University of Gothenburg, Sweden)
remote

The “xeno” – “the other” – is a prefix found in such hyphenated bridge words such as xeno-poetics, xeno-politics, xeno-feminism, or xeno-architecture. Xeno is the other, the not-self. But who or what is this not-self and where is it? In contemporary architectural discourse we find an increasing interest in more-than-human justice, speculation, or realization: ontological or critical posthumanisms and new materialisms explore nonhuman agency and transhumanist or cybernetic posthumanists celebrate the (always coming) fusion of the technological other with the self in order to overcome nature’s imposed (mortal) limits. There seems to be two types of nonhumans, related to the different – and often opposing – understandings of posthumanism: the man-made nonhuman and the non-man-made nonhuman. The xeno seems to mingle with ideologies of the former.
A monad – as found in different monadologies, such as Leibnitz’ or the theosophical – is the smallest part of a metaphysical entity. It is the invisible sibling of the, almost outdated and refuted, physical atom. The monad seems to be exclusively found in life, in the non-man-made.
In my presentation I would like to speculate on the xeno_monad, a conceptual overlap of the man-made and non-man-made entity and where it is physically and metaphysically situated in architecture. Physically, we might find such speculative entity in either the weathering processes of ruination and decay (non-man-made) or in the digital infrastructures which make up an increasing part of our architectural matters (man-made). Metaphysically, we find it in human perception (does the nonhuman even exist without human perception?): it is found in the fear of losing control over (architectural) matters and (information) flows; it is found in the desire to project human shortcomings onto the other, the outside rather than the inside; and it is found in the spiritual, sometimes perversely mixed with the transhuman, desire to overcome human limitations.

Planty Infrastructures and Vegetal Place-making in Coimbatore

Charrlotte Adelina (Freie Universität Berlin, Germany)
in person

How can we think and live with vegetal life in uneven urban environments? The paper proposes to expand our analytical thinking about the making and remaking of urban spaces through the capacities and labour (Ernwein et al., 2021) of the plant kingdom. Majority of literature in the urban geographical domain has focused on understanding vegetal life through the prism of 'greening'. Attention to vegetal spatialities and infrastructural assemblages in a minor key showcases how a smart city project that aimed at rejuvenating the lakes of Coimbatore involved bund "beautification" of the bund with Lantana camara and street vendor's entanglements with avenue trees. Negotiations with the urban space intersect with charismatic more-than-humans that are actively remaking spaces and socialites, with and against the grain of planning (Hinchliffe & Whatmore, 2017) . Further, ruderal engagements (Stoetzer, 2022) and uncivil negotiations (Baviskar, 2020) by human and more-than-human actors intermesh with infrastructural assemblages in urban environments. The paper will thus attempt to lay out a way of thinking and living with plants in cities which emerges from the unruly spillages of the plant kingdom onto infrastructural assemblages in cities.

Healing Anthropocentric Leakages Through Caring Architectural Spatial Practices

Buket Samancı (Istanbul Technical University, Turkey)
Elmira Ayşe Gür (Istanbul Technical University, Turkey)
remote

Leakage means the problem of something, generally referred to as a liquid or gas, coming out of something when it should not. In this article, leakage is reconsidered from a phenomenological perspective and is associated with the undesired consequences of the Anthropocene era, which emerged because of excessive human control over the planet. With gradually increasing consequences in the World, architecture stands on an extremely dominant point. Therefore, the extremity of the architectural spatial design approach is opened to discussion with the leaks caused by Anthropocentrism. With the more-than-human theory, the potential involvement of the neglected non-human entities in spatial outcomes is the focus point of this research. Keeping in mind that the concept of aesthetics, which is highly related to the architecture discipline, is shifted within the more-than-human understanding, this study is framed by the main research question of "how to reflect more-than-human aesthetics on architectural spatial design" and the sub-question of "how to design as a practice of care". It aims to challenge the conventional aesthetic means of architecture and synthesize potential approaches and the leaks caused by the Anthropocene. Within this approach, which can be called caring spatial practices, several case studies are analyzed and discussed by reflecting on the alternative meaning of aesthetics in architectural spatial design. These case studies are selected by their quality of decentering the human on design approach by caring for the non-human through the relations of various entities and their possibilities on the spatial design. Then, a multidisciplinary focus group interview is carried out and the classification of the spatial practices that may cause "leakages" and/or the potential of healing them is made through conceptual thematic analysis. In this manner, this study enables to overview of the more-than-human aesthetics of architectural design and caring practices through different perspectives on the leakages.

Panel #8 Aerial leakages: Entanglements, Knowledges and Interventions in Urban Airs

Stream: Ecological Leakage & the More-Than-Human
ABS/E11
Chair: Maximilian Gregor Hepach (Session 1);
Elisabeth Luggauer (Session 2);
Brett Mommersteeg (Session 3)
Discussant: Nerea Calvillo (University of Warwick, UK)

Air has become a key matter of inquiry, concern and design (Connor 2010, Calvillo 2023). Modern cities have deployed strategies of encapsulating, bottling, sampling, and measuring air, as well as designing and implementing "air-conditioning" systems to envelope life (Sloterdijk 1998; Latour 2005). Yet air remains elusive and is, indeed, a key medium for environmental leakages. Think of pollutants and toxins, methane emissions, smells, noise, light, radiation, heat, viruses, etc. These phenomena not only involve bodies, materials and infrastructures that leak waves and particles into the air, but it is also the air itself that circulates in not fully controlled forms, leaking into bodies, spaces and environments, and challenging modernist epistemological binaries (see e.g. Choy 2011; Adey 2014; Graham 2015; Ahmann & Kenner 2020). Accordingly, this panel focuses on current problematizations of air as flow and medium for a diversity of "environmental stressors". We are particularly interested in bodily, everyday experiences and problematizations of aerial leakages, their vaporous and diffuse character, and the challenge of "attuning" or becoming sensitive to the air. This also entails paying attention to expert knowledge practices that seek to measure and give shape to the air. In both cases, though, a key question is the extent to which these "leaks" provide conditions for knowing and experiencing the urban air.

We are interested in contributions that address aerial leakages from different perspectives. Some relevant questions involve:

1. How leakages prompt scientific, technological and everyday interventions into the air, its flow and composition in order to enable, modulate or mitigate "environmental stressors"
2. How leakages provide ways to "attune" to the air, and the instruments, equipment, and practices through which airy, immaterial and elusive phenomena become graspable and problematizable
3. How leakages generate conditions for developing non-modern or other ways of knowing the urban through its airs

Session 1: Attunements

(09:45–11:15)

Aerial Cosmopolitics: Cohabitation, Leakages, and the Terrestrial

Ignacio Farías (Humboldt-Universität zu Berlin, Germany)
in person

As a way of introducing the panel, this contribution will propose conceptualizing air as a cosmopolitical medium. If cosmopolitics is a shibboleth for the opening of thought and action towards the unknown, the indeterminate and the im/possible (Stengers 2005), then the air is a medium or "plasma" (Latour 2005) through which the not yet known formed emerges, takes shape and confronts actors with their presence. Aerial challenges to established definitions of what a common world of cohabitation might entail become particularly apparent in urban contexts, where aerial flows, leakages and disturbances cut (and slip) through the modern partitioning of space across great divides between the bodily/environmental, the interior/exterior, the residential/non-residential, or the urban/rural. Starting from these observations, this contribution will develop three conceptual propositions concerning (1) the air as a virtual medium of urban cohabitation, (2) leakages as a specific cosmopolitical operation distinct from the disruption of new cosmograms, the

dithering of the idiot or the compositional work of diplomats, and (3) aerial cosmopolitics as a mode of reshaping the terrestrial.

The Whence and the Where of Cool Airs: The Materialization of Stuttgarter Aeroroutes

Indrawan Prabaharyaka (Humboldt-Universität zu Berlin, Germany)
in person

This paper redescribes the stories of climatologists-cum-bureaucrats and concerned citizens in Stuttgart who materialize the aeroroutes, the corridors through which fresh and cool airs are channelled from the forests, gardens, and green areas up on the hill to the lower parts of the city. I am thinking about the aeroroutes with and against Tonino Griffero's characterization of wind as a quasi-thing that "does not properly have a whence and a where." It is difficult indeed to isolate wind as a discrete object. However, just because wind is unisolable and doesn't "properly possess the spatial sides," it doesn't mean it cannot be spatialized and materialized. Instead, the atmospheric routes in Stuttgart are a product of theoretical and applied interventions and conflicts. They are measured, drawn, planned, debated, challenged and maintained as they materialize and function as a citywide cooling infrastructure. By seriously locating the where and the whence of cool airs (if any), I hope, one can begin to locate the aerial leakage (if any).

"La chasse aux décibels" [Decibel Hunting]: Listening to the air in Paris

Brett Mommersteeg (Humboldt-Universität zu Berlin, Germany)
in person

While sometimes called a "Cinderella" pollutant, noise has recently been foregrounded as a leading atmospheric risk (King and Murphy 2016; UNEP 2022). In response, there has been a proliferation of noise maps, sensors, infrastructures, and measurements in which the atmospheres of cities have become sites for knowledge production and the governance of noise. And yet, it is often noted how challenging it is to fix noise as an object of knowledge. As Marina Peterson (2021) has shown, while it can attune us to the air, noise is nevertheless indefinite, elusive and indeterminate, situated uneasily between inscriptions and perceptions. Noise is sound that leaks: physically, categorically, and perceptively. Drawing on ethnographic fieldwork with a noise observatory in Paris called Bruitparif, a principal actor in what a politician in Paris wryly described as "decibel hunting," this talk is situated within two sites of hunting decibels: in noise measurements and neighbourhood meetings. It describes both the "apparatuses of atmospheric attunement" (Choy 2018) – the technologies and practices – for capturing noise, and the ways in which noise resists capture, leaking between decibels and descriptions. Ultimately, this talk explores how noise is hunted in Paris, and outlines the epistemological challenges of listening to the air for noise.

Attunements to Ground-Touching Clouds in Lima: Fog Capture as a Means to Experience the City In a Different Key

Chakad Ojani (Uppsala University, Sweden)
in person

In the Peruvian capital of Lima, conservationists, NGOs, and residents on the urban periphery try to enfold ground-touching clouds in alternative water supply systems and infrastructures of fog oasis ecosystem conservation. While residents and NGOs set up fog catchers to respond to the absence of state infrastructure, local conservationists adopt similar methods to repair damaged more-than-human relations. Amid these attempts to bring about new environmental-infrastructural relations, the fog catchers have re-attuned residents to previously backgrounded features of the city and the atmospheric: whereas the sky now abounds with water in atmospheric suspension, urban asymmetries are rendered conspicuous through the foregrounding of airborne pollutants. However,

as comings and goings that are there in one moment and gone the next, ground-touching clouds present challenges to infrastucturalization. Their ephemerality and enveloping qualities render them leaky, and the clouds also become conduits for various undesirable aerosols. Along the process, fog capture affords novel ways of knowing and experiencing the city through its airs. In this presentation, I describe how the fog catchers have gone from tools of scientific inquiry to alternative water supply systems to untintended, speculative devices for ”explicitating” hitherto imperceptible socio-political and environmental relations in coastal Peru.

Session 2: Inside/Outside

(11:30–13:00)

Material Textures, Sensory Experiences, and Household Interventions in Coping with Heat Leakages

Paz Araya (Humboldt-Universität zu Berlin, Germany)
in person

In the cities of Chile's Araucanía region, the cold season prevails from April to November. Outside houses, air temperatures average 5°C during the coldest days while inside, families huddle close to fires for warmth. However, away from the fire, indoor temperatures are not very different from those outdoors. Coping with energy precarity in everyday life often requires creative explorations with the socio-material aspects of houses and techniques for repurposing materialities. This paper draws from ethnographic fieldwork conducted in the cold seasons of 2022 and 2023 in Araucanía. It aims to unfold the circulations of warm and cold air among different layers, encompassing outdoor spaces, buildings, and human bodies, as well as the interventions made in the leaky building envelope in household settings. I argue that the constant assemblage and re-assemblage of material layers, and the configuration among their thermodynamic properties, textures, porosities, and sensory qualities, play a fundamental role in the permanent negotiation with heat leakages in households. It points out that each layer of material has distinct thermodynamic properties and sensorial attributes, intricately modulating sensitivities, and the experience of thermal comfort and warmth deprivation. This paper describes how these sensitivities are embedded in political contexts, personal narratives, and embodied know-how.

Leaky Micro-Environments: Managing Aerial Leakages in Japanese Architecture and Fashion

Margherita Tess (Humboldt-Universität zu Berlin, Germany)
in person

In an increasingly warming planet, cooling bodies and buildings has become a compelling problem. In architecture, the management of aerial leakages is a fundamental thermal practice. Aeration is a source of passive cooling, but draughts and openings threaten air-conditioned cool spaces. Japanese vernacular architecture has always preferred natural ventilation, and till now, most of the buildings are not insulated. Mirroring architecture, new textile technologies promise to condition bodily-atmospheric leakages through “cool textiles”. As revealed by my ethnography in Fukuoka, a widespread way of dealing with heat is through breathable synthetic clothes that enhance sweat absorption and desorption. Focusing on the link between architectural and clothing environments and their management, I will explore strategies for cooling down through the management of bodily-atmospheric leakages. The socially loaded regulation of bodily leakiness will be analysed specifically through the fashion practices of engineering of the “clothing microenvironment”. This term, borrowed from textile engineering, brings to the foreground the space where bodily-atmospheric leakage management happens and where body and climate overlap. Moreover, I will claim that this thin porous layer between the body and the atmosphere works similarly to a “critical zone”: it is where cooling garments and wearable cooling devices act to allow bodies to effectively regulate temperature in extremely hot and humid environments.

In Atmospheric Fragments: The Dis/Continuity of Urban Care Amidst Heated Humid Airs

Tomás Criado (Universitat Oberta de Catalunya, Spain)
remote

After increasingly scorching years, urban heat is one of Barcelona's main concerns. In the last decade the city has engaged in a massive investment in meteorological measuring (mapping urban heat islands, calculating 'torrid' and 'tropical' nights, considering their health effects) and the implementation of mitigation or adaptation infrastructures, such as: a network of 220 'climate shelters', object of heated public debates due to their discontinuous opening times; 'nature-based solutions', whose maintenance has suffered from recent severe droughts; or an incipient shade infrastructure. Drawing on ethnographic fieldwork in one of these initiatives (a municipal architectural contest to prototype shade solutions for large spaces, narrow streets, and playgrounds of the whole city), I ponder how the leaky features of humid airs pose peculiar conundrums for approaches to climate action premised on either patchy attempts at cooling urban atmospheres by blocking or reflecting sun rays (having no effect in maddening nights where no wind blows) or implementing pockets of total climatic control (turning energy poverty into a crucial environmental justice issue). Revisiting STS works on the dis/continuity of care, what might be at stake here is not searching for the lost whole but more careful attunements to leaky humid airs whilst operating in atmospheric fragments.

Air Purifiers As Psycho-Machines

Maximilian Gregor Hepach (University of Potsdam, Germany)
in person

Air is an everyday medium through which we move and breathe. Being ubiquitous but unnoticed enables the air to function as a medium of displacement in the psychoanalytic sense. Attributing different physiological and psychological states to something (idiomatically) "being in the air" helps explain both how one is exposed and why the source of exposure is ultimately indeterminable; the air provides a transparent model and an elemental logic for intransparent experiences. In "Air purifiers as psycho-machines", I surface how air purifiers at once promise to clean the air and render it opaque. Their advertisement takes advantage of air as a medium of displacement to render air inside and outside a matter of concern, commodifying it. Building on recent work at the interface of phenomenology and psychiatry, I argue that air purifiers are best understood as "psycho-machines", as a "materialised or objectified subjectivity", shaping how one makes sense of and experiences the world. Air purifiers are then not only a response to 'bad air' leaking in from outside or exuding from pets, plant soil, couch cushions, and gas stoves into our porous lungs. The meaning they themselves create leaks and seeps into the very constitution of one's experience and subjectivity.

Session 3: Governing and Caring

(14:30–16:00)

Mobilizing leakages to condition one's air: Inventive and subversive hacks of aerial capsules in Downtown Vegas

Elisabeth Luggauer (Humboldt-Universität zu Berlin, Germany)
in person

Increasing awareness of thermal comfort and air pollutants brings to the fore a perspective on urban spaces as patchworks of different encapsulated and fortified microclimates and airs (e. g. Roesler/Kobi 2018; Nguyen 2020). As such capsules appear e. g. street canyons, vehicles, buildings, or parks. Enforcing the - sometimes more and sometimes less solid and noticeable envelopes around

these micro-atmospheres by tackling and sealing their leaky spots and moments happens primarily with the purposes of thermostating airs and regulating energy consumption. This urban aerial patchwork is particularly experienced in moments when the airs of very different bubbles leak into each other (e.g., stepping out of an air-conditioned supermarket in a busy road on a hot day), but also when the transition between capsules is prevented. This paper unfolds practices of an entanglement of two humans and one dog, experiencing homelessness, of navigating through the patchwork of hot outdoor and cooled indoor urban airs in the Downtown of Las Vegas. Building on their strategies to hack and mobilize leaks in various envelopes to get access to cool air and water, this contribution problematizes the enforcement of closed-up capsules as a mode of governing the city through its air and climate.

Subsurfacing Air. Breathing Bodies, Leaks, Bioindicators

Léa Perraudin (Humboldt-Universität zu Berlin, Germany)
in person

In channeling the elusive and unruly materiality of air as a social, yet precarious medium, scholarship is mostly concerned with spaces above ground. However, practices of sensing, measuring, and metabolizing (contaminated) air as well as techniques of air conditioning to regulate, control and standardize also occur elsewhere: the subsurface. The paper attends to underground air in anthropogenic subterranean hollow landscapes of (urban) bunkers and mines. These sites shed a particular light on how the survival of breathing bodies is delegated to means of technology. In their stabilizing attempts to transform hostile spaces into contained and profitable environments they continue to engender both tropes of terrestrial-infrastructural progress and neo-colonial extractivist technocapitalist logics. Mining and bunker infrastructures have a long history of handling underground air and its potentially life-threatening impact on humans, e.g. through sentinel species to detect toxic carbon monoxide or the monitoring of microbial and particulate contaminants. As such, the vulnerability of breathing bodies becomes a matter of transcorporeal, at times non-consensual, bonds across species, particulates, and infrastructural means. By engaging contributions from the exhibition “Airbound. Sensing Collective Futures” which I co-curated at Matters of Activity, Humboldt University Berlin, I ask what modes of critique these speculative investigations of bearing witness, of sensing and of attuning to compromised air might hold.

Panel #9 Once Upon a Leak:
Technoscience Stories From Museums, Collections, Archives, and Databases

Stream: Tech & Data
ABS/E04
Chair: Lisette Jong (Session 1);
Tahani Nadim (Session 2)

Session 1 (09:45–11:15)

Leak-Proofing the Past

Martina Schlünder (Max Planck Institute for the History of Science, Germany)
in person

The sources from which the historians of science drink do not consist exclusively, but still predominantly of paper and its digitized copy, stored in archives. Since not every form of paper is kept, the question arises as to what we consider so valuable or dangerous that it must or should be kept, and with it the question of what can be tapped and read from the past at all, or what may drip or leak. And how that can happen. In my contribution, I will discuss the different boundaries and

their materiality in archives and among their users that can create very specific forms of leaks. One of the most important boundaries in archives is again a temporal one namely the so-called embargo period. Other boundaries arise from the materiality of the media (paper or digital), which make different forms of leaks possible and are therefore reflected in other politics of containment. From these, in turn, the historian can see how leaks and its consequences are imagined by the custodians of the past.

Archive at Rest

Baldeep Kaur Grewal (University of Potsdam, Germany)
in person

I will present from my thesis chapter titled ‘Not Listening: methods for ontologically damaged archives'. This chapter presents two case studies from the research phase to understand the process of conducting research during a pandemic while both resisting and capitulating to the velocities of academic work. Understanding archival documents as media objects with rules around access and use, the two case studies illustrate ways of obeying or disavowing these instructions. One case focuses on the design of colonial administrative documents retrieved at the National Archives of India.The second study engages with Lautarchiv at Humboldt Forum that contains recordings from Sikh Prisoners of WWI. Here I argue for a method of not listening, of not putting these audio records to the work that has been designated for them by the ethnographers who produced them unethically.

Endangered Tapes, Endangered Species

Sandra Jasper (Humboldt-Universität zu Berlin, Germany)
in person

From the 1930s onwards, animal sound recording began to flourish with the development of portable magnetic tape recorders that facilitated proximity to wild animals. As animal recordings and interest in them started to proliferate, wildlife sound archives emerged as new sites of specialist scientific data sets. Many sounds collected in the two largest wildlife sound archives in Europe—the British Library’s Wildlife and Environmental Sounds collection and the Animal Sound Archive at the Museum für Naturkunde in Berlin—are recorded on magnetic tape, an easy medium to use, duplicate, and circulate. Today, not only numerous of the recorded species are under threat of extinction, but the tape collections themselves are critically endangered. Exposure to humidity, for example, has resulted in the sticky-shed syndrome, which makes magnetic tape fall apart on playback. While being digitized, some affected tapes are played for the very last time before disintegrating completely. Will the contemporary rush to digitization transfer sound collections into a safer, more stable and lasting medium? Curators and archivists have expressed their concerns over digital malfunction, the loss of data, and ethical custodianship with the increasingly unrestricted access to animal sound recordings via online platforms. What happens when recordings of rare species are “leaked” into the world, potentiality getting into the wrong hands? Lastly, with the environmental damage involved in producing and consuming digital media, further “leakages” appear when we consider how digital sound recordings that are archived, consumed, or used in conservation efforts in one place, have produced and continue to produce the destruction of habitats and livelihoods elsewhere.

What’s Up, Z-drive?

Roos Hopman & Tahani Nadim (Humboldt-Universität zu Berlin, Germany)
in person

On August second, 2023, all employees of the Museum für Naturkunde Berlin received an email with the topic “Z Drive: what’s up?”. In this email, a scientist based at the museum asked for clarification

of the current “situation”. Comparing the museum to Deutsche Bahn, he wrote that he felt like he was sitting on a stranded ICE train, waiting for the conductor’s announcements. While the museum is not much of a train, the email did come in the wake of a crash: the museum’s main digital storage space, the Z drive, had become suddenly inaccessible. This “Z-Krise” had catastrophic consequences, as many workers’ projects and day-to-day activities relied on the accessibility of these data. Museum management said during a crisis meeting that the crash had revealed pre-existing “fragilities of the system”. Another attendee emphasized this had been a long time coming, that it was not like the museum “had been struck by a meteorite”. In a public statement on the museum website the crash was described as incidental, a “technical malfunction”. Asking the Z drive what is up, we invite conversation on the distributedness of infrastructural leaks.

Natural Horror Museum

Veit Braun (Goethe University Frankfurt, Germany)
in person

What seeps into, what trickles out of natural history museums? The popular idea about these institutions (and my own fascination with them) is that they serve as nature’s archive: taking in everything from the smallest fly to the biggest whale, never to let go of it. Nothing, we are promised, will leave the museum’s collection, even if it has long vanished from the outside world. Of course the longer you study these places, the more aware you will become of what inadvertently finds its way into their carefully curated collections, and what in turn disappears from them. As if the museum was a haunted house, you will find that specimens, names and histories are constantly misplaced; that strange logics and imperatives take hold of its departments and staff; that you recognize the museum’s peculiar aesthetics wherever you go. Is this why I find them increasingly uncanny? Or is it, perhaps even scarier, that the stories are in fact true: that despite everything that secretly enters or leaves its collections, the museum will forever hold onto them? This is an attempt to come to terms with my growing horror with natural history and its homes.

Session 2 (11:30–13:00)

The Politics of Disarticulation: Legacies of Strategic Concealment in the Traffic of Racialized Human Remains

Paul Wolff Mitchell (University of Amsterdam, Netherlands)
remote

Racialized human remains collections arose when medical-anatomical practices met questions about racial differences constructed in Europe in the 18th century. In this presentation, I will focus on how practices and presumptions filling dissection halls with the bodies of socially marginalized people in Europe, and later North America, in the late 18th and early 19th centuries were extended into an expanding colonial-imperial network to amass racialized bones for study and display. Through attention to cases entangled in anthropology’s disciplinary formation, centered in present-day Germany, Indonesia, South Africa, Scotland, and the United States, I will trace the “politics of disarticulation” - acts of strategic concealment and depersonalization by which anatomists intentionally detached identities and histories from human remains to enable collection of these remains in the face of resistance from the deceased’s kin and community. These cases suggest that much of what is unknown about the history of the bodies in anthropological human remains collections may be by design. Moreover, they show how colonial power relations structure what information anatomists recorded and published about the human remains they collected. I will conclude with reflection on how the emergence and persistence of anthropological human remains collections has always crucially hinged on acts of strategic concealment and depersonalization, and

how these histories complicate present ethical considerations concerning the embodied afterlives of racial science in museums.

A Life in Stones

Regina Sarreiter (Independent Scholar)
In person

The long temporality of stone artifacts made them particularly attractive for the then young disciplines of archaeology and ethnology at the beginning of the 20th century. They regarded stones as witnesses of early life and petrified expressions of craftsmanship and everyday life. Stones can survive millennia without changing their materiality, maybe a tiny piece chips off an arrowhead or a firestone at some point but stones tend to keep in shape. In our panel I want to talk about a group of stone artifacts that are presumed to have been shaped and put to use during what is widely referred to as the Stone Age. The group resurfaced during an excavation in the dri1920s in today's Kwa-Zulu Natal in South Africa. Dug up by a German Catholic missionary and ethnologist, an Austrian archaeologist and a German shopkeeper living in South Africa, the stones began to leak into different collections in South Africa, the Vatican, Austria, and Germany. They were put in paper bags as specimens of the Stone Age in a national archaeological collection, elsewhere they continue to create confusion: shall they be placed in a natural history collection or in the anthropological collection? Another institution deemed them meaningless stones and discarded them altogether during their renovation. But who really cares? I did for a while...

Bronze Tooth Hippo: An Uncomfortable Story in the Natural History Museum

Lisette Jong (University of Amsterdam, Netherlands)
in person

According to Specify, the collection management software used at the Museum für Naturkunde in Berlin, there is nothing remarkable about the hippopotamus amphibius skull with collection number 44088. Placed in a row of hippopotamus skulls on the bottom shelf of a cabinet in one of the museum's storage rooms it sits as one amongst many exemplars of its species. It seems that for long it has gone unnoticed or at least unrecorded that the teeth in the skull are replicas cast in bronze. Until last year the curious gaze of a newly hired preparator was caught by the juxtaposition of bare bone and sculpted metal. Upon their first meeting, he introduced the visiting STS researcher to the bronze tooth hippo and she started an investigation into its trajectory. Now she is here to tell a story about transformation, valuation and neutralizing data infrastructures in the natural history museum. The encounter with bronze tooth hippo instigated a story that addresses the fluid boundaries between colonial hunting trophies and natural history specimens and how uncomfortable knowledge seeps in and out of porous institutions.

Double Poison

Aram Lee (Independent Artist, Netherlands)
remote

I question whether the museum can embrace any living thing so it can become a space that allows life. Can microbes in the museum be these kinds of life, these living cultures? Such a question makes us think about what constitutes life and what can be considered life. Departing from research into the waterway infrastructure of the museum as a potential respiratory system that can 'ecologize' and detoxify the space, I have found life in a place that attempts to keep it out. The micro and macro-scale perspective will re-establish the connection between the surroundings of the museum's enormous, controlled system and an object's positionality in our sensorium. This can activate a liquid interchange between the invisible microorganisms, poisoned objects from the

collection and audiences. The ultimate territorial border of the museum, starting with its microbes, moving to its building, then to the city, to the canal, and to the ocean, will be marked and seen by the uncontrollable, unruly choreography. These organisms can deconstruct the categories of body, species and territories if we let them. It's in this in-between state that knowledge can leak and flow, creating a space where all agencies might be able to participate in the museum.

When the Hidden Objects of Race Science Resurface: The Hauntings of Racism in Anthropology

Thiago Pinto Barbosa (University of Göttingen, Germany)
in person

Scientific racism used to permeate many of anthropology's theories, methods, and assumptions. Although anthropologists have deployed considerable efforts in curating their discipline out of its entanglements with racism, such racial knowledge is also inscribed in objects with a material existence that has persisted throughout discursive efforts to move beyond "race". While some of these objects have been hidden, taken away from the museum or put in the depths of dusty cupboards, they sometimes resurface. But what happens when those objects that are associated with "race" reappear or are brought to light? In other words, what can we learn from those moments when the suppressed past of scientific racism leaks into the present of anthropology? I examine three instances in which objects of race reappeared, following three types of material objects (physical anthropology books, anthropometric measurement instruments, and racial typology sculptures) as well as objectified human remains, in Germany and India. I argue that the hauntings caused by the sights of "race" are a sign of the incomplete move away from scientific racism. This signals towards the still necessary efforts to tackle and overcome such a difficult legacy, which will hardly be solved by tactics of suppression.

Panel #10 Between Intervention, Activism and 'Sound' Social science Research. Challenges in Feminist Science & Technology Studies

Stream: Methods & Epistemologies
ABS/E08
Chair: Martina Erlemann & Petra Lucht

Empirical approaches in Feminist STS often go beyond canonical methods of qualitative social science and cultural studies and leave the safe terrain of the analytical descriptive paradigm. This produces at least tensions between Feminist Research and research that claims rather to be descriptive, value-neutral, "purely academic", in short: apolitical. Nevertheless, feminist approaches have been applied persistently, affecting a leakage in the norms and requirements of orthodox methodologies in sociology and/or cultural studies. From the beginnings of feminist critique of the natural sciences, among them Evelyn Fox Keller's "First Person Account" in 1995, Feminist STS was also an activist, interventionist and political endeavor. In this panel the tensions and challenges that Feminist STS researchers might still be confronted with when pursuing their research interests are addressed. This touches questions of:

- In what ways does Feminist STS lead and has been successful to diffuse the distinct disciplinary paradigms of the natural sciences and technology on the one hand and the inter- and transdisciplinary paradigms that foster STS and Feminist STS on the other hand?
- What does it mean to do self-reflexive research on issues that touch social inequalities, imperialist and neo-colonial practices, urgent global challenges that are ingrained in the natural sciences on the one hand while at the same time the natural sciences hold on to objective research paradigms?

The papers of this panel have in common that they address societal challenges, like e.g. climate catastrophe, in/exclusion in the academy, the robotization of borders and engage with activist agendas or ethical issues.In the first session of this double-session panel four papers will be presented that discuss the issues raised in the panel. The second session is workshop-like where attendants of the panel are invited to share their experiences of the challenges that Feminist STS brings along.

Session 1 (09:45–11:15)

Researching In/Exclusion Processes Within Your Own Department: A Study On Workplace Cultures in a Physics Department in Germany

Andrea Bossmann & Martina Erlemann (Freie Universität Berlin, Germany)
in person

Researching in/exclusion processes in academia is positioned in between gender and diversity politics and academic research which is usually located in the social sciences or cultural studies. In particular when investigating in/exclusion processes in STEM cultures, this is in contradiction with expectations of fulfilling norms of sound research having to be apolitical and value-neutral – norms which are upheld widely in many STEM fields. This can even be more challenging when feminist researchers who set out to study in/exclusion processes in STEM, are institutionally affiliated to the very STEM institution they study. In our contribution we present an ongoing project that studies workplace cultures of physics in an intersectional perspective in order to explore potential in/exclusion processes due to various social dimensions and their intersections. Our case study is the very physics department we are affiliated with as feminists science studies scholars. Beyond this, we apply participatory elements that strive to empower potentially discriminated groups and to ensure or even establish an inclusive workplace culture at our department under study. A special focus of the paper will be placed on the challenges that arise in this particular constellation.

Realities of Lithium Streams: Disrupting Hegemonic Narratives on Green Energy

Tamar Raquel Grosz Moraga (Independent Scholar)
in person

When capitalist ambitions couples with dreams of limitless energy production and omnipotent fantasies, it accelerates bodily, cultural, and ecological processes. Diverse cultural forms are molded into a singular fantasy of a future of green technologies, while various beings passively yet forcefully sustain an extractive system, only to be subsequently disposed of as 'unprofitable,' leaking into a cultural and political void. In the Atacama Desert lithium ancient salt beds are awakened from their aeons dream. Extracted from the brine, purified lithium emerges readily accessible in our hands, as energy resurveys for cars, cellphones, and electric devices. A continuum fuel for a “civilized” society. This perpetuating cycle constrained western cultures to a daily use of energy technologies; lithium, put to hold energy for a short moment, ensuring a continuum surge of energy at any disposition, solidifying the mechanism by creating hegemonic realities. In Latin America, this reality sparks an activist academic discourse that recognizes the necessity to foster a counter-interventionist science, capable of dismantling the prevailing extractive paradigm. The paper addresses an onto-epistemological, ethical examination to sustainability issues by engaging with worldviews of indigenous communities from local Salt Lake regions. This interaction aims to challenge the hegemonic narrative on green energy.

Leaky Boundaries? Intra-acting in the Robotization of Borders in Europe

Isabel Matthias (Technische Universität Berlin, Germany)
in person

"Not least due to the fear of leaky borders, states and conglomerates of states expedite the robotization of borders aiming to selectively facilitate the border management among global lines of power. As a result, borders are being extended to digital spheres and become multi-local. Consequently, the robotization of borders outlines the discrepancy of a physical act and the moving between embodied-material subjectivity and dis-embodied-digital entity. Depending on one's situatedness in the world, this can highly influence one's life (or death). At this juncture, the specific, future configuration of the robotization of borders still depends on the intra-actions of the involved actors: Who will develop the robots? Who will make use of it against whom? How will humans and non-humans live together? And will there be queerings by activists, hackers or artists? Therefore, against the backdrop of the multiple crises of our times, the robotization of borders in Europe shall be examined. The goal is to identify both potential problems of the robotization of borders and open up a space for envisioning livable futures for different entities on this planet.

Schools as Local Climate Actors in the Neighborhood

Petra Lucht & Franziska Kaiser (Freie Universität Berlin, Germany)
in person

Climate change is generally known, but the increasingly extensive knowledge about climate change does not lead to the implementation of climate adaptation measures in everyday practice. The presented project aimed to develop concepts for bridging the 'knowledge-action gap' with a participatory project design and with references to practice and action, thus motivating and empowering heterogeneous target groups to take climate-friendly actions and decisions. The project's starting point is to work with schools translating academic knowledge on climate change into everyday action with reference to a Citizen Science project in class. Scientific findings on climate change become comprehensible in the context of such a project-oriented and research-based learning in order to develop concrete climate adaptation measures. In the second phase of the project, climate adaptation measures were prepared in such a way that high schools cooperated with neighborhood residents as well-connected actors at the local level. These co-operations were initiated through local activities in the neighborhood, e.g. through district offices. Research perspectives of Citizen Science, climate research, participation research, educational research and evaluation research. In this paper, we reflect whether and to what extent critical-reflexive approaches of the Feminist STS were guiding in the context of this inter- and transdisciplinary project.

Session 2 (11:30–13:00)

Workshop/ Discussion with all presenters

"Experiences in feminist STS"

Panel #11 Data Deluge and Leaks in Border and Migration Governance

Stream: Bodies & Governance
ABS/114
Chair: Jasper van der Kist & Paul Trauttmansdorff

In an era characterized by substantial levels of mobility, the rapid circulation of commodities, and the widespread dissemination of information, the governmental imperative for effective border and migration control has become increasingly significant and complex. In light of these challenges, datafication has rapidly emerged as the predominant and favored solution. The implementation of large-scale information systems and digital technologies has made borders simultaneously more fortified and permeable, further complicating the already opaque and ambiguous practices of managing migration. At the same time, this has opened up new opportunities for individuals to circumvent these practices and traverse borders.
This panel brings together contributions that delve into the conference theme of 'leakage' within the context of data-driven border and migration governance. The concept of leakage has an intricate and notorious relationship with borders—reminding us of cracks, liquidity, or porosity that haunt every border, regardless of how securitized and technologically advanced it may appear. Seeking to mobilize the generative potential of leakage further, the panel seeks to generate new perspectives and terminologies for understanding the processes of border datafication: What can we see when breaking with the conventional image of watertight border controls and infrastructures and instead focus on phenomena of 'overflow,' 'excess,' 'spill-over,' or 'data loss'?
Building on science and technology studies, critical border and migration studies, and critical data studies, we invite papers that consider or take as their point of departure themes such as:

i. The fraught circulation of data: How do erroneous information, mistakes, or inaccuracies enter data flows in border control? How do data become corrupted, manipulated, or misinterpreted?
ii. Data overflows or data excesses that exceed the normal/intended frame of calculation: How does sensitive information related to border control practices spill into public domains? What if border authorities fail to keep the personal data of migrants protected?
iii. Leaks as a critical mode of intervention: How do entering data or data escape constitute activities of managing, challenging, or subverting border control?

Imagining Alternative Data-Futures in Migration and Border Management

Jasper van der Kist (University of Antwerp, Belgium) and Paul Trauttmansdorff (European New School of Digital Studies, European University Viadrina)
in person

In light of the ongoing issues and debates on migration management, and the current trend of seeking salvation in datafied solutions, this paper explores the (im)possibilities of imagining different data-futures. It builds on the results from experimental and collaborative methods during a workshop in January 2024, which brought together practitioners, experts, and stakeholders in the field of migration management. The workshop encompasses two different collaborative methods: (i) the collaboratory, in which participants engage in 'concept work'; (ii) 'experimental design', where a hands-on exercise of prototypes seeks to re-imagine relations between people and technologies. Further drawing on research findings from the field of STS on public participation and civic engagement, we explore the question of whether it is possible to apply their ideas and insights to migration and border studies. Is it possible to imagine a new regime of data sharing in migration management that resists today's excessive forms of data collection and repressive forms of data usage? Can we conceive different relations between states and migrants in the production of digital data and methods? Is it even reasonable to think that collaboration between academia and governments would be possible without risk?

Data, Voice, Algorithm: Leakage and Datafication in Media Technological Border Control

Michelle Pfeifer (TU Dresden, Germany)
in person

In this paper, I will look at three instantiations of media technological border control in Europe; database, voice recognition, and smartphone data extraction; to theorize the relationship between datafication and leakage. Drawing on 18 months of multi-sited ethnographic fieldwork in Europe, archival research, and critical documents, policy, and discourses analysis I demonstrate that data-driven projects that are framed as governmental reforms purportedly meant to fix the perceived crises of migration by means of sealing off borders that function to deepen inequalities and enhance border policing. I argue that the operations of data, voice, and algorithms transform where we locate the border both spatially and temporally as I focus on the mundane administrative practices of asylum administration and situate data-driven and algorithmic technologies within genealogies of colonialism and racial science. I mobilize leakage as a concept in two ways. First, leakage allows us to understand the colonial continuities of contemporary border technologies. Second, through leakage we can analyze and track how data points move along, through, in service of, and despite borders. Drawing on these findings I will reflect on the possibilities and limitations the metaphor of leakage offers to analyze and critique datafication in relationship to technological border policing.

Mobilizing Digital Archives in Asylum Procedures: Epistemic Practices and Conflicts of Mobile Phone Usage for Evidence

Ivan Josipovic (University of Vienna, Austria)
remote

In the absence of paper-based means of evidence on asylum seekers in European immigration bureaucracies, mobile phones and other personal devices are increasingly used to make claims about an applicant's identity, travel route or country of origin. Drawing on the cases of Austria, Germany and Switzerland (DACH region), I ask how mobile phone data has been mobilized as a means of evidence in asylum procedures since 2017. Based on a qualitative analysis of 56 decisions by administrative courts in the region and using a sociology of translation, I identify four epistemic strategies through which mobile phone data are mobilized both by asylum seekers as well as immigration officers and courts in order to (de)legitimize applicants' claims. The findings are supplemented by 10 interviews with Swiss practitioners in asylum bureaucracy on the implications of a standardized smartphone data extraction procedure, as has been legislated in the entire region since 2017. In this vein, my contribution will add to a better understanding of how the proliferation of personal digital archives shapes knowledge production in asylum procedures and eventually affects the possibilities and constraints for claiming rights.

'Fixing the Leaks' — On the Role of Synthetic Data in Biometric Identification

Thea Kirsch (Freie Universität Berlin, Germany)
in person

Over the last two decades, the use of biometric identification technologies for border and migration control purposes has rapidly proliferated worldwide. Previous studies have highlighted the key role that techno-scientific promises of efficacy, objectivity, and accuracy play in this context — and how biometrics continuously fail to deliver on these powerful promises. As a result, the datasets used for the training of biometric identification algorithms have come under increased scholarly and public scrutiny. In contrast, relatively little attention has been paid to the newest 'solution' proposed by research and industry stakeholders in response to growing concerns over performance and trustworthiness issues: The use of synthetic data to make bigger and more diverse algorithmic training datasets available. In this paper, I suggest conceptualizing algorithmic training datasets as a type of 'leaky' infrastructure. Drawing on interview material and ethnographic observation from

biometric research and industry events, I tease out how synthetic data is envisioned to 'fix' the inaccuracies that derive from 'gaps' in traditional biometric training datasets. In doing so, I contextualize my findings against overarching themes in the study of border datafication. Fears of leakage lead to new data practices — and thus to new sites of leakage.

Deliberate Inaccuracies and Data-in-the-Making: Accommodation and Orientation Pilot Projects in Belgium

Soline Ballet (Ghent University, Belgium)
in person

Migration governance fundamentally relies on data to create, shape, implement, and modify policies. This article grapples with the concept of leakage, by exploring the case of the unraveling of 'accommodation and orientation' pilot projects for illegalized migrants in Belgium. In the context of a restrictive migration regime, these pilot projects present a collaboration between federal, local, and civil society actors, providing shelter and sociolegal support while aiming to uncover migrants' 'durable future pathways', which is either legal residence or voluntary return. I argue that these projects, as they are under severe political pressure, turn into policy laboratories, where the social context is manipulated to enable smooth experimentation, with actors pushing for positive results. Within this process, the producing, collecting, shaping and showcasing of data becomes essential. To elaborate this argument further, I demonstrate the constant presence of purposeful leakage in data-in-the-making through the pre-selection of the participants based on their cases' success rate and the continuous discussions over the design of particular indicators capable of measuring the impact of the projects. As the case illustrates, this ultimately generates a perpetual loop where deliberate inaccuracies enter into data in order to legitimize the interventions between care and control.

People On the Move Between Technology and Human Rights: Experimentality, Overflows and Interoperability in Senegal and the ECOWAS Region

Alice Fill (École Normale Supérieure, France)
remote

The processes of "datafication" and "digitalisation" targeting people on the move have profound implications, both in security-related and humanitarian contexts. This paper explores the entanglements between human rights law and technology in emerging hybrid forms of border, identity, and migration governance. Focusing on Senegal and on more broad regional dynamics at the ECOWAS level as a case study, it delves into how these co-emergent imbrications contribute to redistributing power, agencies, and subjectivities among different actors involved in the (crowded) field of migration and border control. Drawing on elite interviews, the paper sheds light on the distinct modes of governance that underlie these processes and entanglements, particularly in relation to experimentality and indiscriminate data collection, which can potentially lead to data excesses and overflows. Moreover, by connecting the dots between frameworks of data sharing and interoperability between systems and databases that collect border control data, identity, and migration governance data, it aims to provide an interpretive framework for understanding how boundaries, frictions, and dysfunctions between human rights and technological affordances are renegotiated and sustained on the ground. Through this prism of analysis, the relationships between various national, regional, and international actors are discussed and problematized.

Panel #12 Perspectives on Slow Violence

Stream: Energy Infrastructures and Contamination
ABS/E08
Chair: Lenka Veselá

Leaky Pollutants and Citizen-led Mobilisations

Shachi Mokashi (Vrije Universiteit Amsterdam, Netherlands)
in person

Sudden, violent, and spectacular instances of environmental contamination often garner a lot of attention and produce opportunities for immediate mobilization and action. Contamination, however, may not always be spectacular. Pollutants, such as PFAS, permeate our everyday lives and environments slowly, and often with uncertain consequences. Using case studies of and ethnographic research on citizen mobilization around ‘slow’ (Nixon, 2011) environmental contamination in Europe, I explore the politics of engaging with pollutants that seep and leak; rather than burst or spill. Focusing specifically on citizen-led groups organized around air, water, and soil pollution in Europe, I ask: what sorts of materialities, discourses, and institutions make it possible for citizens to document this type of environmental pollution? How do citizens bring attention to slow contamination? How do these activities translate into mobilization and action? Thinking about citizen-led mobilization around leakages allows us to question the relationship between expert and non-expert practices of knowledge production; tensions and contestations between citizen-led groups and formal institutions; and the governance of environmental contamination. Through this presentation, I would like to draw attention to the complexities between authoritative and ‘expert’ narratives of contamination and the subversions posited by citizens and ‘non-experts’. I hope to present an account of the entanglements that form around leaked environmental pollutants and the kind of action that aims to document and stop them.

Toxic Landscapes: Aesthetic Mediations of Slow Violence and Toxicity in Morocco's Rif Region

Nina ter Laan (University of Cologne, Germany)
in person

Located in northern Morocco, the Rif is a distinct and precarious region on the southern shore of the western Mediterranean that has been subjected to accumulating violence on several intertwined fronts. From chemical warfare by Spanish invaders during the colonial period, to post-independence state repression and a significant exodus of its populace, to an increasingly arid and barren landscape due to climate change, to the violence of the rapid expansion of the EU's externalized border regime. In this paper, I take 'the toxic landscape' as a central notion and lens to explore aesthetic practices, such as images, songs, films, and paintings, that have been created in different eras about landscape, violence, and soil pollution in the Rif, and that have circulated inside and outside the Rif. My point of departure is the use of chemical weapons by Spanish colonial forces against local rebellious tribes during the Rif War (1921-1927). These toxic compounds poisoned farmland, seawater, and water wells, killing thousands of civilians and leaving a legacy of environmental degradation, infertile land, and physical abnormalities-the Rif has the highest cancer rates of any region in Morocco (Balfour 2002, de Boer 2020). By examining how Riffian Amazigh (Berber) activists create and use aesthetic forms to engage with this 'toxic landscape', I trace the relations and publics they produce at local, regional, and transnational levels. I show how these aesthetic practices signal past and present entanglements between people and their environment, and how they (re)shape the Rif today, both as a geographical entity and as a discourse of identity. Drawing on forensic architecture and investigative aesthetics (Weizman 2017, Fuller & Weizman 2021) and literature on toxic substances and the human/non-human relations they produce, as well as scholarship on unsettled landscapes and trauma (Fyfe & Stenberg 2019), I argue that toxicity, which refers to the harmful relationship between a concentrated human-produced substance and an

organism, connects and defines different periods, populations, and ecologies of the Rif as one, revealing how human/non-human relations are already part of a political economy.

Leaky Chemicals, Trespassing Radiation

Nona Schulte-Römer (Humboldt-Universität zu Berlin, Germany)
in person

Negative effects of environmental stressors such as particulate matter in the air, toxins in water and soil and urban heat or noise are often difficult to grasp and impact bodies and ecosystems in 'slow' and 'invisible' ways. Standard risk assessments cope with this challenge by following a protocol of hazard identification and hazard characterization, exposure assessment and resulting risk characterization. Social scientists have problematized such cause-effect oriented approaches that narrowly focus on hazardous molecules, exposure scenarios and scientifically proven bodily or environmental harm. Instead, they highlight sociomaterial entanglements and historical and structural inequalities that produce and sustain environmental contamination on our "permanently polluted planet" (Liboiron, Tironi, and Calvillo 2018). In particular, sociologists have proposed the concept of "residues" to theorize "from chemical domains rather than just applying theory to chemical cases" (Boudia et al. 2021, 20). Residues are thereby understood as not just physical elements, but also markers of political and cultural choices, objects of scientific inquiries and governmental regulation with a past, present and future. However, this conceptualization of chemical leakage and particle pollution misses key features of what we call 'wave pollution'. For instance, while particle pollution tends to produce legacy effects, wave pollution is intermittent and disappears when the source of environmental stress stops emitting. By comparing light pollution and aquatic micropollution, this presentation outlines key differences between these distinct modes of environmental contamination. We conclude that hazardous relationships and risky exposures to wave pollution such as trespassing light, noise or heat challenge STS researchers to not only think with elements and residues, but also with energies and waves as a matter of care.

Panel #13 The Leaking Home: Exploring the Digital Transformation of Domestic Spaces

Stream: Tech & Data
ABS/213
Chair: Nils Ehrenberg

Beyond Security: The Consequences of Continuous Surveillance in Smart Homes

Nils Ehrenberg (Aalto University, Finland)
in person

Smart home technologies continuously collect data with algorithms and actuators that respond to the data. A common concern in smart home discourse is that the introduction of further sensors in people's intimate spaces might compromise security. While security is a valid concern, I believe there is a need to consider how these technologies shape our relation to the home. In my research I make a Foucauldian reading of smart homes as disciplinary technologies that suggests that with the introduction of smart technologies, the home is no longer bound by the physical space, but that as technologies reach beyond the physical space, through cameras that observe our neighbors, checking on our pets during the day, or activating the robot vacuum remotely the boundaries of the home become more porous. Smart technologies reshape the boundaries of the home as algorithms respond to our mere presence or approach to the home without any direct access. Future discourse on smart homes needs to go beyond security concerns and discuss how continuous surveillance reshapes the boundaries and relations to our lived spaces.

Leaky Androids, Leaky Smart Homes in Popular Culture: More-than-human Perspectives of What Counts as Human

Blanka Nyklová (Czech Academy of Sciences, Czech Republic) & Bartosz Hamarowski (Nicolaus Copernicus University in Toruń / University of Lodz, Poland)
in person

Feminist ethics, along with feminist epistemological and (speculative) ontological thought, question the assumed superiority and veracity of the existence and meaning of the modern – masculine – subject, highlighting the incessant efforts to delineate which bodies can be recognised as true political, independent actors. This imperative to reevaluate contemporary notions of subjectivity and political agency extends into the realm of audio-visual representations. The tropes of robots, cyborgs, and androids have become lenses through which the ambivalence between the fascination with non-human sentient entities and the apprehension of their expanding agency are magnified. These cultural motifs resonate in depictions of domestic space as a site for questioning pivotal divisions of modernity: private/public, technological/organic, subject/object. It is within the home, infused with smart and gendered technology, where the very definition of what counts as human gets challenged at the utmost personal level. In our paper, we delve into the imaginaries of 'leakage' – often vividly depicted in popular culture as instances of technological homes and bodies 'bleeding'– and explore its ramifications for the ontoethical status of (smart) technology. We further argue that these ruptures in uncanny mechanical bodies, provide the metaphorical windows through which we can critically examine human condition from a radically more-than-human perspective.

Leaky Boundaries: Exploring Disruptive Gender Practices in Smart Homes

Nina Fárová (Czech Academy of Sciences, Czech Republic), Sandra Frydrysiak (University of Lodz, Poland), Julia Gruhlich (University of Göttingen, Germany) & Petr Kubala (Czech Academy of Sciences, Czech Republic)
in person

The mechanization of the household (e.g. through the introduction of the washing machine, the vacuum cleaner) has done little to change the gendered division of labor in the household, which is still primarily performed by women and the household is still perceived as a female sanctuary. However, the rapid digitalisation of the household in recent years raises the question of whether the automation of the home could have an impact on gender arrangements and practices. On the one hand, technologies can contribute to hacking and disruption of gendered practices such as care giving and domestic housework, but on the other hand contribute to their petrification and reification (e.g. through the emergence of male connoted digital housekeeping practices). From a feminist standpoint, the proliferation of digital surveillance and control features also raises questions about power dynamics over digitally interconnected devices and the role of authority and inequity in smart homes. Based on our ethnographic research on smart homes we want to explore the leakages and/or fluidity in gendered practices in relation to smart technologies. Do the gender boundaries get blurred in the smart home? And where are the potential leaks located? What would a gender equal smart home look like?

Naked House / Naked People - When Buildings No Longer Protect Us and Our Privacy Is Stored in the Clouds

Anna Badyina (Cardiff University, Wales)
remote

While using smart technologies improves our everyday life in some ways, it transforms our residence into ‘a glass/naked house’ as we reveal who we are 'in drips’ to Tech Companies who promise us a series of services in exchange of learning more about us. Algorithms learn how to learn about us, through sophisticated mathematical regressive techniques applied to User Generated Content either

collected through multiple smart devices (e.g., mobile phones, smart meters, etc.) or voluntarily uploaded to the web by us (e.g. Facebook, Instagram, etc.). Through them, we are trapped into spirals of data collection/fit-for-purpose offer loops potentially able to fulfil our most intimate needs. In exchange, our lives become visible to others and are at risk of being controlled, manipulated, and appropriated by powerful institutions or predatory actors. The walls of our houses lose meaning, our data goes to The Cloud and our lives are constantly shaped by exploitative invisible masters.
This part of the panel intends to discuss the ideological and practical aspects technological reinvention promotes in terms of governance and value extraction opportunities for the few thereby the many, and new social relations and constraints it creates. It further exposes the politics of time that are at play in this process, which privilege short-term vs long-term effects, building on the ideas of Technofeudalism and 'cloud serfs' (Varoufakis 2023). The ideas of a world grappling with an entirely new economic system and therefore political power.

Panel #14 Waterscape Imaginaries

Stream: **Ecological Leakage & the more-than-human**
ABS/114
Chair: Ingmar Lippert

Leaky Landscapes: Ethnographic Engagement With More-Than-Technological Action Along the River Spree

Desirée Hetzel (Humboldt Universität Berlin, Germany)
in person

Terrestrial water resources are under increasing pressure from climate change and human-water interactions need to adapt to changing waterscapes to avoid arid landscapes. Rivers as watercourses are not only fluid transport routes supplying water, they also create a landscape of floodplains along their banks. For centuries, the river Spree's waterways in Berlin-Brandenburg have been controlled by engineering structures such as drainage, canalisation and sluice systems. However, hydrological and political uncertainties about how to respond to rapid (planetary) environmental change are undermining confidence in this regulation of hydraulic infrastructure. Recent droughts in the region have made arid futures more tangible and are shaping discussions about sustainable water management along the River Spree. Old and new relations between former mining sites, the tourist landscape and the metropolitan area along the river begin to emerge. This paper discusses on the basis of ethnographic material emerging imaginaries of the water landscape along the river Spree. We use the analytical lens of leakage to demonstrate how imaginaries exceed top down technology-driven solutions and associated problem framings: water physically escapes control, eludes predictions, refuses to be stabilized into particular landscapes, and conjures up feral images and imaginaries. Quite literally, through the premise of a future without leakage(s), of not losing a drop of water between the region of Lusatia and Berlin, sluice systems become central to the discussion of permeability and retention.

Intractable Waters; Examining Dams and Cartography in "River of Stories" (1994).

Anandita Saraswat & Aratrika Das (Indian Institute of Technology Indore, India)
remote

Water in both its material form seen in floods, tsunamis and in its symbolic form seen in terms of the imagery and metaphor of flow resists containment. Visualizing water only in terms of a resource to be extracted overlooks the unruliness of water and eludes the relationality of water to the

surrounding living forms, a line of thought that can be traced back to the colonial settlement patterns to the construction of contemporary large scale mega dams. This paper looks at the notion of leakage through a disruptive understanding of water. Disruptive potential of water is undermined by the "extractive view," a term used by Macarena Gomez-Barris to describe the commodification of natural resources and humans by colonial projects reflected in the mapping and cartographic practices which are further resonated in the recent time in the satellite enabled objective mapping of natural resources that eludes the ecological aspect. The paper through a reading of, The River of Stories (1994), a graphic novel by Orijit Sen, will analyze how the permeability of river water and the identities of the tribe, reflected in the stories and songs of Malgu Gayan, is elided in the construction of dam. The use of water as a lens in the graphic novel becomes functional to highlight the notions of dispossession and inequality arising out of a lack of understanding of the permeable and fluvial nature of water that will be looked at through the notion of "hydrocolonialism." At the same time the graphics in the novel subverts the cartographic practices that sees water only in terms of tractable resource. Thus, the paper will foreground how the suppression of the shared, relational, and permeable understanding of water reflected in the mapping patterns and the construction of dams leads to the notion of social and environmental injustice as reflected in the graphic novel.

Poetics of Permeation and More-than-Human Infrastructures: Pipes, Leaks, and Architectural Malfunctions in Kim Stanley Robison's New York 2140

Eleonora Rossi (Independent Scholar, Germany)
in person

In recent years, scholars across the humanities and social sciences (Neimanis 2017; Mentz 2021; Chen et al. 2013; Jue 2020), have turned to water to provide innovative critical readings of a range of crucial topics, relating in particular to the impact of human activity on the planet and the acceleration of social and environmental crises. They have extended an invitation to think with and through water (Chen at al. 2013) to explore spaces, communities, and imaginaries beyond the reach of anthropos, the human. Within the burgeoning landscape of the Blue-Green Humanities, key studies have examined human relations to various large bodies of water, including oceans, seas, rivers and swamps (Anca Farca, 2019; Boast 2020). With this paper, I provide a specular and complementary analysis of this trend as well as of the topic of leakages, by looking at the microscopic bodies of water that infiltrate in human spaces and infrastructures, producing glitches that trouble human hegemony. Working across literature and anthropology, and focussing specifically on Kim Stanley Robison's New York 2140 (2017), I develop a poetic of permeation in which pipe cracks, condensation, and other aqueous glitches within domestic and urban spaces provide destabilizing but also possibly emancipatory moments of dysregulation. With each drop, anthropocentric patterns of ordering and control become noticeable and can therefore be addressed and rewritten. This paper draws on current studies on water as well as on the work of anthropologists Star and Ruhleder (1996) and Bowker (1995), who describe infrastructures not as defined by a physical location nor a permanent one, but rather by sets of working relations. As Jewett and Kling (1991) also put it, infrastructure is a fundamentally relational concept, which becomes infrastructure in relation to organized practices. As environmental crises intensify, this working relation is subject to dramatic changes, which challenge its anthropocentrism. Drawing on Robinson's work as a case study, I argue that domestic and urban infiltrations uncover and denaturalise the workings of human architecture and infrastructures, exposing their vulnerability to environmental agents and calling for their conceptual and material reconfiguration in post-anthropocentric terms.

Panel #15 STS Meets Dis/ability Studies

Stream: Bodies & Governance
ABS/114
Chair: Mathias Denecke

Autism Leaks: Non-innocent Figurations of Closure and Porosity in the Science and Experience of Autistic Biology

Ombre Tarragnat (Paris 8 University, France)
remote

Autism is a multilevel figuration apparatus which crystallizes diffuse cultural and political phenomena. In neurobiological autism research, normative social representations as well as first-hand accounts of lived experience, it is permeated by and constituted through ambivalent imaginaries and affects around the closure of the organism and its lived world. First attached to the category of child schizophrenia, autism is still often perceived as a disorder of disconnection from the real world, the autist acting as the prototypical model of the rigid, hermetic, inscrutable individual trapped in a sealed, closed-off interior world.
More and more clearly, these figurations of autism as the prototype of boundedness are revealed as unachievable attempts to repress fears and anxieties around porosity, contamination, and excess. This paper thus evidences the dual constitution of autism through a constellation of figures alternating between closure and porosity. Highlighting their role in the biopolitics of ableism as well as in the more progressive regimes of new materialist theory and autism advocacy, it points to the non-innocence (Haraway, 1985) of the leaky (Shildrick, 1997) bodies of autism.
Examples of figures include: so-called autism epidemics caused by vaccines and pollutants, current research on the effect of leaks in brain barrier or gut on autistic “symptoms”, potential channelopathies in synaptogenesis causing autism, the figures of empaths or hypersensitive unable to filter incoming sensory or affective stimulation, the hyperfunctioning of the autistic nervous system in the Intense World Theory (Markram and Markram, 2010), the normative rejection of autistic hyperlexia, hyperfocus and special interests into categories of excess, the recurring atopic condition of autistic bodies suffering from eczema and allergies, the linking of autism with multiple chemical sensitivity (Chen, 2012), and the experience of autistic meltdowns as outward leaks resulting from excessive inward stimulation.

Pain and its Neuro-Digital Control. Exploring Leakages at the Brain-Computer Interface

Benjamin Lipp (Technical University of Denmark, Denmark)
in person

Chronic pain is a leaky phenomenon. It often transcends medical categorisation and is notoriously difficult to manage, let alone treat. It leaks into people’s lives affecting their body, mind and relationships with others. It is leaky in time. It can flare up without warning or retreat only to foreshadow its return. As chronic pain has become one of the most common causes of disability, it has also been rendered a target for technological innovation. Particularly in the United States where the ongoing opioid crisis is wreaking havoc, digital pain technologies are promoted to support people living with pain without the risk of addiction. Such devices combine electrical stimulation and sensing capabilities collecting data for both self-monitoring purposes and for regulating the intervention. Their promise is to control chronic pain’s leakiness through a combination of adaptive automation and datafication without requiring much attention. This contribution presents first results from a diary and interview study with users of such devices in the United States. This qualitative, longitudinal investigation traces people’s everyday ‘chronic living’ practices (Wahlberg et al. 2021) of managing pain through ‘smart’ pain technology as an entry point to explore leakages at the brain-computer interface. It shows that not only chronic pain can be considered leaky, but also the technologies that are used to manage it. Instead of simply sinking into the background allowing

to manage pain anywhere and at any time, I will illustrate how brain-computer interfaces produce uneven, 'chronic' temporalities and new responsibilities for users to maintain the neuro-technical interface. At the same time, I explore a core tension in the neuro-digital pursuit to control pain: the 'neo-liberal' subject so often invoked in critiques of digital health is both bypassed and re-engaged by these technologies.

Panel #16 Fiction and Toxic Leakage

Stream: Energy Infrastructures & Contamination
ABS/214
Chair: Fenna Smits

Between a Spill and a Leakage - Narrative Complexity and Ecocriminality in Breaking Bad (2008-2013)

Verena Wurth (University of Cologne, Germany)
in person

In my paper, I explore the narrative function of repeated oil spills and leakages that are depicted or thematized in Vince Gilligan's Breaking Bad (2008-2013), the neo-Western TV series about Walter White, an Albuquerque chemistry teacher and family father, who, initially fueled by his need to finance his lung cancer treatment, turns into a criminal drug lord.
Firstly, I argue that spills and leakages often serve as a threat, hindrance, or complication to the protagonists' criminal endeavors. Such threats to the criminal plan do not only occur on metaphoric levels, for instance when Walter and Jesse fear that their accomplice is going to "spill" to the police "like the Exxon Valdez" – a rather incongruent comparison between and the severely violent environmental disaster off the Alaskan coast in 1989, and a leak of information with the individual repercussion of temporary incarceration (S2E8). Spillages also occur on the visual level, for instance when White and his sidekick Jesse Pinkman fail to restart their RV after a meth cooking session in the New Mexico desert, because their generator catches fire after gas from the vehicle's tank seeps into it (S2E9) – a death sentence but for White's chemical ingenuity.
In addressing the oscillations between visual and verbal spills and leakages in the narrative, I argue that – in particular – the leakages complexify the serial narrative trajectories (cf. Mittell), using the example of the showdown between the DEA and White in the final season, when the tank of his car is shot, his gas almost imperceptibly drips out, so that it comes as a surprise that he is later stranded in the desert, This specific location of the repeated oil seepings brings me to add the dimension of ecocriminality and environmental injustice to my analysis, highlighting the slow violence (cf. Nixon) involved, especially since the oil contaminates the seemingly unpopulated area of the crime scene, which happens to be, in point of fact, the Native American 'To'hajiilee' reservation.

Toxicity and its Aftermath in Indra Sinha's Animal's People

Nibedita Bandyopadhyay (Shiv Nadar University Chennai, India)
remote

The paper analyses Indra Sinha's Animal's People (2007) in order to delineate chemical abuse and toxicity that cause harm to nature in the name of development and scientific experiments and victimize the marginalized sections. Animal's People is a fictionalized document of the Bhopal gas disaster in the Indian state of Madya Pradesh, which took place on the night of 2 December 1984 at the Indian branch of the American company Union Carbide Factory. Due to inadequate security management, the poisonous gas came out of the factory, killing more than five thousand people and

many livestock. This disaster had an overall catastrophic impact on the environment of Bhopal that continues till now in the form of slow violence (Rob Nixon). The victims of the Bhopal tragedy were denied justice, and they are still fighting for it. The marginalized sections predominantly constitute the poor people, and the non-human others suffer because of the androcentric attitude of the dominant society that subjugates the weaker sections. Their memories remain neglected by the predominant media, as was the case with Bhopal, where some of the media houses posited the incident as a minor one, leading to insufficient compensation for the victims. The fiction critiques neo-liberal globalization by exposing the hypocrisy of the multinational companies that pay no heed to the safety of the workers. The paper identifies a counter-discourse running through the text that overtly dismantles and problematizes unchecked development that leads to the maltreatment of the "othered" parts of society. It applies Stacy Alaimo's "trans-corporeality" to capture how living organisms are affected by the fatal toxicity produced by the poisonous chemicals and the subsequent disability, disaster and systemic injustice that is imposed on a section of people in the form of chemical poisoning.

Res(crip)ting the Waste: An Inquiry into the Dis/ability Aesthetics of Animal's People

Sonakshi Srivastava (Ashoka University, India)
in person

Order precedes existence. Our existence is coded into a system of comprehensibility that allows us to impart meaning so as to impart meaning to the world around us. To understand that there is any meaning at all also pronounces the notion that this meaning generation is a human-made activity. The ability of humans to scribe boundaries and draw lines is a way of asserting differences, thereby creating and maintaining order. Objects, once deprived or extracted of their use (by humans) become waste. Waste, then, is the unwanted, undesired remnant which needs to be expelled from society to maintain order of hygiene and cleanliness. In their afterlives as waste, objects are yet not deprived of meaning but rather come to occupy meaning when examined from a certain vantage point.

This paper takes into consideration Indra Sinha's Animal's People, a novel that unravels the life of the eponymous character, Animal – a victim of a gas leak in his village of Khaufpur (alluding to the Bhopal Gas Tragedy of 1984). The Bhopal Gas Tragedy occurred in the wake of post-war neoliberal order. The result of all these activities, geared towards the development and industrialization (emphasis mine) of India was that more than 300,000 people were suddenly, one night, blighted by a crippling disease, of which more than 2500 died horrible deaths, yet the people declared incompetent and irresponsible were neither the multinational nor the government but the sufferers" (Das 160-1). Animal comes to be known after the gas leak twists his spine, rendering him unable to walk on all-fours. Here, the disabled body qualifies as the "waste" for the State in particular and for able-bodied citizens in general. By building on the works of Jasbir K. Puar, and Susan Morrison amongst others, the paper will navigate through the aesthetics of dis/abled bodies, especially in the Global South which functions as an informal dumping ground for the first world nations like the USA. The paper considers the visibility of the disabled protagonist who refuses to be obliterated and strikes back by negotiating his peripheral and perilous position bringing to the fore the biopolitics, where leaky borders disturb order, and where some bodies are more equal than others.

Panel #17 Energy and the Materiality of Data Infrastructures

Stream: Energy Infrastructures & Contamination
ABS/0017
Chair: Jakob Roschka

Disentangling our Environmentally-Situated Practices of Thermal Control: From Taipei's Overclockers to Accra's Repairers

Cyrus Khalatbari (HEAD – Genève and the Swiss Federal Institute of Technology, Switzerland)
in person

Digital growth and progress is depicted by our major internet corporations as non-materially situated: disconnected from the earth and its processes of extraction, optimisation and waste. Critically reframing this dominant trope and narrative, the contribution proposes to map the environmentally-situated practices of computing power: required to operate the energy and graphically-demanding objects of our platform and algorithmic cultures. Informed by theoretical inquiries and fieldworks, the contribution moreover focuses on one aspect of our sociotechnical cultures: the production - and leakage - of heat generated by our devices; and the consequent need to cool down these devices. The presentation is structured in three parts. First, it contextualizes computing as inherently connected to thermal manipulation and control. From Mel Hogan's inquiries on water infrastructures upon which our cloud platforms operate to Nicole Starosielski's analysis of quantum technologies and their dependence on cryogenic refrigerators and liquid helium, the first section situates the topic inside a network of STS and environmental media-studies academics and scholars. The second and third section of the contribution map our practices of thermal control through two case-studies emerging from different lifecycle stages and sociotechnical contexts of our planetary computing culture. The first case-study emerges from Taipei, Taiwan. It follows a community of hardware hobbyists and gamers interested in practicing extreme overclocking: pouring liquid-nitrogen on their computer chips in order to optimize their performances and benchmark energy-intensive programs and videogames. The second-case study is located in Accra, Ghana. It disentangles a task undertaken by second-hand computer repairers: replacing dusty fans and optimizing their customer's machines' cooling systems in order to avoid them overheating. Placing these two inquiries in dialogue, the contribution proposes to map the environmentally-situated practices of thermal leakage and control. By doing so, it further contextualizes the eco-material implications of our needs and desires for digital growth and progress.

Energy Humanities and Urban Cultural Studies

Antonella De Sena (University of Naples "L'Orientale," Italy)
remote

Constant growth in the context of a limited biosphere cannot be sustained. The faster growth-oriented societies appropriate planetary ecological space by expanding their urban economic metabolism, the faster the living systems of the planet collapse. For this reason, the Energy Humanities is a new interdisciplinary scholarly field that was developed in order to study the entanglements between cultural sensibilities, political power, and energy technologies.
The main goal of this intervention is to suggest that the integration of Energy Humanities and critical urban studies has the potential to illuminate relevant aspects of contemporary culture. In particular, this approach will be used to analyse contemporary Spanish urban culture.
There are multiple ways to apply this frame to literary and cultural analysis. The one that will be used here is reading recent textual narratives as more or less critical reflections of the urbanization of capital by tracking how their "cartographic imaginaries" expose or conceal the dysfunctional infrastructures, wasteful energy dynamics, operational landscapes, and ideological contradictions of the neoliberal urban process.

I will provide a concrete example by applying this frame to El Calentamiento Global by Daniel Ruiz, an illustration of how ecological depletion, petrochemicals, garbage, tourism and concrete entailed by the Mediterranean urban process, are combined.

Saved by the Leak: Bitcoin Mining and the Absorption of Energy Leaks as a Justification Strategy

Martin Tremčinský (Czech Academy of Sciences, Czech Republic)
in person

Bitcoin mining is a notoriously energy-intensive process often described as an environmental catastrophe on its own. The computational process demands large amounts of energy and produces extensive surplus heat. In this context, we conceptualize energy surpluses as energy leaks and argue that absorbing the leakage caused by the energy-transforming systems' inefficiency is often used to obscure the original irrationality of these systems.
Focusing on the absorption of residual energy by Bitcoin miners, we examine how absorptions serve as a justification of Bitcoin mining while diverging attention from its core social and ecological effects. One example of leak absorption is building Bitcoin mining facilities near natural gas extraction sites. These sites came under criticism when unprocessed gas was burned in so-called flaring. Now, it is used to power up Bitcoin mining stations. However, such usage inhibits the improvement in gas capture and transportation. At the same time, it helps to divert attention from the inefficient and toxic gas extraction and obscures the ecological and societal harms caused by the Bitcoin ecosystem. Bitcoin is thus presented as a solution instead of a problem. The second case concerns the surplus heat of Bitcoin mining. Some Bitcoin miners promote this leak as "free" energy that can be utilized in household heating, justifying mining despite its low economic returns and high ecological costs. Both these practices diverge attention from ineffective, highly energy-intensive methods – the extraction of fossil fuels and Bitcoin mining itself – and reframe them as economical resource-management. We argue that the ability to capture the leak thus becomes an end in itself, displacing the question of the source of the leak and systemic solutions. Such "economization" of energy leaks captures and recruits Bitcoin miners into a matrix of further economic and ecological (self)exploitation.

Panel #18 Waste Water

Stream: Ecological Leakage and the More-than-Human
ABS/114
Chair: Blanka Nyklová

Leaking Fertility and Unruly Phosphorus: Sewage Plants as Sites of Control

Lukas Adolphi (Catholic University of Eichstätt-Ingolstadt, Germany)
in person

The Baltic Sea is no longer just the background for Caspar David Friedrich's chalk cliffs, a meeting place for German nudists or the scene of mysterious pipeline explosions - for some years now, it has also been the site of the world's largest death zone and thus the stage for a fundamental planetary shift. Death zones are areas of water where nutrients, mostly flushed out from industrial agriculture, cause increased algae growth, consuming so much oxygen that other marine life can no longer survive there. The lack of fertility in agriculture, compensated by fertilizers, leads to a surplus of fertility in water bodies, shifting the ratio between concrete life forms, but also between life and non-life in general. And there is no end in sight to this development so far: both the use of fertilizers and

the area of dead zones are growing worldwide. An essential nutrient for the emergence of fertility is phosphorus. Phosphorus is used as a fertilizer in agriculture, where it is not fully absorbed by crops, but also forms inorganic bonds and is leached into nearby water bodies or seeps into groundwater. These diffuse leakages lead on the one hand to a loss of productivity in agriculture and on the other hand to increased fertility in water bodies. Sewage plants function in this context as a site of control, where unruly phosphorus is controlled and leaking fertility is recaptured. Building on an ethnographic study, this presentation examines the techniques and strategies used in sewage plants to respond to phosphoric leaks and explores the extent to which the sewage plant may itself be leaky. From a more-than-human perspective, the sewage plant ultimately appears as a site in which power structures are reproduced and manifested, but at the same time constantly subverted.

When does Soap Pollute? Tracing Shifts in Materials and Values by Following Soap in Household Cleaning, Wastewater Treatment, and Ecotoxicology Research

Sam van der Lugt (University of Amsterdam, Netherlands)
in person

This paper highlights the interdependence of materiality and values by tracing how soap becomes environmentally good or bad across three practices. I follow soap in household cleaning, wastewater treatment, and ecotoxicological research (oriented towards pollution in surface waters). In each practice, I ask what makes soap good or bad for the environment. This approach adopts the Discard Studies methodology of 'trash tracing' (Holmberg, 2021) and lines up with recent calls in Valuation studies to outline shifts in value (positive and negative (Thompson, 2017)) of an object across connected practices (Waibel et al., 2021; Greeson, 2020). Soap, however, turns out not to be one object. In household cleaning, 'soap' comes as bars and bottles; in wastewater treatment, 'soap' figures as 'nutrients'; and in ecotoxicology, 'soap' becomes a potential carrier of 'toxicants.' As a result, what is 'environmentally good' is also different in each practice. In households, 'environmental good' is about using the right products; in wastewater treatment, it is a balance of eaters and nutrients; and in ecotoxicology, it is an absence of toxicants that harm organisms. Challenges surrounding the alleviation of soap's environmental bads are notably not about informing human actors (or accounting systems) who fail to take leakages into account—actors often know about damages caused outside of their immediate practice but struggle to incorporate such knowledge within their practice's distinct characteristics. Instead, this case highlights the difficulty of making various goods and bads meet across practices comprised of different materials. In such cases, interfering in practices is a better tactic than informing subjects.

Geographies of Tourism-generated Wastewater in San Andres, Colombia and Gili Trawangan, Indonesia

Yim Ming Connie Kwong (Leibniz Centre for Tropical Marine Research, Germany)
in person

Nonhuman dimensions of a place are usually considered as passive backdrops of human activities. This has led to a call to look at the nonhumans as actors when looking at environmental changes, and recent interests in the temporalities of more-than-human worlds. Given these trends and calls in the literature, temporalities of less visible matters are not yet much attended to. This paper looks at the discharge and flows of tourism-generated wastewater in small tourist islands, San Andres in Colombia and Gili Trawangan in Indonesia. Based on primarily three-month and eight-month ethnography and participatory mapping interviews on respective islands in 2022-2023, this paper analyzes how wastewater is discharged, managed or contained, flows, circulates, or in other forms of being perceived, experienced and understood by tourism stakeholders. It also discusses when and how wastewater becomes visible and comes to people's senses in relation to infrastructure and facilities, seasons and information. This gives some implications for the decision-making process for wastewater and tourism management.

Panel #19 Hegemonic Bias in Science and Technology

Stream: Tech & Data
ABS/E04
Chair: Marta Scaglioni

Transforming STEM - Or How to Leak Feminist Science and Technology

Sahra Dornick (Technische Universität Berlin, Germany)
remote

In my contribution, I would like to reflect on the ontology of leakage using the example of feminist teaching and research in STEM. Feminist teaching and research in STEM is by no means common in the German higher education landscape. Even though there are promising approaches of feminist and postcolonial STS and furthermore many interested students, feminist and postcolonial knowledge, methods and practices are largely ignored.
Nevertheless, there are small cracks, such as a few social science researchers in technology-developing projects or the GenderProMINT study program at TU Berlin, where feminism is seeping into the STEM sector. I would like to deepen the understanding of the interrelation between feminism and STEM by using the metaphor/concept/imagination of leakage by reframing my experiences of teaching and researching as a feminist researcher in STEM subjects and in technology development projects: What can we learn about feminism in STEM using the leakage as a metaphor? What new ontology of learning can we develop from it?

The WEIRD and The Wild: Leaking Interfaces Into Lives and Literacies

Chris Scheidler (Weber State University, USA) & Alex Way (The University of Utah, USA)
in person

This presentation provides an in situ (in the wild) exploration of how WEIRD [Western, Educated, Industrialized, Rich, Democratic]-skewed communication technologies mediate Japanese writing practices. The speaker traces Japanese literacy practices from childhood to adulthood and demonstrates–through curriculum standards, classroom materials, and existing scholarship (Gottlieb 2005; Taylor & Taylor, 2014)–how Japanese language users develop their ability to comprehend and write the three Japanese scripts (hiragana, katakana, and kanji) which make up the Japanese writing system (JWS). Building upon this foundation, he details the results of an IRB-approved mixed method study involving diary writing and interviews (Prior & Shipka, 2003) which he conducted in Nagano City, Japan to explore the literacy practices of his college student research participants. The results of his study demonstrate: how interfaces impact script choice and other semiotic features, how students use software in innovative ways to compose multimodal materials, and how students use social media applications for collaboration and community-building, among other findings. This study highlights how student writing practices counter WEIRD assumptions built into communication technologies. Although the Japanese students in the study utilize Western-developed technology, they effectively absorb their identities into these technologies (Arola, 2017), thereby making the tools their own.
This work critiques communication interfaces as reflections of western, enlightenment, market-exchange based values encoded into design principles. Such principles are extensions of “a science of design” derived from mid-century psychology and developed further in architecture and industrial design (Gropius, 1970). To put it plainly, the interface design flattens and universalizes complex systems of machine, information, and person. Such flattened and universalized interfaces are regarded as aesthetically elegant and intuitive; but, to whom and under what conditions?, we might ask. History points toward the extractive potential and uneven power dynamics (Irani, et al., 2010). Using Rosner’s (2018) method of “Critical Fabulation”, this presenter arrives at potential interferences that fly in the face of a universalism encoded as elegance and trust. And aims to interfere or “disturb a narrative that is privileged within a prevailing design culture” (90). This is to

say, for all the talk on the radical possibilities of technology to offer new and different alignments, the design practices that have become in vogue often restrict such possibilities. It is especially important, given the congealing and universalizing forces acting on global media practices, to reimagine design principles and practices based in 21st century commitments to equity, decoloniality, and antiracism.

Excess at the Interface: On Ephemeral, Affective Dynamics in HRI Settings.

Pat Treusch (Paderborn University, Germany)
in person

Being affected and developing emotions towards robots can be regarded as the core pillar for what has emerged as one central strand of robotics, namely 'social robotics'. 'Social robots' are supposed to become everyday companions to 'us humans'. The social robot appears to be a figure full of promises, providing a techno-fix for apparently all kinds of societal shortcomings that societies foremost of the Global North are facing: from a lack of nurses in institutions of elderly care to a lack of schoolteachers and health care providers such as general practitioners. It constitutes not only the embodied, but also the affective and emotionally engaging interface. Notably, social robotics ties the realization of the affective interface to an anthropomorphic or zoomorphic design of such an HRI interface.
This abstract re-considers the ways in which affects and emotions are regarded as the operationalizable condition for socially meaningful, that is, socio-cognitive behavior in machines. It also revises the concomitant reduction of social meaningfulness to the idea of implementing and evoking a predefined set of emotions through an anthropomorphic or zoomorphic robot design. In line with selected strands of feminist sts that critically inquire the use of standardized emotions as the vehicle and instrument for socio-cognitive skills in robots, I will argue for a re-turn to affects and emotions at the HRI interface. How to re- claim the supra- and intra-individual nature of affects at the interface? I will develop an account of affects as the more-than-human practices of engaging at the interface. With this, I want to acknowledge the ephemeral dynamics and ask how they produce an excess of meanings and feelings beyond the human/like. Finally, I will speculate what kinds of ways of relating and engaging this could enable.

Panel #20 In the Laboratory

Stream: Methods & Epistemologies
ABS/E08
Chair: Roos Hopman

Leaky Laboratories

Freia Kuper (Alexander von Humboldt Institute for Internet and Society, Germany) & Ariana Dongus (TU Dresden, Germany)
in person

More than 20 years ago, Bruno Latour wrote that "the laboratory has extended its walls to the whole planet" (Latour 2001). Indeed, the laboratory, traditionally the primary space of knowledge production in the experimental sciences, based on a particular natural and social order (Knorr-Cetina 1992) and defined by its closure, may have become leaky.
The leaky laboratory is connected to the changing relationship between empirical data and theory as well as to experimentation as a mode of knowledge production: An abundance of empirical data as the privileged driver of research is thought to challenge theory-driven scientific approaches, which Kitchin calls an "empiricism reborn" (Kitchin 2014:3). A revaluation of deductive modes of reasoning

thus changes the status of both experiment and laboratory. Instead of making claims based on inferences from theory, one could now use the mass of data to produce new knowledge. How has the laboratory, as metaphor and as space, and with it practices of experimentation evolved or stagnated?
Building on this framework, we will discuss two examples of 'labification' and experimental culture in times of crisis. One example looks at an idea of collective experimentation and how we might think productively about experimental culture with the intent to solve real world problems. The other example looks at data extraction for predictive analysis of migration movements, to discuss how and why refugee populations become subjects in an experimental setting and how this shows the dangers of the laboratory metaphor. We want to discuss the potential and limits of a critical research agenda when thinking about new forms of labs and scales of experiments.

Governing Life Through Cells

Ruzana Liburkina & Veit Braun (Goethe University Frankfurt)
in person

From stem cells to bacteria, cells have not only become an epistemic key object of scientific research. Increasingly, they are also used across the biosciences as tools and technologies for shaping, manipulating and managing life. Whereas DNA markers, predictive genomics, large-scale population genetics and other molecular technologies have received extensive attention by STS, cells as biopolitical instruments have largely remained unexamined. In this presentation, we want to shed light on cells' double role as centers of calculation and technoscientific vectors of life.

Competing Logics of Leakage and Insulation at a Novel Antarctic Research Station

Spencer Adams (Rachel Carson Centre, Ludwig-Maximilians-Universität München, Germany)
in person

The British Antarctic Survey's Halley Station is, in its self-presentation and media coverage, simultaneously at the vanguard of two seemingly countervailing projects. First of these is the effort to create the conditions for truly comfortable, communal human inhabitance at the extreme edges of the world. This was what came to be so highly touted about the futurist design of the sixth iteration of the station, erected in 2013: that through various socio-technical design interventions, the station provided an internal environment that might insulate its inhabitants from the psychically taxing, extreme conditions of the surrounding environment, all while being re-locatable in case certain cataclysmic glaciological possibilities arose, threatening the station's very existence situated on the rapidly cracking Brunt Ice Shelf. Secondly though is another project, one spoken about more prominently in the station's recent years: namely, the near full automation of scientific instrumentation in the station as inhabitance through the Antarctic winter has been deemed increasingly unsafe in the context of rapid advances in climate change and the COVID-19 pandemic. What I propose in this paper is that these two projects rhetorically draw on two competing imaginaries of leakage rooted in prevailing social logics of the Anthropocene. For Halley as a comfortable home amid remote, ecological extremity, the station's novel design is what prevents the leakage of the Antarctic ice shelf's unique seasonal, environmental, and even aesthetic extremity (and at times hostility) into the station's necessarily human psychic environment. But for automation specialists, a different imaginary unfolds, one that sees the history of Antarctic inhabitance as defined by the leakage of human populations onto the previously pristine continent. Automation has the potential to clean up that spill of the human as excess into Antarctica without disrupting the extraction of observational data. These competing logics, I contend, are two poles to critically assess in evaluating more generally the novel social, technological, and infrastructural techniques of the Anthropocene.

Panel #21 Metabolic and Endocrine Disruptions

Stream: Bodies & Governance
ABS/114
Chair: Melina Antonakaki

The Sadness of the Anthropocene

Lenka Veselá (Brno University of Technology, Czech Republic)
in person

Industrially manufactured chemicals are everywhere. What do environmental flows of chemicals bring to our lives and how can we live well with them despite their potential for harm? With the focus on a particular group of environmentally ubiquitous chemicals called endocrine-disrupting chemicals, my inquiry draws upon the ways in which chemicals can influence our brain chemistry and thus also our thoughts, feelings, and motivations. Using the Endocrine Disruption Tracker Tool (EDTT) — a speculative instrument for a collective investigative practice that I created — I examine the likely influence of chemicals on our emotions and the possible role of these emotions in opposing the chemical violence that systematically impairs life in all its forms. The EDTT is modeled on a tool for diagnosing premenstrual syndrome and premenstrual dysphoric disorder, using the same set of emotional symptoms but adapting the original design by expanding the functional range to cover emotional symptoms that are caused by the production and interplay of both hormones and hormone-disrupting chemicals. I developed the EDTT to help me, my research participants, and a broader community of concerned people to confront the uncertainties and ambiguities of our lives affected by the involuntary chemical exposure. Drawing on the insights and stories shared by the workshop participants and using the visual cues from my short film The Sadness of the Anthropocene, my paper considers the possibility of our emotions being impacted by environmental chemicals, including the prospect of affirming these emotions, as they are affected by anthropogenic chemicals.

From Solid Boundaries to Liquid Relations: A Leaky Look at Non/Consensual Endocrine Disruption

Em Panetta (York University, Canada)
remote

Leakage can act as a material-discursive apparatus describing fluid, molecular encounters. Through a cross-disciplinary pooling of scientific scholarship, personal relations, technoscience studies, and embodied experience, I show that leakage is intra-active and chemically-oriented, in ways that the solid figure of “entanglement” is not. This paper investigates two forms of endocrine disruption: hormone replacement therapy (HRT) used by trans people, and per-and polyfluoroalkyl substances (PFAS) as drinking water contaminants. Trans individuals who use HRT consent to their disruption, and harness the leaky potential already present in our bodies. By embodying molecular/body intra-actions, HRT not only transforms the contours of the physical form but exposes leakages at other scales, especially the instability of medioscientific power. PFAS, however, signals nonconsensual disruption. This form of bioaccumulative fallout, in part from military-industrial wasting, contaminates the drinking water in my hometown, putting my community and generations to come at risk. While the extent of the damage is still murky, leakages among chemical, person, water, and state, make space both to honor and respond to our violated land/bodies, and to mobilize our potential to hold violent systems accountable. In both cases, endocrine disruption is a fundamentally material remodeling, changing the contours of human flesh through hormonal activity, and offering insight about the fungibility of other bodies and systems; just as the divide between self and molecule dissolves, so do those between natural/unnatural, self/other, toxic/pure, damaged/queer.

Attending to violent systems and transformed bodies, leakage is a commitment to care at multiple scales, ultimately oriented toward irrevocably altered worlds. As these and other leaky relations among proximal bodies (human, chemical, water, soil, political, animal) continue to spill forth, they will remain unruly and imprecise. Contentious divides will continuously reappear, suggesting time and time again that solid paradigms are not fit for our current intellectual moment. I argue that leakage examines not only how liquid material may illuminate our understanding of molecular processes, but how such knowledge may also function at other scales, which troubles relationality at large. It is in and through leaky flux that the potential for worldbuilding and new flourishing begins.

Leaking in, Leaking out: Absorbing Environments in the Era of Globesity

Varvara Borisová (Charles University, Czech Republic)
in person

The growing number of people with excessive weight poses a challenge for healthcare systems across the world since obesity is associated with higher risks of developing chronic non-communicable diseases like diabetes or vascular disease. While the narrative of irresponsible behavior, laziness, and fast food as the main reasons for getting fat prevails in the public discourse – a reason why people with obesity are stigmatized by the rest of society (e.g., Trainer et al., 2021; Spratt, 2023) – recent biomedical studies show that factors beyond one's control, including epigenetics or imbalance in gut microbiota, can cause weight gain.
Obesity is a metabolic disease and often signals metabolic dysfunction. The idea of human metabolism as a factory transforming food into energy that had long determined our knowledge about how our bodies work emerged in the 19th century under the influence of industrialization. On this premise, the most popular weight reduction principle "calories in, calories out" builds. However, nowadays, metabolism is conceptualized as a "regulatory zone [...] built by and responding to environmental information" (Landecker, 2013: 496). Thus, human metabolism is not bounded by the boundaries of one's physical body but rather is in constant multidirectional exchange with the environment that our bodies absorb and, in turn, transform (Solomon, 2016). Due to this porosity of human bodies, the boundaries between inside and outside get blurred. Can we thus talk about the non-communicability of diseases at all?
Drawing on the interviews with healthcare professionals and people with obesity, this paper will discuss how obesity treatment specialists perceive the role of (toxic) environments in the unfolding obesity epidemic in Czechia and how people diagnosed with obesity navigate these environments. Using the imagination of leaking and leakages, the paper will critically examine the current obesity treatment approaches and how different substances – like food or medications – are entangled in the treatment process.

Panel #22 (Un)leaking Marine and Coastal Ecosystems

Stream: Ecological Leakage & the More-Than-Human
ABS/E08
Chairs:
Tanja Bogusz (Universität Hamburg, Germany)
Nane Pelke (Universität Hamburg, Germany)
Ramona Haegele (German Institute of Development and Sustainability, Germany)

Do leakages exist in marine and coastal ecosystems? The answer might be "yes" or "no", or both at the same time. Current marine social science research proposes three possible approaches:

1. Thinking through a “wet ontology” (Peters & Steinberg 2015), the ocean embodies a world of constant flows, connections, and liquidities, forever evolving. Such a perspective, stressing the material and phenomenological peculiarities of the ocean challenges to reimagine the world (ibid.) and comprise the ocean as being in a perpetual state of “leaking” or in a state of “unleaking.” “Leakage”, reflected through the lenses of marine ontology, seems, here, either taken for granted or paradoxical to think through.
2. Yet, thinking through marine and coastal ecosystems as a network of materialities, species, practices, and human beings, allows for exploring “leakages” within marine and coastal ecosystems. Unwanted “leakages” in gas pipelines, for instance, stoke fear concerning the potential impact on marine lives and societies. The introduction of so-called “invasive species”, such as the Chinese mitten crab or lionfish “invading new waters”, underscores “leakages” within oceanic habitats resulting from human classifications. Sinking container ships could impact marine fauna and flora by “leaking” consumer goods and substances. The UN High Seas Treaty, has just illuminated targeted “leakages” of oceanic concerns, materialities, species, and governance, thereby exemplifying the political and economic dimensions of leakages.
3. Thinking through knowledge systems, finally, the ocean knowledge system has predominantly been shaped by natural scientists, and academics, primarily from the Global North. While marine social scientists recently seeped into the ocean knowledge system providing important societal knowledge, their approach has often been grounded in “marine empiricism” (Bavinck & Verrips 2020). Reconsidering this ocean knowledge system calls for a transgression of a boundary, for a provoked “leakage”: a “leakage” that enables collaboration between marine social sciences, the natural and engineering sciences, and civic society, but also promotes thinking through the “sea as a theory machine” (Helmreich 2015).
In this interdisciplinary panel, we invite scholars and practitioners from diverse fields to join us exploring the complex and dynamic interactions between sea, science, technology, society, and marine and coastal ecosystems. Our panel will provide a platform for examining oceanic “entanglements of what may count as leakage, leaky, or leaking” through STS-methodology.

Brazil and the Pool: A Story About Leaking Nationalism

Emilian Franco (University of the Bundeswehr Munich, Germany)
In Person

Between July and October 2022, I completed a field research stay at the Center for Artificial Intelligence (C4AI) in Sao Paulo, Brazil. I was able to gain insights into a project called KEML – “Knowledge Enhanced Machine Learning for Reasoning about Ocean Data.” The goals of this project include the mapping of over 7,400 kilometers of Brazilian coastline and adjacent waters, defined as the "Blue Amazon". I want to focus on a subproject that involves conducting small-scale experiments in a wave simulator housed at the University of São Paulo (USP) , the Tanque de Provas Numérico (TPN). Experiments in deep sea exploration are carried out here, but also playful exercises such as the creation of the Brazilian flag on the surface of the pool. This "event" (in the meaning of Anna Tsing), resulting from a more-than-human/pool assemblage, may tell us more about Brazil's efforts to become one of the “big players” in the field of AI development. It also informs us about wicked promises of deep-sea exploration and sheds light on techno-sociological structures in a (post)colonial world. Moreover, it may reveal something about banal nationalism that leaks into and out of the technological advancements at the C4AI in São Paulo.

Anticipating Marine Carbon Storage – Exploring (Knowledge) Leakages

Hansjörg Graul (Goethe University Frankfurt, Germany)
In Person

Since land-based endeavors of carbon dioxide removal (CDR) are not successful enough and face various challenges with land pressure being continuously high or increasing, the Ocean is being

framed as the new frontier in Carbon Dioxide Removal. From the perspective of Marine Social Sciences this contribution will critically engage with concepts of "blue carbon" – more specifically with attempts to amplify (natural processes of) oceanic carbon sequestration through technologically enhanced cultivation of macroalgae. These aspirations might be described as multi species assemblages of (economic) practices, technologies, and life forms. Fundamental to these are then contested knowledges, questions of scalability and dynamic temporalities. Hence, the main goal of my talk will be to discuss questions of knowing if and how carbon could be stored (un)leaky through macroalgal cultivation (1), to what extent storages as well as leakages could be quantified or be accounted for (2), and in which as well as for which temporal horizons this could be achieved or if leakages will occur over time (3). I intend to illustrate this in analyzing cases of enterprises engaged in macroalgal carbon storage, promoting such solutions, and promising to durably keep carbon stored.

Exploring "Situated Modelling" for Ocean Knowledge System(s)

Anja Klein (Humboldt University Berlin, Germany)

Reconsidering the "ocean knowledge system(s)" in research contexts quickly engenders calls for more inter-and transdisciplinary research. Such projects often use various quantitative and qualitative modeling methods as tools for integration. However, models also afford dissent and disagreement, as they require collaborators to be very specific with their assumptions about the dynamics, boundaries, entities etc. of the system in question. Models can handle some uncertainties in these assumptions better than others, especially "natural variability", but less so ambiguities or unknowns. In the relational framework of "Situated Modelling" (Klein et al 2023, submitted) we built on the notion of "situated knowledges" (Haraway 1988) and used the notion of "epistemic agonism" to move beyond the logic of integration (Barry/ Born 2013) and make space for difference and dissent. I will tease out this double role of models based on my fieldwork with such an inter- and transdisciplinary project. The project used statistical and conceptual modeling for integration between different disciplinary work packages to research the connections between Climate Change, flooding and diarrhea incidence in coastal communities in Ghana. This faced particular challenges for a situated use of models due to differences between disciplines, scales, methodologies and familiarity with the modeling methods as well as the concrete coastal contexts.

View From Nowhere vs. View from Everywhere? Empirical Insights into the Knowledge Practices of ICES

Kurt Rachlitz (Norwegian University of Science and Technology, Norway)
In Person

My talk will focus on what Borie et al. (2021) call "institutional epistemologies" within scientific advisory organizations. Drawing upon their study, I will provide insights into the knowledge practices of the "world's first intergovernmental marine science organization" (Rozwadowski, 2004, p. 42, 2002) – an organization that De Donà & Linke (2023) even call "the IPCC of the oceans", namely ICES. I will elaborate on ICES' scoping practices (how is legitimate participation in knowledge creation enacted?), its standardization practices (how is commensurability, e.g., of concepts or futuring modes, enacted?), its representational practices (how is knowledge given form?), and its public practices (how are knowledge and data shared?). My data come from 22 qualitative interviews with representatives of ICES as well as its neighboring organizations (e.g., advice requesters, NFIs, ministries, EU), that I conducted in 06/07 23, ethnography of scientific as well as committee meetings conducted as part of the ASC 2023 (09/23), and another 20 or so interviews I will conduct with other types of neighboring organizations (e.g., NGOs, industry associations, RSCs) by February. Building on previous STS work on ICES (e.g., Nielsen, 2008; Wilson, 2009), I theorize that ICES is currently experiencing a fundamental controversy over its knowledge practices: The legacy

of the stock assessment approach pulls toward a “view from nowhere” (similar to IPCC), while the ecosystem approach calls for a shift to a “view from everywhere” (similar to IPBES).

Panel #23 Technopolitics and Counter-Leaks

Stream: Data & Tech
ABS/0017
Chair: Julia Valeska Schröder

Leaky Networks & Bareback Browsing: Technopolitics of Leakage

Arianna Ferrari (Independent Artist and Scholar)
remote

All bodies are open and therefore leaky. Bodies have nevertheless been long conceived as closed and sealed off. Leakyness has therefore become a negative quality, particularly addressed to the female and queer bodies/subjectivities often interpreted as too open, excessive, dangerous, to be controlled and sealed off again, in order not to leak anymore. Similarly Internet Networks have always been leaky. Every wireless network card downloads/reads in all packets it can then erase those not directly addressed to it. A closed (monogamous) network card - a network card that only read and wrote your traffic - would be inoperable. (Chun, 2015) Nevertheless our networks are conceived as private and closed, when they have never actually been that way. Consequently openness and leakage become negative and dangerous, and they need to be controlled.This presentation calls into question the current technopolitics of privacy and closure as a fake bubble which conceal control and surveillance and proposes to embrace the leakyness of our networks, by initiating forms of bareback browsing, to counter the corporate security of Web 2.0, which is in fact no security, by building new forms of interaction that cannot “leak” because they do not seek to create imaginary bubbles of privacy between users in the first place. (Chun, 2015)

Can Fish Sweat?

Veronika Varga (Independent Scholar)
in person

While nuclear discourse is mostly centered around the spectacular failure of technology, I turn my attention to the construction of what I termed as containment-fiction and its inconsistencies. The rhetorically constructed fantasy of nuclear containment – one of the objects of our political desire – is an obstacle to the flourishing promised by its image. I aim to avoid reproducing damage-based accounts of nuclear technology because such narratives fail to account for how nuclear power plants leak uncontrollably, ubiquitously, and in unremarkable ways. In this project, I study how the foundational myth of nuclear containment made space for the production of other, politically legitimizing fantasies. Technological objects that organize social relations, like energy infrastructure, cannot be separated from the political goals and logics enacted through it. Channeling the ideas of other feminist scholars, I propose a methodology for studying the technopolitical qualities of an infrastructural object through its quasi-failures, to which I refer to as leaks. By highlighting this unexamined category of leakages, I show how the once seemingly rigid boundaries and thresholds of the industrial site are remapped.

Audio Leakage Community

Budhaditya Chattopadhyay (Critical Media Lab, Switzerland, remote), Bojana Knezevic (Independent Artist, remote) & Yati Durant (Independent Artist, UK, remote)
in person

The panel presents case studies where sonic data leaking becomes a form of ground-up knowledge generation as well as a methodological tool for artistic intervention and activism. Drawing from this perspective, the collective Audio Leakage Community investigates the phenomenon of sonic leakage as a conceptual premise to carry the socio-political agency emerging from such trespassing / permeating / spilling over / bleeding / infiltrating the dominant systems of governance, surveillance and control. The collective thought and action are based on gatherings, workshops, seminars, improvisations, performances, public interventions, artworks and publications. The emergent plurivocality is a cumulative impulse in the research, and artistic methodologies of the collective that forms a symbiotic assemblage between sounds, meanings and actions. In four acts the panel discusses different positions dissecting the phenomenon of audio (data) leaking in various contemporary contexts, like censorship, whistleblowing, surveillance, wars, and ecological disasters are of particular concern for the panelists. They unpack the significance of leakage, which they argue is vital for life forms and systems, enabling processes of growth and exchange through symbiotic cohabitation beyond their revelatory capacities and agencies.

Panel #24 Transdisciplinary Methodologies

Stream: Methods and Epistemologies
ABS/0017
Chair: Andreas Bischof

On Becoming an Infrastructure: Personas of Participation and Intervention

Mace Ojala (Ruhr University Bochum, Germany), Fabian Pittroff (Ruhr University Bochum, Germany)
in person

Because there are multiple ways of doing research, one of the pressing challenges of transdisciplinary research clusters is their internal integration. Even within the humanities, disciplines speak and work differently. Especially in long-term research institutions, collaboration under a common program requires an ongoing process of linking diverse interests and practices. Additionally, contemporary research environments are virtual in many ways: every discipline uses at least some digital tools for coordination and data handling, and remote work is the new normal. This layer of virtuality is not just one more problem to solve, but a possible starting point to inquire about and foster integration.
Our contribution draws from ongoing projects within a Collaborative Research Center (CRC), a long-term research institution consisting of dozens of subprojects. From within this context, we investigate practices of data handling and their infrastructures. The goal is to build a virtual research environment that enriches collaboration across the CRC. What distinguishes this infrastructure endeavor is the use of an ethnographic approach following an STS tradition (Suchman 1987, Star/Ruhleder 1995). Besides workshops and user studies inspired by participatory design (Simonsen/Robertson 2013), a key element of our inquiry is an embedded praxiography (Mol 2002) of the multiple ways in which research is being done at the institution in question.
Our presentation will review how participatory and interventionist research such as this can benefit from the deployment of practical personas—virtual characters that STS ethnographers and infrastructure caretakers can enact to become effective and creative agents of a common research environment. This includes the question of how data and knowledge can be leaked beyond the day-

to-day work of small teams to benefit the cluster as a whole. For this purpose, we link our inquiries on virtual research practices with personas from STS literature and beyond, like the Idiot (Stengers), the Trickster (Haraway), the Parasite (Serres), the Server (Full Stack Feminism), the Flâneur (Benjamin), or even the Leaker (Snowden).

Thinking with Leakage in Transdisciplinary Sociotechnical Spaces

Daniela Zetti (Universität zu Lübeck, Germany)
in person

Both historians and philosophers of technology emphasize that contemporary spaces of life and work are socio technical spaces. They argue that, under socio technical conditions, development processes can profit from a sound knowledge in historical trajectories and a critical reflection of today's responsible research and development practices. In our contribution we want to make such historical and critical dimensions productive by reflecting on knowledge transfers in transdisciplinary processes of development and participation and by exploring the concept of leakage and its potential to further the analysis of transdisciplinary environments. Knowledge transfers from ethicists to technologists and vice versa (to name two possible and prominent actors in development projects) follow an ideal of mutual exchange. Yet, each actor will have to take into account several characteristics of sociotechnical environments that challenge, disturb or enrich the ideas of expertise and reciprocity, that shape processes of co-production and open up new fields of action. A number of AI ethicists thus has recently suggested to profit from an updated definition of concepts such as explicability, trust and opacity in order to take individual and collective challenges in digital societies into account. In our contribution we are going to situate and explore the concept of leakage with a view to historical and procedural notions of sociotechnical development. How can porosity be understood and explained? What counts as leakage in contrast to what? When do containment strategies work towards (technical) stability, when are they perceived to work on (social) boundaries? Which kind of figurations of third parties have emerged since the end of the 20th century? We will base our explorations on examples from project experiences as ethicists and on an analysis of computer, network and AI based ecologies of cooperation and information.

Knowing Citizen Science: Social Epistemologies and Epistemic Practices in a National Citizen Science Contest in Germany

Julie Sascia Mewes (Museum für Naturkunde – Leibniz Institute for Evolution and Biodiversity Science, Germany)
In Person

As citizen science gains acceptance, there is an increasing demand for more reflexivity in the field. Social science and humanities research has long described the involvement of non-academics or "amateurs" in academic knowledge production, demonstrating that current activities grouped under citizen science have the potential to move the discussion on the relationship between science and society forward with regards to knowledge acquisition and transfer.
The paper explores recent developments in Science and Technology Studies regarding the social epistemologies of Citizen Science and its diverse actors and practices. Using this perspective and based on the analysis of documents and qualitative interview data, this paper aims to further the understanding of how Citizen Science is promoted and conceptualized in "Auf die Plätze! Citizen Science in deiner Stadt", a national citizen science competition. Various documents created for or related to the contest, including jury statements and informational materials for participants and the public, are examined as tools that embody different epistemologies of citizen science - what it means, what it can be, and what it should be.
Critically reflecting on which epistemologies and associated practices are apparent and reinforced will provide transferable knowledge for Citizen Science advocates, scholars studying the emerging field of Citizen Science, and beyond. It also means promoting further diversification of epistemic practices in terms of actors, objectives, and expertise in various modes of participatory research.

Panel #25 Epistemic Leakages: Configuring the Politics of Epistemic Uncertainty

Stream: Methods & Epistemologies
ABS/E11 Auditorium
Chair: Markus Hoffmann & Alexander Schniedermann (Session 1);
Markus Hoffmann & Melina Antonakaki (Session 2)

The panel engages the notion of 'epistemic leakages' primarily through empirical and/or engaged work on the replication crisis, vis-a-vis epistemic disputes in collaborative knowledge production, as well as during the study of how the scandalization and standardization of scientific reporting are shifting epistemic priorities. The stories, experiences and arguments presented focus on consequential leaks that draw the conduct of scientific work to public contestation. Panelists explore the narratives and political programs regarding the detection and extent of crisis, or the value of epistemic activisms and the political necessities of academic reform and ask how do the epistemic leakages observed, namely the spillover effects connecting different (techno)scientific domains and contexts, contribute in strategically reshaping scientific integrity? Other contributions shed light on how new (human and non-human) actors populate "stories of porosity, boundary transgression and fantasies of closure and containment" (Leakage cfp, §1), for example during re-negotiations over what counts as the truth, at instances when at stake is the co-stabilization of degrees of separation or protection/regulation between the state and the realms of privacy (Thompson 2005: 20, 211). Overall, the panel questions the view of leakage as a negative event and extends previous discussion on the relation between empirical demonstration and epistemic uncertainty (Karlsson, 1990), by factoring in the role of power asymmetries. We aim to understand how regimes of (movement) restriction and regimes of (ethical and political) protection (Hilgartner, 2012; 2017) come into force in the programmatic claims of epistemic interventions. Our STS attentiveness to leakages pertaining to the politics of method (re)directs the lens onto hidden instruments (legal incorporation, administration, expertise) and refracts established arguments by exploring the relation between i.e., experiment and the practical management of uncertainty through contemporary dynamics of technoscientific crises and public contestation.

Session 1 (09:30–11:00)

Leaks, Crises and Scandals: How Science was Broken

Bart Penders (RWTH Aachen University, Germany / Maastricht University, Netherlands)
in person

Science is broken. Science is in crisis. These and other claims about the state of science form the moral panic underpinning the scientific reform movement. The realization that science is not pure, was never modern, and scientists consider naïve views of the scientific method as mere suggestions are no stranger to the social study of science, yet was built up into a scandal of massive proportions in the hands of metascientists (Peterson & Panofsky, 2023). Which epistemic transfers (i.e. leakages) helped construct this scandal and which social movements help sustain it? In this paper, I will approach the aforementioned scandal as the result of articulation work. This encompasses both discussions of epistemic diversity and moral plurality around reform. I will focus on epistemic transfers or lack thereof and ask how 'epistemic leakages' and 'artful contamination' (Zuiderent-Jerak, 2010) can help us understand the process and politics of how science was broken and intervene in how it is remade.

Replication as a Social Movement: On the Community Evolving Around Replication Science and Meta Research

Sheena F. Bartscherer (Humboldt-Universität Berlin, Germany)
in person

The current ‘replication crisis’ shares many of the concerns of older debates and promises to improve research practices across different fields by clarifying replicability as a concept and by updating methodological standards to an increasingly interdisciplinary and digital research environment. However, the current efforts to improve replicability have grown to a level that surpasses earlier debate. Replication studies are a central part of what many are seeing not just as contributions to improving research quality within a specific field but as a new, cross-disciplinary field of research sometimes called ‘replication science’ or ‘meta research’. In line with previous research in science studies about emerging research fields we conceptualize replication initiatives as a social movement. As part of our research project, we are in the process of collecting information on all replication initiatives in the social, behavioral and cognitive sciences and provide a ‘cartography’ of the field and its community. To survey their epistemic and institutional properties, two main data sources are being used: published materials from these initiatives (papers, protocols, websites, etc.) and in-depth, qualitative interviews with participating researchers. A summary of our findings thus far will be presented.

The Laboratory as a Governance Puzzle: How the Stem Cell Research Community Learned to Stop Worrying and Love Epistemic Priority Disputes

Melina Antonakaki (Technical University of Munich, Germany)
in person

At the turn of the century, research with human pluripotent stem cells (hPSC) seemed plagued by controversy—attacks by religious and political groups, lengthy ethical deliberations across countries, targeted reform to accommodate new ontologies, wildly contested policies and research restrictions, unregulated clinics, misconduct allegations and public ir/replication debacles. In the meantime, the heterogeneous, international community assembled around the governance object called ‘human pluripotency’ grew ever more politically sophisticated and resilient, pushing along an agenda of strategic commercialization of hPSC research, while advocating for the epistemic urgency of an Embryo 2.0 public debate. How/where reverberate today the effects, both epistemic and sociomaterial, of such lively politics, like the original Embryo debate? And what are we to make of the current positions held? I approach the questions as attending to epistemic priority disputes—a synthetic model of two science studies literatures rarely examined at their intersections: A) works on the dynamics of epistemic priority in collaborations & knowledge procurement ecologies (Galison, 1997; Knorr-Cetina, 1999); B) works on how scientists mediate the status of discovery events during priority disputes and govern its political effects (Merton, 1957; Brannigan, 1981). I focus on epistemically advantageous moves identified, starting with the extension of the stem cell lab.

Are Market Designers Epistemic Activists?

Sebastian Koth (Weizenbaum Institute Berlin, Germany)
remote

To answer this question, I will outline the change of the concept of market in economics over the last few decades. This shift from market fundamentalism to market instrumentalism entails that markets are not merely transactional spaces based on default laws, but tools for disseminating information, shaping beliefs, and catalyzing social transformation in the domain in which they are deployed. Accordingly, markets do not simply determine the flow and exchange of goods and services but organize knowledge and thus decisions and behaviors of individuals and collectives. I argue that market designers are epistemic activists in at least two fundamental respects. Firstly, they aim to alter the context in which individuals and collectives claim knowledge such that these individuals and collectives tend to make decisions and act in ways that align with improved objectives. Secondly, market designers believe they are in a sufficiently good epistemic position to discern how these knowledge contexts should look like. By relating this argument to the science reform movement "decentralized science" I want to provide resources to better understand the

challenge to distinguish between metascience and epistemic activism. However, the resulting argument holds significance for digital design and policymaking more broadly.

Session 2 (11:30–13:00)

Standards, Stages and Sideshows in Biomedicine

Alexander Schniedermann (German Centre for Higher Education Research and Science Studies, Germany and Leiden University, Netherlands)
in person

Over the recent decades, there have been a variety of reform proposals and social movements that aim to change scientific policies, practices, and culture. Often, these movements center around promoting particular practices or provide concrete standards for day-to-day tasks. Impregnated with epistemic values originating from some core disciplines, proposed standards have been criticized to risk scientific diversity and pluralism (see Leonelli 2023), in a way that one may call “epistemic leakages”. But research on standardization has shown how tinkering with standards is rather the norm and working with standards is "an active act of allowing oneself to be transformed while at the same time transforming the standard" (Timmermans and Berg 2003, 73). Now, if we reject the notion of a standard-driven ‘cookbook science’ and presuppose the complexity of scientific practice as irreducible, what remains as the actual effect of (re-)standardization? Based on empirical mixed-method analyses about how scientific authors, thus recipients and users of standards, deal with reporting guidelines in biomedicine, I argue that standardization creates “epistemic leakages” by shifting the venues of action rather than changing or foreclosing the latter. Instead of reducing epistemic uncertainties, standards realign and redistribute attention to the various tasks involved in scientific practice.

Any Chance for Leakage in Embedded Ethics?

Anja Pichl (University of Potsdam, Germany)
in person

Embedded ethics has recently been called for and experimented with in different fields such as organoid research or medical AI development. This could be interpreted as an attempt to introduce an ethics leak, to open up the process of science and technology development to ethical and societal concerns as it is often presented. A closer look at two exemplary studies, however, reveals rather narrow and a-political concepts of ethics and technology, as well as idealized understandings of the relationship between science and society at work in these projects. Embedded ethics often doesn’t live up to the promise of making a difference in science and technology development towards greater accountability and contribution to the public good. In effect, it tends to contribute to containment and a problematic tradition of co-opting ethics for the purpose of promoting unquestioned visions of scientific progress and technology development. This talk explores the conditions under which embedded ethics could lead to leakage (if at all) and become a site of subversion. Integrating STS approaches and concepts such as co-production or technoscientific imaginaries could play a crucial role in enabling a leaky embedded ethics.

Leaking Health Ontologies in Practice. Ontological Pluralism and Epistemic Injustice

Ozan Altinok (Leibniz University Hannover, Germany)
in person

Against the backdrop of arguments in the philosophy of science that have accounted for the motley of the sciences (Hacking, 1992) and others in the anthropology of illness that have attended to the

multiplicity of disease (Mol, 2002), the received perspective on disease positions the unit of care, the diagnostic unit, and the intervention unit as indiscriminate and follows a disease essentialism. At the intersection of philosophy of science in practice and global justice, I follow recent developments in biology, especially evolutionary medicine, to develop a much more socially differentiated and individualized understanding of diseases. I critique the new paradigms of explanation and intervention, born to contribute to the discussions around the new entities within biology, as they run the risk of causing epistemic injustices through epistemic leakages. These epistemic leakages in the form of decontextualization / deterritorialization are particularly prominent in the unification of these different practices of research for different purposes of health-related topics. In these cases, the position and the role of ethics in managing this risk and responsibility should also include the potential leakages of different hierarchies in the unification of healthcare perspectives. This issue is particularly pressing for the members of the non-hegemonic epistemic community.

Navigating Epistemic Leakages in Citizen Science

Libuše Hannah Vepřek (University of Tübingen, Germany)
remote

In biomedical sciences, citizen science is seen to present opportunities for solving analytical bottlenecks by inviting volunteer participants to analyze scientific research data. In projects based on human computation, human participants are combined with algorithmic tools and AI technologies to achieve unprecedented capabilities. However, such projects also create epistemic leakages destabilizing and disrupting established practices and trust-building mechanisms in scientific knowledge production. Based on three years of collaborative and ethnographic fieldwork on a human-computation-based citizen science game in the field of Alzheimer's disease research, this paper analyzes how these leakages are turned into productive impulses through continuous motions of "intraversions" (Vepřek 2023), which I understand as circular forward movements and shifts in agency and roles within human-technology relations. With the introduction of new "epistemic agents" (Ponti, Kasperowski & Gander 2022) and alongside their intraverting relations, trust, counter to its understanding as a primarily social relationship (e.g., Luhmann 1988; Hardin 2006; Weingardt 2011), should be considered as sociomaterial practice. As such, it is mediated via technologies (Corsín Jiménez 2011; Vepřek 2023). By focusing on the changing human-technology relations I show how epistemic leakages are negotiated by different actors and navigated towards hybrid and distributed modes of knowledge production.

Panel #26 Leaking (Into) Homes: The Integrational Work Within Everyday Smart Homes

Stream: Tech & Data
ABS/E04
Chair: Andreas Bischof (Chemnitz University of Technology, Germany)

As technological development continues to advance at an unprecedented pace, the process of integrating smart technologies into domestic environments is undergoing a transformation of its own. This panel therefore takes an interdisciplinary perspective to explore how IoT-devices are gradually being integrated into our homes and daily lives. The proposed panel discusses the intricacies of technological leakage within the context of everyday life in smart homes. The central thesis posits that contemporary smart home technologies permeate home dwellers' lives not as sudden floods but rather as steady, step-by-step integrations akin to a slow leak. We consequently understand leakage as a term that speaks to the gradual integration of smart home technologies into the home. From domestic technologies to home dwellers, marketers or infrastructures, various actors and their practices (willingly or not) contribute to this process; a process that we call

integrational work. Drawing from various empirical material, this panel will investigate the dynamic interplay between users, smart home technologies and their domestic environments. Through a combination of qualitative interviews, ethnographic field research and data analysis this panel will offer rich insights into the lived experiences of ordinary smart home dwellers as they navigate the process of technological leakage in domestic spaces. Especially the stages of experimentation, adaptation, and normalization are of interest to trace the transformation of domestic spaces and daily practices. Our interdisciplinary approach promises the examination of the manifold and heterogeneous ways in which new technologies are 'leakingly' integrated into smart homes.

Session 1 (09:30–11:00)

Unboxing Convenience: Exploring Voice Assistants Gradual Integration in Smart Homes Through User Valuations

Lukas Schmitz (TU Dresden, Germany)
in person

The ongoing integration of smart technology, particularly IoT devices like voice assistants, is reshaping domestic environments, creating networked, smart homes. The motivation for integrating voice assistants thereby often follows certain ideas: people are convinced that voice assistants can make their everyday life more convenient. This contribution aims to describe these notions of convenience as valuation processes that are successively undertaken by users depending on their attachment in material constellations and discourses (Hennion 1999). Drawing on ethnographic field research and interviews with voice assistant users, it is shown how these valuations differ depending on the specific attachment of the users. Thus, a differentiated picture of the gradual integration of smart assistant speakers via convenience valuations in the domestic environment is provided. In particular, irritations and uncertainties in the integration are described as central for the users' valuations. In my talk I will outline the complexities of valuation concerning the use of smart speakers processes and suggest a broader concept of convenience, that is taking users' perspectives into account.

The Smart Things That Leak Into Your Home

Dennis Eckhardt (Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany)
in person

Households in Germany integrate a huge amount of IoT-devices in the last few years. People are interested in getting new 'smart features' for their homes and buying them for less money. They get their information about those devices on platforms like YouTube, Telegram or Discord and integrate the devices into their homes step by step. After first contact and autodidactic tinkering with those devices, dwellers are getting more and more into the fascination of the smartification of their own homes. This talk presents insights from ethnographic field research done in 10 households and on online platforms. It draws together theories from 'making homes' (Pink), integrational work and practice theory to show how IoT-devices are made leaking into homes. By that, the paper shows how dwellers take care of their family members and own needs. Annemarie Mol's concept of tinkering is helpful to understand the very ways of integrational work of and with their homes as careful ordering. Homes need to be understood as powerful and reflective actors that have societal impact.

Digital Care Work in the Smart Kitchen

Katharina Graf (Goethe University Frankfurt, Germany)
in person

When imagining the smart home, we commonly depict it as devoid of humans: Smart technologies and IoT have taken over the work of the cook, the cleaner or the carer. The human pursues other, more valued work, or finally has the time to enjoy leisure activities. Because of these rather futuristic imaginations, we fail to see how contemporary homes are already smart. Indeed, in line with the call to attend to leakages, my ethnographic research around digitalization in domestic kitchens suggests that while our kitchens are much smarter than we realize, humans remain central to the work therein. This work could be conceptualized as digital care work. On the one hand, smart devices leak into the everyday life of the domestic in a multitude of (often hidden) ways. Especially smartphones and digital kitchen robots are now ubiquitous in provisioning, preparing and consuming foods and contribute to digitalizing everyday practices. On the other hand, human care work remains central to these practices, from the touch of the finger to the negotiation of the likes and dislikes with other household members. Based on an anthropological analysis of this gradual smartification of domestic kitchens in Germany, my talk takes seriously the human component in this process by honing in on the (gendered) care work that it takes to integrate smart technologies into ordinary homes.

Session 2 (11:30–13:00)

Glitches and the Work of the Unreliable Digital Domestic

Eva Isselstein (University of Bayreuth, Germany)
in person

Digital technologies are often less reliable than their manufacturers promise. Far from providing a seamless integration and smooth automation, smart home realities frequently fail: robotic vacuum cleaners map rooms incorrectly, voice assistants do not understand commands, cameras detect insects on the lens as human intruders, and devices suddenly stop responding. With Legacy Russell (2020), I conceptualize these instances as glitches. Glitches are systemic and can shed light on built-in normative orderings, while they also transgress these and can be generative (Leszczynski & Elwood 2022). My research delves into these glitches, which regularly prompt smart home dwellers to engage in unanticipated acts of digital housekeeping, “babysitting” or repairing devices. Drawing on my online ethnography on Reddit and interviews with smart home dwellers as well as feminist perspectives on care work, this paper utilizes the glitchy instances of the smart home to make visible the work required to integrate digital devices into everyday life. I thus illustrate how the gradual “leakage” of digital technologies into the home is an often arduous process, entailing collaboration and communication between household members. This exemplifies the smart home as a relational space where technology shapes and is shaped by social relations.

Developing Alexa in the Face of ChatGPT: Investigating Technological Leakage

Niklas Strüver (University of Siegen, Germany)
in person

The smart home constitutes a set of technologies and ideas that are advertised to facilitate a more efficient or pleasant everyday life. The interconnected nature of smart homes allows for the seamless exchange of information between various devices, enabling efficient control and automation. At least that's what advertisements used to promise. However, in the wake of ChatGPT, developers like Amazon seem to think that voice assistants are in need of some adjustments. This presentation will focus on the development of the voice assistant Alexa in the face of ChatGPT. This will be achieved by analyzing interviews conducted with developers of the Alexa Price competition. Here socio technical problems in the development can be contrasted.
The typical narrative of Alexa in the smart home can be seen as a device and service that establishes infrastructural connections between various devices and platforms. This can be contrasted with the changes Amazon strives to make as an adaptation to large language models. The perspective of leakage helps to understand how Alexa and its new features serve as gateways for new AI-tools to

find gradual integration into the (smart) home, exposing users and third parties to change in their known practices.

Panel #27 Regional Extractivism: Lusatia and Ore Mountains

Stream: Energy Infrastructures & Contamination
ABS/E08
Chair: Alena Bleicher

Leaking Extractivist Logics: Dispositif Engineering of Cultural Leakage

Ingmar Lippert (Goethe University Frankfurt, Germany)
in person

STS has a longstanding interest in the shaping of society through technology, and as part of that in the way cultures are shaped through science and technology (consider the (re)production of masculinity in physics (e.g. Traweek 1988, Lucht 2015)). Cultural leakage typically denotes, in the humanities, mutual shaping of cultures or containers of cultures. The premise is an, at least, implicit concept of bounded cultures or their containers, where elements cross the boundaries, such as across Caribbean cultures (Brathwaite 1974), planetary or galactic civilisations (Voros 2014). Positing such a boundary, this talk focuses on a corpus of materials containing artifacts of the interaction of Lusation extractive industries, specifically lignite mining, and Sorb cultural associations. With this corpus I link to recent STS conversations on extractivism, land, soil and the underground (e.g. Kinchy et al 2018). I analyze the material with respect to framings of Lusatian environments, landscapes and geographies. I exemplify this analysis with textual, film and human-machine-interaction data. I argue that the mining industry effectively engineers a form of cultural leakage, in which in the interaction, the extractivist logics of the mining industry is leaked into artifacts that belong (also) into the universe of the Sorbs, constituting a form of cultural subversion. I theorize this finding as a form of dispositif engineering (resonating with Law's (1986) heterogeneous engineering), that achieves cultural leakage of an extractivist logics, that comes to then characterize the Sorbian territorially claimed Lusatia.

Lithium in the Ore Mountains: The Green Sacrifice Zone of the Low-Carbon Transition

Karolína Žižková (Czech Academy of Sciences / Masaryk University, Czech Republic) & Tereza Stöckelová (Czech Academy of Sciences, Czech Republic)
in person

Lithium has been recently assigned a crucial role by the EU in both low-carbon and digital transition for its energy storage uses in electronics, electric vehicles and the renewable energy sector. This allows lithium mining proponents to frame its extraction as a climate change mitigation strategy, a paradoxical turn critically referred to as “green extractivism” by political ecology (e.g.: Tornel 2023; Voskoboynik, Andreucci 2022). This paper offers an analysis of the lithium mining legitimization narratives and their local contestation in Cinovec, where underground deposits of “European significance” have been identified across the Czech-German border. The paper critically analyzes how the legitimization builds upon the emphasis on 1) mining continuity and tradition in the region, 2) jobs creation and sociotechnical imaginaries of regional transformation into the hub of innovation and research, and 3) ethical aspects of “responsible” mining and taken-for-granted compliance with environmental standards in Europe in comparison to the Global South. This framing obscures the nature conservation efforts and land reclamation already taking place in the proposed mining site; the lack of demand for jobs in the mining sector among local residents and their fear of low- skilled labor migration; and unequal distribution of risks (such as pollution, noise, water shortages and infrastructure damage) and benefits of the extraction. To account for the situation in

Cínovec/Zinnwald area, we employ the concept of the “green sacrifice zone”, suggested by Zografos and Robbins (2020) to study the logic of cost-shifts and coloniality underlying some of the low-carbon transition agendas including the European Green Deal. We conclude that rather than sites of unfortunate mismanagement, green sacrifice zones constitute an inherent leakage of European climate politics which is based on the fantasy of endless economic growth and raw materials supply security.

Re-framing Mining Spillovers: A Case of Sociotechnical Imaginaries of Pit Lakes in Lusatia

Alex Hok Nang Tam (Brandenburg University of Technology Cottbus-Senftenberg, Germany)
in person

This presentation discusses the sociotechnical imaginaries of post-mining landscapes in Lusatia by delving into the transformative potential of ecological aesthetics as a catalyst for experimental heritage-making. It demonstrates how the notion of leaking is useful to shift the focus of heritage practice from a boundary-setting, change-averse endeavor to one that dissolves sharp anthropocentric confines and connects with materialities and temporalities beyond human perception. The fall of communism and the subsequent reunification of Germany in the early 1990s brought about substantial changes to the mining landscape of East Germany, including Lusatia. The privatization and reorganization of lignite mining led to the closure of mines and a profound transformation of the open-cast mining areas in Lusatia. As mining operations ceased, the abandoned pits gradually transformed into large lakes, and this recultivation model became widely recognized for its success in restoring disturbed landscapes. The late 1990s witnessed a transformation of former mining areas in Lusatia in the context of Germany's renewable energy development policies. The region began to shift its approach to land use and resource management. Wind turbines started to be integrated into the landscape, showcasing Lusatia's commitment to sustainable energy practices. This transition toward renewable energy sources reflected the region's responsiveness to environmental concerns and its efforts to build a more sustainable future. This is exemplified in Herman Prigann’s “Yellow Ramp” (“die Gelbe Rampe”) (1993–94), a land art project undertaken at an open-cast lignite mine near Cottbus, emphasizing the concept of ecological succession and continuous growth. Through examining the shifting artistic concerns for the Lusatian industrial landscape, this study posits that the post-mining landscape entails a constant re-drawing of boundaries of the definition of aesthetics, where it is viewed as a set of inter-relationships between biological, cultural, political, and historical aspects of ecosystems.

Panel #28 Thinking with Leaks - On the Potential and Limits of an Epistemic Resource

Stream: Methods & Epistemologies
ABS/105
Chair: Alysse Kushinski (Harvey Mudd College, USA)

Leaks are rarely desired: internals or secrets spilled to the public, a broken water pipe, or human bodily fluids. Leaks disrupt the ordinary and address what’s taken for granted. In cultural and media studies, the leak therefore lends itself to be employed as a critical thinking technology (Haraway; Kushinski). Regarding technological infrastructures, it allows us to focus on what is better not to leak. It makes the supposed invisibility of infrastructures an issue and points to the porosity of hegemonic imaginaries of efficiency and frictionless traffic. The leak, however, goes beyond being merely a pendant to disruption, granting a glimpse behind the scenes of what’s hidden from view. As a departure point for scholarly reflection, it’s possible to ask for what reason leaks don’t seem desirable, who determines what’s considered desirable, and how such assumptions have become historically ingrained. Accordingly, the leak offers itself as a thinking technology to unravel power

relations. In our panel, we take the talk about leaks seriously and understand it as an epistemic resource. It irritates imaginaries and bodies of knowledge and offers a rich metaphorical field to develop counter-depictions to common understandings of infrastructures. Hannah Schmedes takes a closer look at technological infrastructures' imaginaries. Before the backdrop of the leak, she unfolds gendered assumptions of containment. Matti Denecke then investigates what the leak is directed against, if it is meant to trouble present depictions of an infrastructured world in flow. Eventually, Alisa Kronberger focuses on how the leak relates to Barad and Haraway's feminist thinking of diffraction. To what extent does a thinking of diffraction differ from that of the leak and how can they inform each other? Altogether, the panel interrogates the potential and limits of the leak as an epistemic resource for theory building.

A Leak Can Be a Feminist Lead

Hannah Schmedes (Ruhr University Bochum, Germany)
in person

"Leaks and eddies are everywhere" (2016, 105) writes Donna Haraway in Staying With the Trouble. Indeed, leaks are omnipresent: in discourses about the bodies, in closed systems sciences as the failure of flow in material infrastructures, but also in discursive constellations of knowledge production in the form of leaked documents or whistleblowing. Leaks, in their becoming-apparent, symbolize a failure to contain, a dysfunction, a seepage. Following up on these forms of leakage, the notion of containment and its porosity are revisited, regarding their gendered and sexualized implications. From this point of view, I will develop a reading of the leak, which departs from this very localization of technologies of circulation and containment. In following various infrastructural lines and their porous points I will pursue Sarah Ahmed's thesis as to whether and to what extent a "leak" can become a "feminist lead" (2017).

With Leaks Against a World in Flow?

Matti Denecke (Ruhr University Bochum, Germany)
in person

What does the talk of leak offer when juxtaposed with a contemporary diagnosis of the world that is not in flow, but "messy"? (Kushinski 2022) A world in fluid-like flow, characterized by the unimpeded and uninterrupted movement of data, capital, goods, and people in globally networked infrastructures, does not exist. Data is routed and must buffer (Soon 2021); capital does not flow but circulates and is accumulated (Gänger 2017; Harvey 2020); fleeing people are pushed back and detained (Bojadžijev 2021; Hart 2011); travels can be sticky, as one must expect waiting times (Costas 2014); supply chains rely on planned interruptions to synchronize shipments of goods (Gregson et al. 2017; Tsing 2009). My assumption is this: The use of the leak as an epistemic resource to theorize against a globalized world of flows simultaneously affirms that very world of aqueous flows. This is because the leak is directed against idealized notions of flow, which historically can be credited to management discourse (Denecke 2023; Dommann 2023; Roberts/Joseph 2015).

Diffracting Leakage

Alisa Kronberger (Ruhr University Bochum, Germany)
in person

As a metaphor from optics, Donna Haraway introduced the notion of diffraction as an important thinking technology and a different kind of critical consciousness (Haraway 2008). Her aim with the figure of diffraction is to perspectivise an interplay between literal and figurative meaning for the construction of new models. In contrast, Karen Barad uses diffraction to propose both a new

relational ontology of world-making and a methodology of “thinking through one another” (Barad 2007). While Barad understands diffraction patterns as components of a processually conceived worlding, for Haraway diffraction holds an elemental potential: it allows us to think of differences not as the other of one, but as the effect of a relational assemblages. Like leaks, diffraction stands for ongoingness and hidden complexities. Both figures offer ways of thinking critically about the constitutional conditions of human and non-human bodies and systems. In my contribution, I will first discuss the idea of a diffractive critique and method with Haraway and Barad. Diffraction as a critical figuration (Thiele 2020) will then be confronted with the conceptual figure of the leak, in order to ask for differences differently so as to make a difference.

Panel #29 Leaks, Cracks, Spills, Overflows – Waste, Pollution and Toxic Substances in Motion

Stream: Energy Infrastructures & Contamination
ABS/E04
Chair: Sven Bergmann, Johanna Kramm (Session 1);
Baldeep Kaur Grewal, Yusif Idies (Session 2)

We will trace the movements and relations of materials and substances that have supposedly been disposed of, buried, dumped or submerged. The supposedly safe containment of substances and materials seldom works as planned, and often the sink becomes the site of a new leak or spill. For example, the management of waste, pollutants and toxic chemicals rarely runs smoothly – incomplete disposal and recovery processes are the rule rather than the exception. This can manifest itself in "disruptions", for example, when pollutants leak from landfills or waste and toxic substances escape from infrastructures built to manage them. The components and relations of such infrastructures often only become visible in the event of disaster or collapse.
In addition to the spatial diffusion of waste – especially through water and air – leakages usually also have a particular temporal dimension. Modern waste streams are increasingly complex and are constituted by substances with varying rates of decay. Certain materials rot away more quickly or become the object of entry themselves, while others need a longer time span. The emerging materials always carry a temporal trace that marks the time of their input into an environment either via planned dumping or via unexpected leakage. The specific relation of pollution to land is characterized by Max Liboiron as a form of colonialism and identified by Rob Nixon as “slow violence”, the effects of which humans and other living beings on the contaminated land are often still exposed to decades later. The planned removal of waste away from sites of privilege and power is done to encourage disposability while the pollutants are shipped elsewhere to be managed by inherently leaky structures like landfills or incinerators. Due to the complex, emergent structure of these discards, their flows can be understood as "wicked problems“ that systematically elude their own containment and clear problem-solving strategies.
We want to explore practices, structures and entities that generate waste, discards and pollutants as such and set them in motion in the first place.

Session 1 (09:30–11:00)

Infrastructures of Leakages. The Case of Migrating Radionuclides in the Final Disposal of Nuclear Waste

Christiane Schürkmann (Johannes Gutenberg-Universität Mainz, Germany)
in person

In the development of repositories, scientists assume that radioactivity escapes in the course of hundreds to thousands of years. At the same time, scientific research is being conducted into how

best to counter leaking radioactivity. In this context, the development of multi-barrier systems is being worked on, in which engineered and natural materials are used in combination to retain radionuclides in the long run. The paper discusses repositories for radioactive waste as infrastructures of leakages. Regarding their development and construction, they center the treatment of leakages of their toxic, built-in waste inventory. In this way, the paper illuminates how leakages are inscribed into disposal scenarios as scenarios of future infrastructures of waste management. It further asks how far this goes along with a politicization of dealing with leakages. Thus, a field of tension opens up with regard to these infrastructures: They are supposed to counteract leaks of toxic and hazardous matter; at the same time, in their function as repositories with barrier effects, they assume and enable leaks in the underground. Moreover, this is not a passive leakage of materials – on the contrary: what leaks out are radionuclides being in motion and migrating into and through the materials opposing it.

Slow Violence and Rapid Testing: Speculative Modes of Estimating Risk in the Mold Multiple

Christoph Schemann (University of Bayreuth, Germany)
in person

Mold fungi are one of the most familiar and most frequent microbial beings in our ecosystem, the urban environment, houses and homes. Given that, it is all the more astonishing then that the crucial nexus between domestic contaminations of mold and human health is highly contested and unstable and opens the door for often unpredictable and speculative effects in different directions (cf. Schemann 2020). In this presentation I will question three ontologically different modes of estimating risk in the “mold multiple” (cf. Mol 2002) that are situated in three different local contexts: first, an inhabited home that has been exposed to mold damages several times in which risk manifests in the form of “slow violence” (Nixon 2013, Davies 2018); second, a cleanly home with no visible mold damages that has been rapid tested with an apparatus called “Sporometer” (a purchasable mold testing device designed to measure the load of indoor mold spores in the ambient air for laypersons on a scientific basis, comparable to a coated Petri dish) where risk gets progressively produced through a “market device” (Muniesa et al. 2007); and third, a professional laboratory for environmental analysis where risk gets cut “together-apart” (cf. Barad 2014) several times by different laboratory procedures and socio-technological fixes that aim at containing and governing mold in a certain direction, yet working with “leaky knowledge”.

Session 2 (11:30–13:00)

Leaky Bodies, Leaky Theories: Discussing Engagements with Chemical Bodies in Power Relations

Annika Troitzsch (Goethe University Frankfurt, Germany)
in person

Farming, pharmacy, mining, building, energy suppliance, plastics - modernity seems to be chemical. And so are bodies: They encounter chemical residues and live with them – some (often involuntarily) more than others. In my contribution, I will explore different empirical and theoretical engagements with the relation between bodies and chemical pollution. They range from rather exposure-oriented approaches that can be found in critical perspectives on unequal exposure in environmental justice scholarship, environmental health scholarship and political ecology to rather relational approaches questioning the normativity of uncontaminated, “natural” bodies and understanding the body as always leaky and changing with chemicals, as found in perspectives of queer ecologies, critical disability studies and new feminist materialisms. In mapping the different approaches, I aim to trace how elements of approaches following a substantial ontology of exposure leak into those following a relational ontology – and the other way around. I analyze how different approaches of understanding

bodies in relation to chemical pollution in power relations can be brought together. Thinking with these entangled approaches, I suggest to look at chemical bodies in power relations in terms of vulnerability, considering the selectivity in forming chemical bonds and pursuing a repoliticization of chemical pollution.

Leaky Cleans: Transposing and Transforming Micropollutants in Dutch Water Cleaning Practices

Fenna Smits (University of Amsterdam, Netherlands)
in person

Wastewater treatment plants struggle to deal with new pollutants that were not accounted for in in their original design, particularly the so-called micropollutants like pharmaceuticals, chemicals, and microplastics. The persistent nature of these pollutants undermines traditional biological treatment systems, resulting in continuous spills and leaks into surface waters. Water engineers are tasked with 'solving' these problems, but they face limitations: they cannot prevent micropollutants at the source nor control their distribution. Consequently, they strive to ‘take out’ micropollutants from wastewater flows. To do so, they explore various treatment technologies, knowing that none of these approaches is impervious to leakages. Cleaning one substance might inadvertently lead to the leakage of another harmful contaminant. Cleaning in one site may lead to dirt elsewhere. Never quite clean. This paper delves into how water engineers navigate the problem of micropollutants in the Netherlands by trailing how these pollutants are shifted around within messy cleaning practices, all the while wondering with the engineers whether it can be transformed into something innocuous. By doing so, the paper analyses the varied ways in which pollutants not only shift place but also change form. Instead of viewing cleaning as a linear problem-solving task, moving from bad to good, this paper proposes the terminology of 'transposing' and 'transformation' to specify how each shift in place and change in form brings about different goods and bads, along with various approaches for handling them.

Leaky Legacies – Toxic War Remnants in the Seas

Sven Bergmann (German Maritime Museum, Germany)
in person

Even decades after wars, their legacies are not only particular cultural heritage, but also a current and future environmental problem. Toxic explosives such as the carcinogenic trinitrotoluene (TNT) escape into the sea from corroding munition shells (inside war wrecks or at specific post-war dumping sites). As with many other hazardous materials and substances, the sea was long considered a safe dumping ground. On the other hand, the risk of a spill was downplayed because the sea is large and therefore the substances should disperse well (dilution approach). However, compared to a direct accident in which munitions explode, the slow corrosion and leakage of munitions in the sea is not particularly spectacular. There are no images of mass fish kills, but rather of liver damage in localized flatfish living near war wrecks – a ‘slow disaster’ that requires elaborate marine research to document. Research on changes and pollution in the marine environment thus often faces uncertainty, indeterminacy, speculative impacts and unpredictable temporal consequences. Knowledge production is therefore complicated when it comes to determining what constitutes a risk, hazard and/or threat to the marine environment. This presentation draws on a transdisciplinary research project in the North Sea.

Panel #30 State Power and Regimes of Control and Containment

Stream: Bodies & Governance
ABS/213
Chair: Martina Erlemann

Is There Anybody Out There? Communication From (and In) Prison

Aaron Bielejewski (Center for Criminological Research Saxony, Germany)
in person

Prison has often been considered a “total institution” with an emphasis on the enforcement of daily routines. The modern prison has, for a variety of reasons, delegated a great deal of organization to prisoners themselves. The increasing normalization of in-cell television and telephones both empowers prisoners in terms of giving them access to information, media, and (still restricted) communication with the outside, while simultaneously reducing the burden on administration and staff to organize prison life from minute-to-minute. Prison media and communication can conversely increase the tolerance of “lockdowns” and extended periods in which prisoners are restricted to their rooms – often attributed to a lack of personnel. As part of a larger project exploring the role of media in prison, interviews and participant observation was conducted in JVA Chemnitz, a prison for women in Saxony. Prisoners described both their experience with differing forms of media and communication and also how it impacted their experience of prison and daily routines. In Germany, competing trends toward prisoner autonomy (in the form of free-time activities and resources) and towards formalized structures of work, education and therapy. This raises the question of how “the prisoner community” is established and maintained in consideration of on-going contact with “the outside” among prisoners, such as family visits and video chat. Additionally, prisoner communication – specifically telephone and letter writing – generally require financial resources provided by the prisoners, typically requiring prisoners to calculate the financial value (in terms of their own labor or limited resources provided by family) of varying levels of interpersonal contact: while many prisoners report having daily conversations with outside friends and family, they are equally often quick to describe this communication as a financial burden. Prisoners – narratively and interactively – attempt to cultivate prison relationships while still maintaining and defending a “normal” self. It remains necessary to then explore how on-going changes in prisoner’s access to the outside – both physical and communicative – impacts their sense of self and the relationship between their future-oriented post-release identity and their integrated prisoner self.

Making Visible the Brutality of the EU Deportation Regime by Tracing the Leaks in Its Infrastructure

Jakub Crcha (Radboud University, Netherlands)
in person

Last year alone, almost 100,000 people were removed from the EU following a legal order. In order to facilitate such an extensive volume of forced removals, the EU and its member states have constructed and are continuously expanding a robust deportation infrastructure spanning across multiple sites and spilling way beyond the con1nes of the EU territory. Deportation is a process that benefited politically from being invisible to the public eye. Not only would visibility expose practices of deportation to the potential popular resistance, but the inherent brutality of forced removal would also disrupt the curated public image of European space as that of ‘security, freedom, and justice’. As such, the deportation infrastructure in the EU is constructed as an invisible, securitised, and sanitized negation of the democratic public space. As such, it is increasingly rare that an uninitiated EU citizen would come to witness the inner workings of this brutal mechanism. How then, can we as citizens, activists, and researchers make the brutality of the deportation regime visible and politicized? In my proposed contribution, I argue that there are moments within the deportation infrastructure that are prone to leakage. Whether it is a sight of a person being forcefully boarded through the back of a plane, an entourage of immigration officers entering a mysterious back door

of an airport, or a testimony of a witness in an regional newspaper of a small country, I propose that the seemingly inaccessible infrastructure leaves traces of visibility that can be picked up by a careful observer. With the increasing focus on militarisation of border policing and the growing obscurity of the EU border agencies, it becomes ever more difficult for us to properly understand what is happening within the sanitized corridors of state power. We must, therefore, learn how to pick up the leaked pieces that evidence the brutality of the EU deportation regime. In my contribution, I will combine ethnographic insights with media archaeology to discuss the difficulties of researching deportations in the EU, and propose a methodological approach that traces the leakages of the deportation infrastructure into 'everyday life'.

Reverse Search Warrants: How Leaky Legislation Facilitates Google Locative Data Surveillance

Renée Ridgway (Aarhus University, Denmark)
in person

In the 21st century user data has become the world's most valuable resource, as raw material exploited for 'data colonialism'––the appropriation of human life for profit (Couldry and Mejias 2019)––and as a commodity which is sold, traded and reused, facilitating new leakages. Since around 2009 all 'locative data' including detailed location records involving at least hundreds of millions of devices worldwide (Valentino-DeVries 2019), is collated and stored in Sensorvault. This massive proprietary database is part of the larger Google server complex, which 'maps' users' habits and actions for private use, although relying on public infrastructures such as cell towers and GPS satellites. During the past 5 years there has been an explosion of criminal investigations where US law enforcement has been requesting that Google release Sensorvault data through 'geofence warrants' or 'reverse search warrants'. This presentation sheds light on these 'geofence warrants' by reversing the data- gathering process––collecting data on those collecting data on us(ers) through investigative journalism reports in the media, legal documents (court orders, subpoenas, affidavits), US governmental committee letters to Google and recent congressional developments regarding legal 'loopholes'. Instead of appealing to a judge for a search warrant, the NSA and other members of the US intelligence committee are lobbying to oppose an amendment that would stop them from paying companies to obtain location data. Predicated on this confluence of methods between law enforcement's dragnet policing practice and corporations' surveillance capitalism (Zuboff 2015, 2019), locative data from mobile phones facilitates the profiling of (innocent) bystanders, who, by their proximity to crimes, become 'surveilled and surveilling subjects' (Lyon 2018, 6). Parallel and recursive' (Coté 2016), this research contributes to feminist STS infrastructure literature by highlighting some consequences of geofence warrants: mission creeping of user data and the resulting 'chilling effects' (Büchli et al 2020, Fussell 2021).

Panel #31 Leakage and Post-Colonial Critique

Stream: Ecological Leakage and the More-than-human
ABS/214
Chair: Chris Scheidler

Conduction and Memory Lapse: Leakage in Ta-Nehisi Coates' The Water Dancer

Timothy E. Stroud (Martin Luther University Halle-Wittenberg, Germany)
in person

The Water Dancer by Ta-Nehisi Coates uses the metaphor of leakage to examine the ways in which enslaved people resisted and undermined the system of slavery in the United States. The novel's

protagonist, Hiram Walker, a complex and nuanced character who does not conform to the stereotypes of either the docile or the rebellious slave, is born on a Virginia plantation. From his mother he inherits Conduction, allowing him to transport himself and others through space and time by tapping into his memories and emotions. Coates draws on various interdisciplinary fields to articulate a vision of bodies and systems that are open to the contingencies of environmental exposure and socio-technical enmeshment. He uses the metaphor of leakage to describe the phenomenon of Conduction, which defies the boundaries of space and time imposed by slavery. He also uses it to describe the ways in which enslaved people subverted the system of slavery by creating alternative forms of communication, knowledge, and community that escaped the control of their masters. By exploring the politics and materialities of leakage, The Water Dancer raises questions about vulnerable and permeable ecologies, architectures, technologies, and social systems that are constantly subject to change, disruption, and resistance. This paper will discuss Leakage as a metaphor for the slow and steady subversion of slavery and the role of memory in creating new futures.

"Cracking the Surface" – Flows Between Above and Below Ground?

Livia Cahn (Rachel Carson Centre, Ludwig-Maximilians-Universität München, Germany)
in person

Where does the underground begin and end? The notion of the underground relies on a surface that marks a separation between above and below ground. This paper will unpack an empirical case to interrogate the work that is done to maintain this difference. The example centers on a seemingly solid geological sample that was used to prospect resources in the province of Haut-Katanga in the Democratic Republic of Congo (DRC). This small sample was used to inform a comparatively expansive extraction from underground. Leaving behind a hole, it was first kept aboveground in a collection, until it was deemed too radioactive to be stored and was buried again. The refusal of this radioactive rock sample to be contained in the collection and then underground, manifests the connections between above and below ground through contamination.
Even when attempts are made to constraint contaminants, the lasting effects of uranium-bearing rocks are impossible to determine. Radioactivity has long life spans and lasting effects. And yet, this has been no reason not to try to measure and contain radioactivity. From this one example of a sample, a plethora of economic and political projects and campaigns can be identified that have sought to manifest control over. I argue that these mechanisms of control have a flattening effect on understandings of the surface. They require the surface to be neat and stable. To give depth to the surface, I therefore propose a shift in attention from control to unintended movements to change perspective from containment to flows, spills and leaks. The shift from containment to flows applies not only to the radioactive matter but also the material available about it. Traces (archival and spoken) are often kept secret, quiet or out of reach, just like the contaminants themselves. In the face of these attempted concealments, porous knowledge production raises methodological questions about empirical work, that I would like to address too.

Panel #32 Off-Site Walk: Wireless Walk and Talk – Tracing Mobile Communication Signals Through Dresden I

Meeting Point tba
Part II: 11:30 – 13:00 in ABS/105
Nona Schulte-Römer, Ignacio Fariás, Alexander Harder, Brett Mommersteeg & Elisabeth Luggauer (Humboldt-Universität Berlin, Germany)
in person

The number of participants is limited. The tour details will be shared with the registered participants. To register, please contact: Nona.schulte-roemer@hu-berlin.de.

This three-hour 'wireless walk and talk' traces electromagnetic signals through Dresden, engaging STSing participants in the invisible presence of wireless telecommunication technology and its leakages. We are interested in how 5G, the fifth generation of mobile communication technology, creates new physical connections in the city, but also future visions, societal opportunities and opposition. On our tour, we will trace these connections and make the electromagnetic wave spectrum tangible by using measurement devices and through site-specific social-scientific reflections and interventions. Dresden offers a great site for our exploration. The capital of 'Silicon Saxony' was once known as the 'valley of the clueless' indicating that the hills around Dresden blocked West German TV and radio signals preventing the leakage of information during the socialist period. Today, the city government points out that ‘the mobile communications network of tomorrow is being researched’ at TU Dresden where specialists develop novel technologies and applications and run a ‘5G lab’. However, the current roll-out of 5G is also creating "leakages" in the form of doubts and fears among the citizens. In January 2020, a group of civil society actors submitted a petition to ‘stop 5G’ to the Lord Mayor of Dresden pointing to potential environmental and health effects caused by electromagnetic radiation. Official scientific risk assessments, in contrast, show no worrying health effects from wireless connectivity. The tour is organized by Nona Schulte-Römer, Brett Mommersteeg and Ignacio Farías (PI) of the ERC project WAVEMATTERS – how physical waves come to matter in contemporary urbanism.

Panel #33 Biopolitics of Marginalization

ABS/E08
Chair: Jakub Crcha

Leaks in the Machines of Colonial Overcodification: Parasitological Knowledge Production in Early 20th Century China

Dominik Merdes (Technische Universität Braunschweig, Germany)
in person

From the viewpoint of the so-called ‘tropical medicine’, China was a very interesting knowledge space in the late 19th and early 20th centuries. On the one hand, this space played an important role in the institutionalization of tropical medicine at the turn of the century. On the other hand, knowledge production was more dispersed than in other regions that were under direct colonial rule. Officers of the China Maritime Customs Service, medical missionaries, and Japanese colonial doctors were among the protagonists. The archives of ‘tropical medicine’ for these years are based on the writings of doctors of Western medicine who were convinced of the superiority of their school and civilization. With the work of the philosophers Gilles Deleuze and Félix Guattari, their writings can be understood as an expression of processes of overcodification, which—due to colonial

violence—constituted a specific kind of epistemic rupture. This rupture marked knowledges that did not comply with the biological binomial classification or with the elements of Western chemistry as irrational and superstitious. Nevertheless, Western knowledge production drew on those very kinds of knowledge. As Deleuze and Guattari have argued, the power centers of overcodification are also dependent on their own impotence. The marginalization—or even effacement—of knowledges by the machines of overcodification relates to 'quanta it can only convert without being able to control or define' (A Thousand Plateaus, 2005: 226). These uncontrolled and deteritorrializing quanta may leak from imperial worlding-processes (Spivak). Such leaks can be found in the texts of parasitological doctors, regulated through the rules of discourse. Occasionally, unforeseen encounters materialized in their formal writing. In my talk, I will discuss such leaks that deterritorialized Western knowledge grids and flew into the production of parasitological knowledge.

Whence the Gaze? Observing Race Within a Tunisian Microbiome Research Project

Marta Scaglioni (Ca' Foscari University of Venice, Italy)
in person

In spite of its deconstruction as a valid biological category, both by social sciences (Silva 2007; Wade 2015) and by genetic research (Cavalli-Sforza, Menozzi, and Piazza 1994), race is still operational in clinical analysis, as it is used, for example, to predict disease susceptibility. However, its uncareful application may encourage forms of biological essentialization and typological thinking (Meloni et al. 2022), as well as reproduce disturbingly racist narratives and practices, ignoring social and environmental contributors to diseases (Brody and Hunt 2006). As I argue in this presentation, the usage of racial categories in life-sciences is particularly problematic when science is enacted in the Global South, both because of uninformed conceptualizations of racialized populations and lifestyles, and because race is mostly conceived of as the product of a specific geo-political context, that is Western – mainly US-based – world. I draw from an ethnographic study of a microbiome research project in Tunisia, addressing the ambivalence embedded in the racial categories at use, delineating how they consist in the imposition of Western racial "commonplaces" (Bourdieu and Wacquant 1999), on the one side, and integrate local racial constructs, on the other. I conclude outlining how problematising racial constructions within microbiome research, and clinical sciences at large, in non-Western worlds may help give scientists the opportunity to avoid methodological pitfalls and theoretical biases, rendering a more correct and more equitable science.

Seeking Optimal, Creating Marginal?

Kristína Jamrichová (Brno University of Technology, Czech Republic)
in person

Diagnostic Centre for Youth with Behavioural Disorders is a kind of crossroads in the world of residential facilities for youth, through which flow bodies, diagnoses, substances, documents, traumas, expectations, and negotiations. In the research that led to this paper, I worked with this facility within the Slovak special education system, as with an assemblage of discursive-material technologies of subjectivation (Rose 1998). I will first present a brief insight into the corresponding genealogies of expert discourses in practice. Then, I confront these with (selection of) specific practices, i.e. repertoires and modes of action as practiced by the actors of the center at which I arrive through ethnographic research (or "praxiographic" to use Annemarie Mol's term, 2002). Analyzing behaviorism's place in diagnostic and therapeutic practice and the related meritocracy of scoring systems, I suggest that these tools instead shape young inmates as economically acting subjects, which does not have to mean at all they also become socially emancipated. In this regard, I will also briefly touch on the practice of deploying psychiatric medication in residential care for children and youth. The latter is widely understood as an appropriate adjunct to Cognitive-behavioral therapy. The question is whether, rather than being an auxiliary therapeutic strategy, it is a flimsy (potentially dangerous) band-aid solution to systemic problems with no vision of other solutions.The

metaphor of assemblage (DeLanda 2006) is a valuable tool to capture the porosity and sites of possible leakage from these bio-political strategies at different facility levels. How differently is the behavioral disorder or childhood created within the diagnostic or educational technologies?

Panel #34 Workshop: Editors Meet Authors and Reviewers: Re-figuring STS Publications Together

Stream: stsing
ABS/0017
Chair: Martina Klausner

Estrid Sørensen (Ruhr University Bochum, Germany), Ingmar Lippert (Goethe University Frankfurt, Germany), Endre Dányi (University of the Bundeswehr Munich, Germany), Uli Beisel (Freie Universität Berlin, Germany)
in person

Publishing is the collaborative effort of authors, reviewers, editors, typesetters, designers and many more. We would like to use this workshop to exchange experiences, expectations and wishes for the publishing processes within STS. The organizers of this workshop are editors of EASST's (European Association for the Study of Science & Technology) house journal Science & Technology Studies (https://sciencetechnologystudies.journal.fi/) and the publishing house Mattering Press (https://www.matteringpress.org/). We have two aims with the workshop: On the one hand, we want to inform the STS community about the profiles of our publications, our publications processes, what we expect of authors and what authors can expect of editors and journals/publishers. To some extent, publishing organs police the field, deciding what goes in and what is excluded, what counts as quality, etc. When preparing a manuscript, authors are well advised to consider how editors and reviewers may read the manuscript. The workshop seeks to make this valuation process more transparent. On the other hand, criteria for what is good quality are not all too fixed in STS outlets. Although authors and editors focus in detail on one specific manuscript, these discussions are also always opportunities for renegotiating the field, its valuations, and boundaries. How can we as authors, reviewers and editors create the best possible publications together? This is not only a question of writing, but also of various ways a many-way exchange unfolds. We are committed to caring publication processes and to work with authors in ways that do not reproduce problematic structures, styles, etc. within academia. Thus, the second aim of the workshop is to discuss what it might mean to refigure the author, the editor, the publisher, and the reviewer during the publishing process.

Panel #35 (Un)durchlässigkeiten kompostieren in/durch/mit Ruiniertem des Anthropozäns

Stream: Ecological Leakage & the More-Than-Human
ABS/213

Rheinische Sektion der Kompostistischen Internationale
Fedora Hartmann (Heinrich Heine University Düsseldorf, Germany),
Ian Boes (Independent Scholar, remote) & Lisa Krall (University of Cologne, Germany)

Mit Donna Haraway (2016) sind wir dem Denken im und als Kompost verbunden, einem Verantworten (auf) anthropozentrische(n) Ontologien, Epistemologien und Politiken: „We are compost, not posthuman!“ (Haraway 2016, 55). Als Figuration und Verknotungspunkt (vgl. Haraway 2008) bereitet Kompost nicht auf eine Welt für ‚den Menschen‘ nach ‚dem Menschen‘ vor, sondern verlangt vielmehr nach neuen Praktiken der Wissensproduktion, des ‚worldings‘, des Lebens und Sterbens in und mit mehr-als-menschlichen Welt(en) – „an ever arising and perishing world, a world of the many, of many worlds, a pluriverse“ (vgl. Escobar 2018). Was passiert, wenn wir Praktiken des Kompost(ierens) (um)schichten, durchlöchern, aufweichen, sedimentieren und durcheinanderhindurch-lesen (vgl. Barad 2007)? Wie lassen sich Un/Durchlässigkeiten mit einer queerfeministischen Kompostierung des Anthropozäns zusammenschneiden und welche neuen Geschichten produziert das? Wer wird dadurch (erstmals) sichtbar und wer fällt (erneut) durch die Raster? Diesen und weiteren Fragen wollen wir in einem inter- und intra-aktiven Workshop nachspüren, indem wir uns verschiedenen Berührungspunkten kompostistischer Kontamination in Ruinen des Anthropozäns zuwenden.

Panel #36 Unruly Everyday Metabolisms

Stream: Ecological Leakage & the More-than-Human
ABS/E11 Auditorium
Chair: Tereza Stöckelová

Everyday encounters disclose how metabolisms contaminate would-be discrete scales from the microbial to the planetary. While the study of urban metabolisms—how cities manage and are constituted by flows of energy and materials—aims to systematically analyze the socio-ecology of cities through economic and ecological quantifications, such macro-scale approaches can be ahistorical and perform an elision of the everyday (Gandy 2022). Through imaginative ethnographic sensibilities (Elliott and Culhane 2016), attention to more-than-human associations, and alter-economies of knowledge production, our panel attends to otherwise metabolisms that compose (and decompose) cities as lively formations. From urban ecology to animal geography and multispecies ethnography to feminist STS studies of infrastructure, we attune to everyday leakages of bounded forms like ordinary|exceptional, life|nonlife, organic|built, wild|urban, field|research. Our four papers variously stage more-than-human metabolic encounters across unruly sites and theoretical formations to explore Vancouver’s “Leaky condo” crisis, urban gleaning practices, the alchemical possibilities of plastic-eating fungi, and to witness crows as they feast on the leakages of urban detritus. Attending to permeabilities of scale across material rhythms and s/pacings, we hope to move from organizational ecologies and ossified methods toward a wider sense of interspecies and urban metabolism that accounts for multiple stories (Murphy 2006, 150).

Gleaning Urban Shorelines: Curbsides as Sites of Leaky Disposal

Lily Demet (University of British Columbia, Canada)
remote

Dissecting urban blocks, streets and alleyways extend like asphalt riverbeds along whose shorelines the excess of human consumption accumulates for disposal. Residential curbsides are porous boundaries of property where belonging is in flux. Not only municipal waste compiles at the curbside, however. Operating on denizens’ tacit knowledge of the city, matters deemed no longer fit for purpose are offered up as open-ended invitations for recirculation by passersby. Half my wardrobe I gleaned from cardboard boxes left on the sidewalk or by back-alley dumpsters, pinned to trees or tossed over fences. Found items incorporated into my everyday include a coffee mug, desk, stapler, and Epson WorkForce Pro 4740 printer found with a note reading: I STILL WORK. How do gleaning practices disrupt hegemonic economies of academic knowledge production in ways that

matter? This paper explores curbsides as sites of leaky disposal where use and value are differentially conceptualized through serendipitous, curiosity-driven Baradian 'intra-actions.' It proposes an accounting for the entangled nature of field and research, one responsive to the ways in which boundaries are dynamically figured, the field relegated to a site separate from the research findings through the iterative writing out of desks, staplers, printers, etc. with which scholarship materializes.

Plastic Promises and Fungal Alchemy: Commodifying Fungal Metabolisms for Mycoremediation

Morgaine Lee (Simon Fraser University, Canada)
remote

Microscopic particles of plastic seep into umbilical cords, rivers, and vegetal bodies in the wake of everyday excess. Mycoremediation meets the problem of plastic waste with a solution; the seemingly ceaseless presence of plastic in its afterlife can be broken down into its most intimate compounds by a hungry fungus. Fungi are alchemists. As the planet's great decomposers, they transform themselves and the world around them just as plastics have carved their mark into Earth's history. There is a pull of hope in the promise of fungi as both a means to clean up the mess of plastic waste, and as a material to replace it. Plastic too, was once regarded with a gleam of optimism as it was promised to solve human problems including the overconsumption of elephant ivory. Today we (re)turn to the more-than-human in the form of fungi to solve concerns such as plastic pollution and oil spills. This paper teases out parallel affects in our historical and present relations with plastics and fungi to re-imagine resourcist relational sensibilities and extractivist attitudes that ooze from the temptation to engage and profit from mycoremediation.

Corvid Rhythms and Metabolic Co-compositions: Vancouver's Anima Urbis

Ceall Quinn (University of British Columbia, Canada)
remote

At twilight, flows of crows fly to and fro between the Still Creek Rookery and surrounding Greater Vancouver Area (GVA). A murder numbering thousands, these Northwestern crows (Corvus caurinus) are a common reference for human inhabitants who observe their daily migrations as an expression of local ecology. Crows and other autonomous forms of urban nature represent a leakage in modernist imaginaries of cities as curatorial sites of organized landscape (Gandy 2022). As urban scavengers and specialists on humans (Marzluff 2005), crows thrive in the extensively disturbed habitats of cities. Adapting to novel infrastructures—evinced by cacophonous presence along railroad tracks, telephone lines, and the recently built Costco—corvid territories sprawl. Further, crows perform metabolic labors of sanitation and nutrient cycling as they feast on the leakages of urban detritus and decaying matter. This paper will explore how corvid rhythms co-compose the GVA's anima urbis—"the breath, life, soul and spirit of the city" (Wolch, 2002, 722)—in which animals' mobilities, agency, and metabolic labors are performative of multispecies urban space. How might considering crows' flight ways in the colonial city invite an expanded view of multispecies cohabitation and the rhythmic co-composition of urban worlds?

Panel #37 Sociotechnics of Machine Learning and AI Ethics

Stream: Tech & Data
ABS/E04
Chair: Peter Schulz

Ontological and Epistemological Shortages in Ethics of Autonomous Vehicles

Liliia Zemnukhova (Deutsches Museum, Germany)
remote

The umbrella term “ethics of AI” captures a variety of social, legal and moral problems. Scholarly literature discusses a wide range of ethical issues in the context of AI (Floridi, Sanders 2004; Verbeek 2011; Koniakou 2023), for example: privacy vs security (Solove 2011), data exploitation (Golebiewski, boyd 2019), digital surveillance (Zuboff 2019), discriminatory algorithms (Belenguer 2022), biases (Broussard 2023), or morality of autonomous vehicles (Beck 2016). The latter – autonomous vehicles (AV) – as a technology is a complicated sociotechnical network that refers to everyday experience and represents the views of the future (presumably less problematic in the far future). The dominant discourse on AV ethics, both regulatory and public, is still taking shape within the larger domain of AI. AI ethics and AV ethics exist concurrently and are co-produced by various actors. So the communication of AV ethics between these various actors becomes less transparent, smooth or reachable for minor actors. Production of technology in line with ethical principles is already a challenge, but including non-technical expertise as co-producers makes the process appear impossibly difficult. Although independent documents, standards and discussions around AV ethics have not yet formed a fully separate field, it is already possible to identify certain shortages. The talk aims at auditing AV ethics through the (techno)feminist lens: epistemologies expressed in papers and discussions, ontologies manifested in actions and decisions, and engineering and non-engineering practices. Reinforced by feminist criticism allows epistemological reconsidering the background of the production of knowledge on AV ethics and to revise the key policies with the help of feminist ontological concepts.

Leaking Rules: On Machine Learning’s Sociotechnical Hypernormativity

Carsten Ochs (University of Kassel, Germany)
in person

Although algorithmic systems are still associated with „claims to objectivity” (Kitchin 2017: 9), they oftentimes have normative impact beyond developers’ intentions. While this is true for algorithmic systems in general, Machine Learning (ML) techniques aggravate the problematic. As ML systems’ (MLS) output hinges on statistical probabilities it is next to impossible to gain semantic understanding of their operations (Burrell 2016). The limits to MLS’ social negotiability are due to their leaking of social rules. I call MLS’ tendency to produce norms without consideration of general moral standards of society „hypernormativity“: the hypertrophic extension of normative rule(s) triggered by MLS’ tendency to grow exuberantly rules that only operate in terms of impact, not of intent. Hypernormativity raises the conceptual-empirical question of how to conceptualize MLS to account for said tendency, as well as for the social consequences this has; and the ethico-political question of how to negotiate measures that help to contain hypertrophic norm-leakage. My contribution starts with a conceptualization of MLS’ impact (as normative, executive, opaque, and volatile) that draws on Actor-Network-Theory and Berlin based social studies of technology insights. I will next specify the structural consequences MLS thus introduce as hyper-nomy (as opposed to AI autonomy, heteronomy, social anomy). Drawing on empirical research that investigates the social negotiation of hypernormativity, I will briefly discuss the role of ethics within the negotiation; and the ethical standard response (deontological, teleological, virtue ethics) to the problematic. Citing evidence of the non-effectiveness of ethics guidelines, I will conclude by proposing the introduction of public “valuation boards” that allow for the mutual perception, public articulation, and sharing of

experiences of those who are affected by MLS. Thus paving the way for response-ability (Haraway) a “community of the affected” (Dewey) emerges that is able to give voice to the ethical concerns of MLS’ ‘targets’ themselves.

Leakage, AI, and Lie Detection: Harnessing Problematic Yet Pervasive Indices of Cognitive Processes

Jo Ann Oravec (University of Wisconsin-Whitewater, USA)
remote

The notion of leakage plays a strong role in how many lie and deception detection strategies have been characterized and implemented, from the polygraph to advanced facial recognition devices. Some of the human expressions associated with lying behavior are often known as “leakage” or as “tells” (Levine 2018). Leakages or tells reportedly “slip out,” possibly with some level of intension, in the course of human interaction that involves expressions of cognitive activities. The autonomic aspects of lying behavior are often characterized by researchers as having a “life of their own” or “second self” quality (Crane 2019). Related technologies that interpret individuals’ mood and emotional states are currently proliferating, many of which distill and analyze such autonomic input. Problematic or biased processes in the interpretation of such input can provide distorted feedback, with a related lack of validity in the system involved. Effective lying with these lie detection apparatuses often entails some level of personal control over one’s biometrics, including stress levels and their autonomic outcomes. This presentation provides critiques of the models of mind that are involved in the consideration of leakage as well as explores the growing assortment of artificial intelligence (AI)-enhanced lie detection technologies that are purported to capture and analyze leakage. Voice stress analysis systems for lie detection have been used for decades in business as well as medical and crime contexts, with the leakage involved being certain patterns in individuals’ utterances that supposedly signal stress. An assortment of smartphone apps with voice-analysis capabilities that are labeled as lie detection tools are widely available, providing an often-unreliable way of detecting lies in everyday conversations. Artificial intelligence capabilities along with new varieties of sensors are expanding the kinds of leakages being collected (Oravec, 2022), with new kinds of false positives rooted in bias emerging. Unlike the early polygraph which was strapped to the subject, some systems collect leakages remotely (without direct contact with the subjects involved) which can be problematic in producing reliable and valid assessment.

Panel #38 Leaky Bodies and Trans Strategies

Stream: Bodies & Governance
ABS/E08
Chair: Benjamin Lipp

Against the Flow: When the Gift and Sexual Biopolitics Bleed Together

Killian O'Dwyer (Goldsmiths, University of London, UK)
remote

The language of ‘leaks’ and ‘flows’ suggest different registers of intentionality or violence when framed alongside the body. In ‘Sanguine Resistance,’ Lynn Turner echoes the words of Derrida and his objection to the innate cruelty within the desire to make blood flow, a decisive act that floods both philosophical and literary archives with a masculine libidinal economy of enforced law and order. The ‘skill’ in making blood flow is situated throughout history as an inalienable right for Homo Faber to sacrifice and spill in the name of progress and impermeable sovereignty. However, in

‘Sanguine Resistance,’ Turner counters this desire for bloodshed by drawing on Derrida’s aporia of the gift, to retheorises a future for blood that does not enter into a sacrificial economy of reciprocity or indebtedness, but instead manifests as a form of sanguine resistance to the phallocentric regimentation of sexual difference. Similar to Turner, this paper turns to Derrida’s bloody notion of ‘making flow’ to expose a different register of cruel intent behind the purposeful leaking of bodies: the biopolitical management of fluids in the era of sexual wellness. Far removed from the sanguine hue of blood, the capitalist focus on making genital fluids flow feeds into what Paul Preciado describes in Testo-Junkie as a disciplinary sexopolitical regime that controls bodies from within. By relying on the growing technologization of hormones, antidepressants, porn and lubricants, the reigning masculine libidinal economy has found a new method of sovereign control over bodies by inhabiting the inner networks of blood itself. Returning to Derrida’s aporia of the gift, this paper extends Turner’s project to the orifices of the body, to consider how the secreting or leaking of fluids might represent a form of genital resistance to the colonial-patriarchal fantasy of ‘natural gifts.’ In this paper I ask simply: ‘what is the future of sexual flow?’

When Man Becomes Leaky: Testosterone Tales

Marlene Hartmann (Leibniz University Hannover, Germany)
in person

Past feminist research on sex hormones has overwhelmingly attended to the hormonalization of femininity, the way femininity has been reshaped by hormonal practices. While there is still a strong association between hormones and femininity, there is a rising tendency to figure masculinity in hormonal terms: Feminists speak of ‘testosterone fueled’ politics to call out the unduly influence of men, while manospherians insult liberal men as ‘soy boys’ who have been rendered effeminate by the consumption of phytoestrogens in soy products. Older men, ‘fueled’ by the marketing strategies of the burgeoning global testosterone replacement market, urge their doctors to tend to their declining testosterone levels; meanwhile trans men face their own battles with the medical system to access testosterone. The association with hormones, first and foremost testosterone, has ambivalent effects for masculinity. While its hormonal figuration provides yet another somatic anchor point for masculinity, hormones are also slippery entities which trouble the modern ontology masculinity relies on. Hormones occupy an uneasy position as both natural and cultural entities. They turn seemingly solid bodies into leaky vessels, blurring the boundary between body and environment. Sex hormones are furthermore not stable entities loyal to gender binarism; testosterone may become estrogen through the biochemical process of aromatization. Haraway, feminist materialism, and a touch of Latour in my backpack, I venture out to follow the well-traveled path of testosterone as the ‘essence of masculinity’, all the while staying open to being lured sideways and off the beaten track to discover testosterone, the trickster, capable of dis-figuring man. How does the hormonalization of masculinity, the translation of masculinity in hormonal registers, transform masculinity? How does hormonal masculinity handle the leakages described above? But also, what other stories is testosterone telling us if we listen in closely?

Creeping and Leaking as Trans Creative Methodology

Sam Godfrey (University of Exeter and Bath Spa University, UK)
remote

My practice-based research uses sloppy craft techniques, abject materials, and digital environments to explore the idea of ‘creeping’ as a trans creative research methodology. The etymological connection between the word ‘smock’ and ‘creep’ forms the basis of my inquiry, and my research examines how folding, using craft and digital materials, can reimagine the relationship between trans visual art, and the wider textile and trans studies fields. In my practice, I create immersive digital underwater environments and textile installations that one must ‘creep into’ and find ways to move through. The work intentionally resists the categorisation of trans artists within visual arts studies

and offers the possibility to leak out across taxonomies that cannot contain us. The audience is invited to move and adapt their bodies to the space in order to interact with it, and this intentionally places focus on tactile interaction as well as exploring the idea of leaky and porous bodily boundaries. In this conference presentation of my work, I aim to examine the role of creeping and leaking as a trans-specific theoretical and artistic strategy, and its possible contribution to the fields of trans and craft studies. This will be highlighted through images and analysis of recent specific artworks and installations, including 'love letters to watery others' (2023) and 'kelp reflection room ii' (2023 – currently in progress).

Panel #39 Datafication of Health & Food

Stream: Tech & Data
ABS/105
Chair: Jamie B. Smith

Growing Despite Friction: How Agtech Startups Overcome Difficulties in Agricultural Data Collection

Cornelius Heimstädt (Humboldt-Universität Berlin, Germany)
in person

Agricultural technology startups, or agtech startups for short, are becoming increasingly important actors in the collection of agricultural data. However, as agricultural systems are alive, unruly, and leaky, this process does not always go smoothly. Situated at the intersection of science and technology studies (STS) and critical data studies, my article explores how agtech startups engage in the collection of agricultural data, and how they cope with the socio-material difficulties posed by the agricultural systems they seek to turn into data. In conceptual terms, my article analyzes the data frictions that a single agtech startup developing an AI-based mobile application for diagnosing plant diseases encountered in collecting the plant pathology data needed for this technological project. Along with this, the article analyzes the startup's efforts to overcome these frictions. Specifically, the article identifies five interrelated practices that the agtech startup developed over time in response to the data frictions that the agricultural systems in which it operated confronted it with. These findings add to a growing body of social science and humanities research on the datafication of agriculture, by contributing to a more nuanced understanding of the role of agtech startups in this process. More precisely, the article's findings challenge narratives that portray agtech startups as companies with an almost unlimited capacity to extract data from agricultural systems, and instead highlight the fragility of these companies' data collection efforts.

Leaking like a Sieve? Food Data, Big Tech, Big Agriculture and Big Finance

Joachim Allgaier (Fulda University of Applied Sciences, Germany)
in person

Digital transformation heavily impinges on the food and nutrition system. How food is grown, industrially produced, distributed and consumed today is strongly affected by digital technologies, e.g. in order to make these processes more efficient but also more profitable. The global food chain, starting with agriculture, moving over to the industrial production and distribution of food products and ending at the plate of the consumer at home or in a restaurant is affected by various digital technologies such as robots and drones, big data, blockchain(s), artificial intelligence, as well as by ecommerce and food delivery, apps and digital media. Food now not only exists in its material form

but also in the form of digital traces and data that some authors describe as the food code of a digital society (e.g Deininger & Haase, 2021). Food is being dematerialized, reducing its physical substance while increasing its market value and immaterial dimensions. Digitalization has automated and informatized food production and commercialization, with individuals becoming subjects of data-gathering mechanisms by big corporations. Financialization has given financial markets significant power in determining which foods are produced and how, leading to speculation and the transformation of agricultural resources into financial assets. This presentation is concerned with the production and management of food related data. It is based on a secondary analysis of food related data reports
and focuses on the leakage of food, nutritional and agricultural data from various production contexts to contexts of financial exploitation.

Leaking Translations: Verantwortungsentlastung und -übertragung am Beispiel von KI-basierten Gesundheits-Apps

Anja Trittelvitz (University of Applied Sciences Emden/Leer, Germany)
Heike Gerdes (University of Applied Sciences Emden/Leer, Germany)
Jannis Steinke (Technische Universität Braunschweig, Germany)
in person

Hintergrund:

Ein Ziel soziotechnischer Transformation im Gesundheitswesen ist es, Patient*innen informierte und objektive Entscheidungen zu ermöglichen. Aus der Perspektive der (feministischen) Wissenschafts- und Technikforschung (STS) untersucht unser Forschungsprojekt die Herstellung und Nutzung von KI-basierten Gesundheits-Apps zur (Selbst-)Diagnose, den sogenannten Symptom-Checkern. Wir erforschen empirisch, wie Gesundheitswissen, das als objektiv gilt, sowohl bei der Entwicklung als auch bei der Nutzung soziotechnisch erzeugt wird.

Methodik:

Auf Grundlage eines ethnographischen Ansatzes und theoretischen Konzepten der Akteur-Netzwerk-Theorie sowie des Neuen Materialismus (Barad) beobachten wir teilnehmend und befragen mittels teilstrukturierter Interviews sowohl Nutzer*innen (derzeit: n=17) als auch IT- und medizinische Expert*innen (n=7) auf Seiten der Entwicklung.

Forschungsfragen:

Auf der Entwicklungsseite gehen wir folgenden Fragen nach:

- Wie werden medizinische Klassifikationen in technische Infrastrukturen übersetzt und wie werden diese durch diesen Prozess verändert und konstruiert?

Auf der Nutzungsseite leiten uns folgenden Fragen:

- Wie wird körperliches Wissen in das Klassifikationssystem übersetzt?
- Inwiefern verhandeln die Nutzer*innen befähigende oder ausschließende Momente?

Ergebnisse:

Mechanistisch-biomedizinisch-statistisch-genderdiskriminierende Körpervorstellungen und neodarwinistisch-evolutive Fortschrittsannahmen seitens der Entwickler*innen führen zu verzerrten Klassifikationssystemen, die Übersetzungsprozesse von Nutzer*innenwissen bedeutend erschweren. Dies beeinträchtigt in erheblicher Weise den Wunsch nach Selbstbefähigung der Nutzer*innen. Diese

gewaltvollen Schnitte fassen wir als (ab)lenkende leakage. Nutzer*innen leisten hierfür sorgende Grenzarbeit (Star und Bowker), da zusätzliche Denk- und Abfragewege gelaufen werden müssen, weil eigene Körpererfahrungen innerhalb des Übersetzungsprozesses deflektiert (ablenken) werden oder heraustropfen (leak). Response-ability (Ver-antwort-ung) wird von der Entwicklungsseite auf die Nutzungsseite transferiert. Im Strom dieser Exklusion entgegenwirkenden und Partizipation ermöglichenden Klassifikationsarbeit wird Nutzer*innenwissen zur leaking materiality, die sich als Kippfigur zwischen Selbstbemächtigung und Disziplinierung verdichtet.

#40 Workshop: Memories of future histories—
Situating "best practices" and Speculating about "good relations"
–Pre-Registration Required–

This workshop organized by Melina Antonakaki, Baldeep Kaur and Anja Klein, deals with a sensitive topic. The organizers kindly ask interested participants to register via this link by noon on March 14th:https://docs.google.com/forms/d/e/1FAIpQLSeGzV4UinXpgjlvOMxa77osik9TH5UiMMOhFNL3eESJj3w0XA/viewform.

For any further questions, please contact anja.klein@hu-berlin.de

Chairs:
Melina Antonakaki (Technical University of Munich, Germany)
Baldeep Kaur Grewal (University of Potsdam, Germany)
Anja Klein (Humboldt University Berlin, Germany)
in person

This workshop seeks to platform an experience that is seldomly shared and/or explicitly named: being in good relations in academia. We want to invite stories and memories of solidarity, collegiality, safety, care and mutual protection that coexist alongside oppressions. In doing so, we want to ask the question: how shall we go about naming the things that offset, resist and refuse power injustices? Care often looks like individual gestures of reaching out, giving time, making space. It also manifests in multifarious ways, i.e., through whisper networks and daily, mundane negotiations with power. Just like the abuses of power, care can be net- worked, strategic, structural; its effects reverberating throughout the politics of academic knowledge production and circulation. A shared insight becomes our starting point: The good relations established with so much of our dedicated efforts and care amongst participants/collaborators in a certain shared space, cannot unproblematically transfer onto other spaces or collaborations. Even their renegotiation demands to stay put and tap into local resources. The notion of best practice, on the other hand, builds on imaginations of ubiquitous application, implies transferability onto and seam-less integration into sites elsewhere from where they were first put together. Even within visions of continuous 'update', lessons are to be abstracted from many testing sites and seldomly account for local circumstances. When examined together, the labors invested into good relations fulfilling their promise are inversely proportional to the corresponding labors in support of best practices; namely, locality often proves the weakness of best practices ("if they cannot travel, then what's best about them?"), whereas replicability proves the Achilles' heel of good relations. We approach this seeming dichotomy as a generative 'epistemic leaking space'.

Panel #41 The Black Triangle – Sensing Borderland Coal Ecologies

Stream: Energy Infrastructures and Contamination
ABS/214
Chair: Caroline Ektander & Carlina Rossée

The proposed roundtable approaches the geo-politically and ecologically contested border region between Poland, the Czech Republic and Germany, also known since the 1980s as The Black Triangle—a once sensationally polluted industrial complex during soviet rule. The landscape and its histories are deeply fragmented by extraction and shifting nation state borders—the contamination of air, soils, and water bodies, acute ground water scarcity and soil subsidence, but also diverse ecologies and collective cultural histories. Inspired by alchemy and recent thoughts on the elemental, we come together to pickle local ingredients and revisit family recipes, and share stories on the different mutations of living and non-living matter and how they come to be in soils and in air. Through sensorial approaches—smells, sounds, touch and tastes— we will think through and with landscapes of extraction and transboundary contamination.

The Black Triangle

Carlina Rossée (Bauhaus-Universität Weimar, Germany)
in person

This impulse situates us in the Black Triangle, namely the border region between Poland, the Czech Republic and Germany, a landscape where the mining and processing of rich deposits of metal ores and lignite goes back into the 11th century. Volcanic, metamorphic, and erosive processes as well as anthropogenic transformations have shaped a region geologically characterized by its mountain ranges. The mutations of living and non-living matter, from plants into peat, and peat becoming the moist, soft rock that is lignite, can be traced through the sediments, just like the centuries-long history of transboundary contamination. As the most air polluting and energy inefficient combustion fuel, lignite – or brown coal – embodies an undead fossil regime that culminates in the current situation of energy transitions. Affecting in particular this border region's human and non-human inhabitants, the continuation of mining at the open pit of Turow, at the Polish tip of the triangle, has led to a highly contested situation leaking the borders of the three post-socialist nations, with decisions by the new Polish government being highly anticipated.

Toxic Commons

Caroline Ektander (Bauhaus-Universität Weimar, Germany)
in person

Violent processes of extracting fossil fuels, and particularly brown coal, have toxic effects that distribute unevenly in time and space and across species boundaries in our profoundly interconnected planetary commons. Smoggy skies and scarred earth transform over time into polluted and mismanaged groundwater and altered vegetation. The appearances of harm continuously shapeshift. Characteristic of toxicity is its propensity to disappear, out of sight, and out of mind, as it challenges dominant western, cartesian assumptions about the world. At the same time, the indistinct contamination of life continually interconnects us in new and unknown ways, slowly forcing linkages between supposedly separate and stagnant beings into an ever expanding toxic commons. Grounded in an understanding that humans are poorly equipped to register and to think meaningfully about the timescales of slow violence embedded within histories of capitalist extraction, this impulse proposes toxic commons as an analytical framework for sensing and making sense of toxicities permeating the Black Triangle border region surrounding the Turów Mine.

Sky Inside the Soils

Alexandra Toland (Bauhaus-Universität Weimar, Germany)
in person

Who, or what, mediates pollution? In this multi-phase, co-authored research-creation project we explore the social, economic, and aesthetic dimensions of plants as mediators between soil and sky in atemporal, more- than-human communities living in environments of extreme toxicity. Following the seasonal cycle of ruderal plants growing around the Silbersee, we imagine through them the historical trajectories of leakage on a former mining pit once used for wastewater from the Agfa film factory and chemical textiles plant in Bitterfeld- Wolfen. Using pigments derived from plant biomass, we reflect on our collective experiences through the medium of silkscreen – a tightly woven nylon mesh, not unlike the synthetic nylon products once manufactured in Wolfen. We track changes in viscosity over time and invite an embodied understanding of such changes though the repetitive process of silk screen printing: pour, place, pull, lift, sweep, repeat. Sending plant residues through a screen, we work-with and think-with leakage as a creative process that bodes productive as well as destructive forces. Using archival images of Bitterfeld-Wolfen over several decades, we re(con)textualize ideas about remediation and regeneration—what it privileges and how—and what that says more generally about the toxic legacies of industrial cities.

Jasmina Al-Qaisi (Independent artist)

The leakages into biomes and biographies break together during a vocalized fermentation process finally feeding into the gut microbiome. Jasmina composes repetitive patterns of speech and sound from anecdotes of chemical disasters, cooking approximations, phantomatic symptoms of autoimmune diseases and salt tales.

Panel #42 Art, Poetics, and Technoscience of Microscopic Vision

Stream: Methods & Epistemologies
ABS/213
Chair: Patrick Bieler

Recording the Atoms: On the Porousness of Mattered Bodies in Late 19th Century American Fiction

Franziska Wolf (Heinrich Heine University Düsseldorf, Germany)
in person

Gin a body meet a body Flyin' through the air, Gin a body hit a body, Will it fly? and where? […] Scottish physicist James Clerk Maxwell was not merely concerned with physics and the sciences, he also wrote poetry. In one of his most famous poems, "Rigid Body (Sings)", the speaker wonders what happens to two bodies colliding in the air. Will they continue flying? Will they deviate from their path, change directions? How does matter behave in matter and in relation to it? Maxwell was one of the few who expressed his curiosity and understanding of the characteristics and structure of matter, its different states, both in scientific and artistic terms, but he was certainly not the only poet intrigued by the plethora of new insights and discoveries about the structure of everything – matter, animate and inanimate; life, nonhuman and human – that spawned in the 19th century. The atomistic hypothesis found its way from more specialized debates and expert communities into wider public discourse and the arts, stimulating new ways of conceptualizing the world and the universe on the grand scale – as well as the human and the mind on the small scale. There are, for instance, literary references to the human's "shell" (James 225) and "nucleus" (Wharton 39), its flowing and crystallizing character. Naturally, this would also engender reflections about the

permeability of (human) matter, questions of human overflowing, of societal, urban, or organic containment or contagion. This paper seeks to explore more comprehensively the extent to which atomistic and material notions of (human) life were probed and negotiated by late 19th and early 20th century literature as well as the forms and functions these reflections and discussions could take on and fulfill. What are the prevailing materialities and states of matter that characters move in and to? How do they imply chances or threats of leakage? Are there gender-related differences in the representations, connotations, and meanings of respective states and degrees of imperviousness? Addressing questions like these, this paper will try to disentangle the mesh of (porous and leaking) materiality, psychology, and affect that is at the basis of such considerations.

Visualizing Leakage: Reconstructing the Super Sight

Dominique Ellis (University of Nebraska–Lincoln, USA)
remote

Light leaks in a camera and water leakage are the visuals that open up new modalities of seeing the slow accumulation manifest in lens-based media over time. My current project, Reconstructing the Super Sight, examines Abbott's camera developed during WWII. My motivation for studying and reconstructing this camera and technique is my interest in making visible what we do not see on the surface, where microscopic and invisible elements are magnified and illuminated. As a second-year MFA graduate student in photography, my interest lies in the properties of light leaks within a camera and water leakage from pipes to investigate the camera's potential to illuminate scientific research. Photographer Berenice Abbott's (1898-1991) science manifesto in 1939 ushered a pivot in her career, both aesthetically and conceptually. After her decade-long documentation of a modernizing New York City, she turned towards an investigation of scientific properties. In particular, one image of penicillin mold magnified from a sample of one inch in diameter to a photograph of 16 x 20 inches is one way of understanding leakage. From 1942-1962, she focused on the intersections of art and science, making highly detailed images with her invention, the Super Sight, often described as a camera obscura in reverse. Historically, the camera obscura is a technique where the outside environment enters an interior space, and the image is inverted. A small opening in a wall or window allows light to come into a dark room. By reconstructing the Super Sight projection camera in the 21st century, I will place this unusual process in the broader context of scientific and artistic innovations in photography. This paper will redefine Abbott's direct photographic projection process by studying light projection, the double pinhole, and magnification, which were integral parts of how she was illuminating the leakage of her time.

Leaky Things

Pauline Hafsia M'barek (Independent Artist and Scholar, Germany / Belgium)
in person

Leaky Things is an ongoing artistic research project that began in 2019 and includes various formats that mutually produce each other, such as an exhibition (2021), a lecture performance (2023) and a publication project (2024). Drawing on the history of microscopy, ancient natural philosophy, and feminist materialism, the project is a material and phenomenological approach to the body and its porous nature. Starting from the smallest human openings, the pores of the skin, I investigate the way in which we are in constant contact and exchange with the environment through the skin as a membrane, and are thus penetrated, impregnated and touched. Bodies are understood as fleshy, oozing and overflowing, their boundaries prove to be permeable and vulnerable. Through various self-experiments I try to perceive how things and ourselves become mutually subject and object of experience. The focus is on processes that aim for proximity, processuality and exposure. By assembling sensitive surfaces, microscopical phenomena and captured processes, the research deliberately chooses experimental forms, transitory stages of an artistic research in flux. Just as bodies are conceived as sites of exchange and traversal, so too the works and different formats

cannot be understood as clearly delineated and self-contained entities, but as an open arrangement of communicating forms – as leaky things.
As a proposal for the conference Leakage, I would like to propose a performative lecture about the porosity of an artistic research process and its multiple entanglements between the research object, its technical dispositve and the researcher's body. With the help of a usb-microscope, I embark on a journey into the microscopic shallows and contaminations between computer and body, screen and eye, digital and touch. A constant oscillation between observation and experience, distance and proximity, where categorical boundaries blur, dimensions reconfigure and questions multiply.

Panel #43 Planetary Data

Stream: Tech & Data
ABS/E11 Auditorium
Chair: Leman Çelik, Stefan Laser & Estrid Sørensen

With the panel, we hope to produce "Leakage" by establishing an ongoing discussion about the planetary dimension of data, asking researchers: "How can we live well with data on a damaged planet?" The panel will take a conventional paper presentation format with methods added to facilitate the linking and combination of the papers' arguments and conceptual takes on planetary data.
Our contributors inquire about the planetary ontology of data. Both local data practices and the global distribution of data across large, hard-to-fathom infrastructures are well-explored areas of Science and Technology Studies. Data circulates through global networks, and heterogeneous expertise goes into developing and processing digital data. Data practices and analytics are closely interwoven with global value chains of the IT industries. We will follow those environmental trails of data through facial analytics, computer programming languages, knowledge infrastructures, epidemiological studies, and data centers.
Data are global, social, (geo-)political, and fundamentally living off the planet as they draw on metals, fossils, plants, and the like. The panel explores how we, as STS researchers, attune to the planetary relations of and in the sites where data practices and infrastructures are shaped, maintained, used. Thinking of digital data through the notion of the planet, we focus on "critical zones" where the thriving of humans and companion species on the planet have become fragile. Following diffuse – sometimes obvious, oftentimes invisible – traces of data practices and digital infrastructures, their multiple environmental entanglements with polluted air, water shortages, climate violence, data injustice, land grabbing, habitat damage etc. emerge. The contributions to this panel address very specific instantiations of exploitation. Neither of them merging comfortably into one coherent narrative or experience, the panel explores the heuristic and critical potentials of the notion of planetary data and their shaping and emergence out of multiple planetary worlds.

Leaking Facial Data

Assia Wirth (Université Paris-Saclay, France)
remote

Machine learning-based facial analysis technologies require training datasets to operate. Yet how do facial datasets come to being? Data needs to be produced, collected, classified, and often these processes rely on human-intensive labor outsourced through colonial patterned supply chains. This data work is usually undertaken by workers operating in precarious settings, in which leaks seem to form. These are crucial in exposing the systems of production from which they emerge, opening up the 'black box' of these technologies. When it comes to face analysis technologies, these leaks become particularly crucial for critiques which might be faced with structural difficulty in accessing

relevant sites of production due to the sensitivity of the data and the secrecy of the industries partaking in this technological development.
The presentation explores the analytical possibilities of identifying the potential leakage which may occur at the annotation stage of production of facial analysis technologies. While signaling the most vulnerable parts of these infrastructures of production, these leaks create a 'way in' for critical perspectives and outsiders thus fostering an unplanned transparency – thus becoming sites of knowledge production. Identifying a leak implies uncovering invisibilized areas of the system of production, thus exposing the system's failures.

Climate Impact of Programming Languages

Pierre Depaz (Université Sorbonne Nouvelle, France)
in person

Programming languages are an essential component of digital technologies. As a mostly invisible interface between social users and industrial, material assemblages, they act as hybrid systems (Latour, 1993), while they nonetheless shape experiences and affordances of everyday life (Kitchin & Dodge, 2011). According to the french Ademe report, 21% of the carbon footprint of the IT sector is linked to uses of the digital (as opposed to 78% due to the production process) (Perasso et. al., 2022); since programming languages define the application processes that users interact with, they are an essential component of such footprint. This contribution proposes an overview of the ecological implications of programming languages, and particularly how the dual concepts of data modeling and levels of abstraction point at both infrastructural requirements and user practices. To do so, we examine three different approaches to thinking of programming languages as ecological actors, examples of how local individuals interface with the rest of the (digital) world: Google's use of SQL for database work, Ink&Switch's use of JavaScript for real-time collaboration and Scuttlebutt's use of Rust for their social media. We therefore propose a comparison of different ways of programming, from local to global.

Planetary Knowledges: How Data Infrastructures Link Science with the Planet

Sandra Abels, Leman Çelik, Stefan Laser, Estrid Sørensen & Lynn Werner (Ruhr University Bochum, Germany)
in person

Knowledge is epistemic, social, and material, yet it is also planetary: tearing up the ground, diverting flows of fuel, water and gasses, affecting living conditions. Data infrastructures use vast amounts of energy, just as they are made of material components extracted from the planet. They form key links between knowledge production on the one hand and the planet on the other. We coin the notion of "planetary knowledges" to conceptualize and discuss the interrelations between planetary matter and knowledge, such as between organic and energetic flows, bodies, and landscapes on the one hand and representations and inscriptions, data points, data processing and visualizations on the other. The presentation takes stock of current STS literature and how it engages with (1) extractive and other planetary relations including their local and geopolitical effects, (2) scientific data infrastructures, and the care for planetary matter in operation and maintenance, and with (3) scientific data practices and their dependencies on data infrastructures. We hope to ignite a discussion of data and data infrastructures as intermediaries between knowledge production and the planet.

Panel #44 Water as Leaky Matter

Stream: Ecological Leakage and the More-than-human
ABS/114
Chair: Sören Becker

Leaky Pipes and Watery Bodies

Ina Dietzsch (Philipps-Universität Marburg, Germany)
in person

Leakage ist ein Buzzword bei der Einführung von (smarten) Wasserzählern weltweit und Teil eines Versprechens, mit dem digitale Wasserzähler zu eine nachhaltigeren Ressourcenverbrauch beitragen. Wasserinfrastrukturen weisen sowohl Trink- als auch Abwasser klare Orte in einem Transportnetz modernen Wassers zu, dessen Ziel es ist, fließendes Wasser unter Kontrolle zu bekommen und es zu zwingen, sich mit möglichst wenig Verlust von A nach B zu bewegen. Leakage ist, was es zu bekämpfen gilt. Es verursacht Verluste und Kosten. In diesem Verständnis wird es auch zu einem Kampfbegriff, mit dem Menschen in Abhängigkeiten von (daten)kolonialen Ordnungsstrukturen mit dem Argument gezwungen werden, dass sie zur Bearbeitung von Klimawandel beitragen. In meinem Vortrag werde ich auf der Grundlage von Arbeiten wie von Nikhil Anand oder Charlotte Lemanski und anderen aufzeigen, welche sehr unterschiedlichen Leakage- Assemblagen sich in unterschiedlichen geographischen und historisch sedimentierten Kontexten bilden, wie sie von Fragen von Nationalisierung, Modernisierung, Citizenship und (Neo-)Kolonialismus durchdrungen werden und dies vor dem Hintergrund hydrofeministischer Überlegungen von Astrida Neimanis diskutierten.

„Wenn wir zum Realwasser gehen, das macht dann auf einmal keinen Spaß mehr“ Wie Wasser gegen seine eigene Materialität abgedichtet wird.

Peter Schulz (Friedrich Schiller University Jena, Germany)
in person

Water, sealing, leakage – a constellation of terms that inevitably conjures up images of leaking pipes from which water spills, or from which oil or other chemicals seep into the water. In contrast, this paper aims to approach the relationship between water and its sealing from a different perspective: water as fluid materiality, in which many substances are easily soluble. The concept of solubility in water already alludes to a specific notion of water, 'modern water' (Linton 2010), dominant in modern Western societies, which understands water on the one hand as a quantifiable chemical substance, H2O, and on the other hand as an available or scarce resource (Linton 2010, Linton and Budds 2014). Understanding water as 'H2O' seals water against both its socio-material hybridity (Latour 1995) and the diversity of the empirical reality of the materiality of bodies of water. However, this purely nomological sealing alone cannot successfully prevent the diffusion of diverse materiality into water. Modern water requires constant sealing practices. Therefore, this paper examines two of these sealing practices based on initial findings from ongoing empirical surveys: First, the fabrication of 'model water' (as opposed to 'real water') for chemical investigations in the laboratory will be scrutinized. This can paradigmatically demonstrate how scientific reference to water necessitates ongoing material sealing practices to maintain the modern, sealed concept of water in contact with its empirical reality. Subsequently, based on the observation of self-presentation in a sewage treatment plant, where treated water is described as 'optically pure', the practice of norming the boundaries of water sealing against 'contaminants' outside the laboratory situation will be accessed. In particular, the current European debate on the establishment of a fourth purification stage for trace substances points to ongoing societal negotiation processes and conflicts of interest regarding what constitutes 'modern water'.

Containing Leakages: Post-Disruptive Organizing in the Ruhr

Armin Beverungen (Leuphana University Lüneburg, Germany)
in person

The Ruhr area was for more than a century dominated by the coal and steel industries. In its aftermath, the disruptive effects of these industries came to the fore in the ecological devastation it wreaked. As the last coal mines were winding down, the RAG Foundation was set up in the early 2000s to deal with what are called the 'perpetual obligations' stemming from coal mining. These include a set of technical and logistical matters: pit water management, polder measures and groundwater purification. Without these, much of the region would be underwater as a consequence of the subsidence caused by extensive and deep coal mining, and groundwater would be contaminated from the pollution wrought by pit water and coking plants. Inspired by Laser and Sørensen's exploration of the Emscher river, the paper explores the perpetual obligations as a case of post-disruptive organizing. Drawing on Hecht's (2018) reflections on interscalar vehicles, it focuses on the deep times of the Ruhr valley, which are marked by attempts to contain leakages of rivers spilling over, of industrial waste washing into groundwater, and of pit water contaminating groundwater. The paper is based on some preliminary research in particular on the Leitwarte Pluto, where the water management systems coalesce around a dashboard fed with data from thousands of sensors left underground in now closed coal mines. What emerges is a view of the Ruhr area as a region profoundly shaped by leakages and their containments, of a complex socio-technical setup that contains the fresh water which finds its way from the source of the Ruhr in the east to the Rhine in the west, and the rainwater that descends from the hilly south into the pits.

Panel #45 Digital Statecraft

Stream: Bodies & Governance
ABS/E08
Chair: Julia Valeska Schröder (Humboldt University Berlin, Germany)

The state celebrates a grandiose comeback in digital imaginaries and practices of digitalization. Proliferating enterprises of the "digital state", "smart city" and the "digitalization of administration" revolve increasingly around the notions of "digital sovereignty" and "public innovation". Moreover, governmental tools and mechanisms reclaim front stage, being discussed as "Smart Governance" and "GovTech". In Germany openly proclaimed as "Neustaat" (CDU/CSU), this renewed intersection of "the state" and "digitalization" demands a critical examination.
In this panel, we explore in what ways the proclaimed "digital transformation" is enacted as digital statecraft: The perpetuation and reinforcement of state sovereignty over digital infrastructures, the reinvention of sovereign institutions through digital infrastructures as well as the infrastructural reinterpretation of the ideas of 'the state" and "sovereignty'. Practices of digital statecraft mobilize digital technologies and infrastructures to medially translate the existing political-institutional system and complement, expand or replace it technologically. Moreover, they negotiate the idea of the state and sovereign politics itself: "the state" is reframed as an infrastructural power, as infrastructural practice or infrastructural experiment. In this statist or governmental turn of digitalization, the state is reconfigured as subject and object of transformation. This panel centers ethnographic research, focusing on practices of digitalization, its tools and modes of operation, shedding light on how "the state" is (re-)arranged, (re-)imagined and re-produced at various levels of the political. It brings research into conversation that analyzes digital technopolitics beyond the dazzling virtualization, mechanization, and modernization master narratives. By challenging the dichotomy between "human-centered realpolitik" and "hard-headed thingpolitik" (Latour) theoretically as well as empirically, the transgression of boundaries and the steady subversion material conditions ('leakage') becomes apparent.

Digitalpolitical Intrastratecraft

Julia Valeska Schröder (Humboldt University Berlin, Germany)
in person

Following the current formation of “digital transformation”, one will come across new professionals: "public innovators," "public service designers" and “creative bureaucrats.” According to this approach, “deep change" does not necessarily come from “disruptive technologies,” but is also achieved through digital political transformation practices. Especially around the arena of "Public Innovation Labs," a version of digital politics is enacted that is to a large extent concerned with forms of political organization, cultural conditions, and institutional structures of the state. Here, the state administration is focused on as an object and instrument of digitalization. Public Administration is sought to be transformed and to be reshaped as an "enabler" of societal and institutional transformation. “Public Innovation Labs” do this with their own means; inventing and adopting multiple more-than-technological instruments and tactics through which the relationship between the state and digitalization is negotiated and where the state’s meaning and its modes of the political are attempted to be practically recalibrated. To capture this phenomenon, I propose the concept of ‘intrastatecraft’ as analytical lens to foreground first, the administrational politics in statecraft and second, the usually neglected component of the word ‘craft’ to attend to the use of instruments, materials and tactics of relating to the nonhuman.

Data-based Governance and Partial Data Connections. Tracing Data Politics of and for a „Verkehrswende“

Martina Klausner (Goethe University Frankfurt, Germany)
in person

The digitization of public administration and governance infrastructures is often framed as data-driven efforts to enhance more efficient and transparent ways of governing. While public administration could be considered a fertile ground for datafication due to its long history of data collection with regards to its territories and populations, in practice, datafying public administration turn out to be rather complex and at least not straightforward. In my contribution, I trace data practices within public administration and ask how and which data are generated/captured and how these data circulate or not. Doing so, my talk attempts to bring together STS approaches to the study of the state and politics with approaches in Critical Data Studies. The talk will draw from a case study on implementing Verkehrswendepolitik (politics for a mobility turn) in Frankfurt am Main and discuss what we can learn about politics by tracing and tracking such data, but also by the absences of certain data and hindrances to their journeys. This allows, I argue, to attend to data politics as both politics done with data, but also the inherent “politics of data.”

On Passages and Events in Smart Urbanism

Julio Paulos (Eidgenössische Technische Hochschule Zürich, Switzerland)
in person

The imperative for cities to be “smart” too often coincides with the desire to systematize urban environments. This ambition to regulate through intelligence is not new, as urban planners seek to regulate, predict and control urban space, development and infrastructures. Yet, what does this imperative to systematize urban environments through new technologies do to traditional urban knowledge? In my paper, I will focus on how the ambitions to develop smart city guidelines are constitutive for creating new passages in the knowledge loop to make urban expertise adaptable for present and future cities. Following an empiricist tradition and taking the city of Vienna as a case study, I look at two meetings that took place in the framework of a European smart city project. I argue that the smart city strategy, when implemented, is not only a discourse or a mandate, but primarily a purification of two separate realms: the traditional mode of governance through

rationalization, combined with the need for innovation due to environmental issues and democratic demands. The result of this hybridisation of modernist expertise and growing expectations leads to conditions of possibility for new modes of knowing.

Infrastructural Imaginaries of 6G. European Sovereignty and the Development of Next Generation Telecommunication Networks

Alexander Harder (Humboldt-Universität Berlin, Germany)
in person

The EU's "Digital Decade" entails not only the expansion of fifth-generation mobile networks ("5G"), but also the development of sixth generation ("6G") networks. EU and German-Funded 6G Networks are expected to further increase bandwidth, lower latency, and make Networks "smarter". Crucially, they are also expected to secure "technological Sovereignty" (EC 2023) in the face of geopolitical and geo-economic conflicts, throwing the politics of infrastructure into sharp relief (Glasze et al 2022, Musiani 2022). Attending to the development and deployment of mobile communication networks reveals a complex arrangement of state and supra-state actors, scientific institutions, network operators, hardware manufacturers and standardization bodies. Which visions of what these networks do, whom they serve and what politics they enable prevail in this context? Based on an interests in the regimes of difference that underlie discourses and practices of sovereignty (Bonilla 2017, cf. Balibar 2004), my contribution draws on fieldwork at Research and Development meetings, interviews and document analyses to explore these questions. Investigating these "Infrastructural imaginaries" (Mattern 2017) of next-generation mobile networks might illuminate if and how statecraft is reconfigured when sovereignty becomes a question of digital infrastructure development.

Panel #46 Pollution and Im/Possibilities of Containment

Stream: Energy Infrastructures & Contamination
ABS/105
Chair: Shachi Mokashi

Forever fleeting – On the Im/possibilities of Knowing and Managing Eternal Environmental Burdens

Janine Hauer (Martin Luther University Halle-Wittenberg, Germany)
Ruzana Liburkina (Goethe University Frankfurt, Germany)
in person

The current moment entangles resource shifts with energy transitions in unprecedented ways. Elements such as hydrogen and lithium promise a greening of industries and technologies, whilst the extraction of other suspicious minerals and fossil fuels is phasing out (at least in Germany). Yet, recent shifts and changes do not prevent the accruement and mobilization of dangerous and toxic residues that need to be safely contained and, in many cases, infinitely monitored and managed. In our presentation we provide ethnographic insights into two research projects dealing with the im/possibilities of knowing and managing eternal burdens. Whilst Janine’s project focuses on the controversies related to the concession and operation of a range of landfill sites across Saxony-Anhalt, Ruzana looks at the continuous and recently renewed endeavor of managing a mining waste settling pond in the Saxon Ore Mountains.
In juxtaposing these two cases we focus on how matter and manner of the element(s) and substances in question are known by different actors; how (contested) facts about their nature and behavior are established, negotiated, and debunked over time; and how these shifting knowledges and different registers of perceptibility (Murphy 2006) affect the governance, utilization, and

materiality of toxic matter. Even though the matter itself is widely acknowledged as requiring eternal liability and care, its state and status are otherwise volatile, and so are the social worlds, ethics, and politics assembled around it. By tracing and emphasizing the significance of temporary configurations for dealing with eternal problems, we explore the potential of ethnographic research amid and on slow disaster.

Malleable Subjects in the Plastic Capitalist Anthropocene

Zhou Fang (University of Amsterdam, Netherlands)
remote

Scientists have proclaimed “the age of plastic” after that of bronze and iron (Thompson et al, 2009). Meanwhile, accumulation of plastics in the geological sphere, i.e. pollution that entered the fossil record, serves more than satisfactorily as indicators for traces of human activities and therefore has been considered as the “golden spike” of the Anthropocene (Davis, 2022). Moreover, the microplastics widely and consistently detected in the vessels under our own fleshes further validate our very bodily transformation as “a plastic man”(Yarsley and Couzens,1945). Such is far from the only form of leakage of plastics. Revisiting the Deleuzian account of capitalism as “the radical decoding and deterritorialization of the material flows that previous social machines had zealously coded on the earth or the body of the despot”(Deleuze & Guattari, 1988), one might argue that plastics-as “the substrata of advanced capitalism”-have surfaced as a new and crucial component in the assemblages that actualize the surplus value extraction from the deterritorialized flows of labor and capital in the Anthropocene in light of the exponential growth of plastic production globally. The (micro)plastic debris buried in landfills and traveling across water and air would be the material imprints of capitalism’s “organic” reproduction, reminding us of its paradox of growing barrenness (Tynan, 2016) while expanding the manners in which its form of growth shorn from life through its ceaseless accumulation and emission of menacing toxins. Even more radically, this artistic research attempts to propose a further spilling of the logic of plastic to the aesthetic and subjective terrain as it facilitates imaginaries and inhabitation to a certain Neo-liberalist principle of elasticity, resilience and replaceability. Last but not least, this research offers a critique on how the contemporary industrial and consumerist practices around plastic matters sustain the aesthetic categories (Ngai, 2010) for subject (de)formations.

Biofuel Leaks

Marie Widengård & Elin Hallmann (University of Gothenburg, Sweden)
in person

Biofuel leaks draw attention to the complex and controversial processes through which economies emerge. While proponents try to frame biofuels as beneficial for society and climate, critics have exposed biofuels’ many problems and ‘leaks’ over the years, from leaky arguments to leaky regulations and carbon leaks. Drawing on biofuel developments in Sweden, this article highlights how leaks can be used to challenge the ideas, interests, institutions and infrastructure of an economy, and in turn, how they can be ‘repaired’ to sustain an economy. Revitalizing Callon’s framing, overflow and leak argument, we highlight that thinking with leakage promotes a shift in analytical focus towards the power of leaks: who and what is benefitting from leaks, and how leaks may be handled to enable or disable change. Special attention is given to the practices that can be used to mend or attend to leaks, and the power of leaks as a counter-hegemonic tactic to reframe a controversial economy.

Panel #47 Visual Cultures of More-Than-Human Ecologies

Stream: Ecological Leakage and the More-than-human
ABS/105
Chair: Alexandra Toland

Towards the More-than-Human World in Film: It's Night in America and Il Buco

Botagoz Koilybayeva (Charles University, Czech Republic)
in person

In the cultural practice of film - the visual medium built on the aestheticization of human and non-human subjects, beings and phenomena par excellence - recently emerging epistemological and ontological drifts toward the realms of the more-than-human, the posthuman and ultimately the non-anthropocentric are gathering a certain momentum. These shifts have been substantiated by a growing body of research and evidence suggesting intelligence in nature, intersubjective experiences of non-human beings and the embeddedness of human bodies in the collective ecosystems.
In my presentation I would like to focus on the aesthetics of two recent films which are built around non-anthropocentric narratives, thus transcending the human, and more critically approach the portrayal of the more-than-human world. Moreover, the films expose a dialectical rupture between humanity and animality gesturing at the interconnectedness of human and non-human bodies and their equal embeddedness in the biotic environments. For example, Ana Vaz's documentary It's Night in America (2022) casts a critical eye on Brazil's colonialist past, hinged on the extraction of mineral riches, deforestation and taking away land from the Indigenous communities and animals, and focuses on the interconnectedness of human and non-human lives in the wake of the pandemic that made some animals reclaim their land back while humans were forced to hide in the comfort of their homes. In Il buco (2021), Michelangelo Frammartino renders caves a powerful physical and metaphysical presence. The film reenacts a real-life exploration of the Bifurto Abyss, a 700 meters deep cave examined by a group of young speleologists whose scientific endeavor disturbs the life of a solitary shepherd. The shepherd and the young researchers are interconnected by the complexity and the mystery of the cave which, for the old shepherd, is imbibed with subjectivity as the rest of the Calabrian valley, yet for the young speleologists becomes an object of research, a hole in the ground pertinent for study and analysis.

Hyperobjects Leaking Through: Unraveling Symbolic Ontologies in 'First Reformed' – An Ecocritical Analysis

Maria Kuzina (Universitat Politècnica de València, Spain)
remote

From an ecocritical perspective, to think about leakages is to think in terms of hyperobjects. This abstract presents a study that endeavors to assess the concept of hyperobjects as introduced by Morton (2013) and it attempts to unravel the multifaceted manner in which leaks assume an ontological symbolic role within the movie First Reformed, directed by Paul Shrider (2017). This psychological drama chronicles Ernst Toller's (Ethan Hawke) struggles as the Reverend of the First Reformed Church during the outset of its upcoming 250 anniversary and offers a reflection on how climate justice, politics and religion get closely concatenated. The present study thus hypothesizes that leaks, represented in the film as a physical phenomenon and material metaphor, also serve as an analytical tool to reveal and deconstruct obscure, viscous, mesh-like social, political, and interpersonal dialectical dynamics. In the era of the Anthropocene, which, according to Morton "binds together human history and geological time in a strange loop" (2016, p.7), leaks problematize the space, thus transforming it into a conceptual entity where the plate-like politics of coexistence begin to collide. Leaks disclose cracks in the reality strata, thus emphasizing the inherently political quality of those holes. Perceiving space as the one constrained to the cinematic conventions within

the movie and the medium itself, the study attempts to illustrate that the leakage-sensitive reading prompts insightful scrutiny of the socio-political contemporary situation.
Lastly, the present study views leaks as phenomena occupying a hyper-dimensional space. In order to trace the noumenon the leak alludes to, it appears imperative to uptake an interdisciplinary perspective. Hence, locating itself on the cusps of environmental studies, critical discourse studies and film studies, the current research thus contributes to the burgeoning field of leakage studies and underscores the urgency of transformative discourses and analyses that pave the way for a more sustainable and inclusive assessment of the intricate tapestry of life on Earth.

Living as Disease-able: Hand Hygiene Posters, Vital Circuits and Multispecies Entanglements

Sheryl N. Hamilton (Carleton University, Canada)
in person

Both before and after COVID-19, public health hand hygiene posters cajoled, commanded, and coached us in washing our hands. Yet despite their ongoing use in national and international public health campaigns to improve hygiene and thus reduce disease risk, these low-tech media of disease management are often overlooked by scholars examining the visual culture of health (see Cooter 2008; Cooter and Stein 2010, 2013). Drawing upon an evolving corpus of over 300 posters, primarily from Canada and the United States, from 2010 to the present, I suggest the hand hygiene posters are vital disease media (Hamilton 2021, 2019). Emplaced in sites of 'passing through' where we experience heightened somatic awareness, particularly of our hands, handwashing posters ask us to attend to, and manage, what I am calling our manual schema. A manual scheme is a specific dimension of the broader notion of body schema, or "the systems of possible movements, bearings and comportments which effectively invite the sensory and motor aspects of the body" (Paterson, 2007: 22). In seeking to make us aware, again, of behaviors and practices that have long been habitual for many of us, I am interested in the ways in which these visual media work to figure our hands as mediating a vital circuit of leaky orifices. The skins of our hands and those of others, animate and inanimate, are figured as environment; our movements as communicable; and our subjectivity as always already symbiotic.This figuration works in tension with the surface level immunitarian discourse of these media and their purveyors anchored in a "bio/necropolitical hierarchy" (Haris 2022; also Ironstone 2019; Donn 2021) to open up possibilities for a diseased mode of being grounded in an orientation of exposure, the material effects and affects of porosity, and enactment of multispecies entanglements.

Panel #48 Sociological Approaches to Data Leakage

Stream: Tech & Data
ABS/213
Chair: Britta Acksel

Hack - Patch - Fix? Living with Data Leakage

Laura Kocksch (Aalborg University, Denmark)
Dennis Eckhardt (Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany)
Nelli Feist (Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany)
in person

With the increased digitization of life at home, at work and in community spaces, more areas become susceptible to cyberattacks. Cybersecurity researchers typically respond with three types of patches: (1) they identify technical vulnerabilities and create updates, (2) they target humans as the "weakest link" and educate or train them, or (3) they identify organizational processes and strength-

en regulation (NIST/2, ISO27001). This logic of patching purports that data leakage can be located in independent areas (technologies, humans, organizational dynamics) and encountered with tai-lored solutions. In comparison to this logic of patching, STS has pioneered a perspective on leakage as slowly emerging, distributed and not conclusively solvable. In this talk, we draw together insights from three independent ethnographic and sociological fields: Smarthomes, small companies and community spaces and medium-sized companies. We portray how cybersecurity is negotiated through (tacit) tinkering, subverting, and re-inventing standards and "good" advice by families, companies, and communities. Cybersecurity, in this por-trayal, is not aimed at patching up leaking data, but at situated work-arounds, enduring ambiguity and anxiety through long-term engagement and collaboration. Different moralities are at play when encountering cybersecurity practices in the home, a family company or among colleagues. Instead of a logic of patching, our interlocutors notice and care for cybersecurity. However, enduring potential data leakage is not without risk: Data leakage can have a vast impact on families' privacy, businesses or critical infrastructures. While avoiding the logic of patching, what does 'good' cybersecurity look like? We propose to unfold the characteristics of (data) leakage in cyber-security. This shifts our attention to what it might mean to endure data leakage without being naïve about its consequences. With our talk, we aim to further open the ontological politics of cybersecurity and engage in a discussion of the role of STS in researching hacks, patches and fixes.

Leaky (mis)conceptions: An Ethnographic Exploration of Social Media, Illusions of Anonymity, and the Lived Experience of Infertility.

Poonam Kamath (University of Tübingen, Germany)
in person

This paper explores the concept of leakages in the form of fear of involuntary disclosure of one's infertility journey on the social media platform of Instagram. Based on my digital ethnographic research on Instagram on the topic of Kinderwunsch (the wish to have children of one's own) in Germany, I demonstrate how the fear of leakage of intimate information about one's private life regarding one's reproductive difficulties and not being able to conceive with or without medical intervention is perceived as an individual shortcoming and one that needs to be hidden from public view. Women with Kinderwunsch use Instagram mainly to gain information regarding fertility therapies, find other women undergoing similar reproductive difficulties and to not feel alone and stigmatized. Many women tend to be silent observers (lurkers) with non-public or anonymous profiles. They fear that people within their own social circles: at work, or friends or familial circles may stigmatize them, pity them, and perceive them as failures if their reproductive struggles become known. To prevent such inadvertent leakages, some women put in quite the effort to remain and maintain their anonymity online. This points to the illusion of anonymity social media users may harbor. It reveals the previous nature of social media and its usage as well as the juxtaposition of privacy and digital footprints and the implications of anxiety regarding the possibility of social media leakages. It also sheds light on the perception of stigma infertility still holds in Germany.

The Leakiness of Synthetic Data: From "Non-belongingness" to Remains

Louis Ravn (University of Copenhagen, Denmark)
in person

Synthetic data, denoting artificially produced data used for data science tasks, are becoming more widespread due to the growing availability of generative artificial intelligence (AI). As part of this trend, synthetic data are increasingly employed to train machine learning (ML) algorithms. Synthetic data are not only used to boost the technical performance of ML algorithms, they have also come represent a crucial transformation of ML ethics: since synthetic data are said to not refer to any objects, events, or persons in the real world, ethical risks that attend ML are believed to be erased. Jacobsen (2023) calls this the alleged "non-belongingness" (p. 8) of synthetic data: the sense that they do not belong to anybody and that therefore they de-risk machine learning. This paper

challenges this emergent synthetic data imaginary of “non-belongingness” by mobilizing the critical data and archival studies concept of remains. This notion, and attendant empirical research, highlights that datasets carry within them a plethora of human remains – to ethical and political consequences. By drawing on recent controversies around generative AI models whose data outputs reveal remains, this paper demonstrates that synthetic data, in spite of their alleged “non-belongingess”, often leak the remains of human subjects. By so doing, it provides a fresh perspective on the emergent technology of synthetic data, pushing back against the notion that they eradicate ML risks. Instead, it is argued that the ethics of human remains in datasets remain as relevant as ever in the age of synthetic datasets.

Panel #49 Leakage and Philosophy

Stream: Methods and Epistemologies
ABS/214
Chair: Alexander Schniedermann

Leaky Re(lation)alities: Containment, Leakage, Interiority and Exteriority

Stylianos Zavos (Tampere University, Finland, remote)
Olli Pyyhtinen (Tampere University, Finland)
in person

In this paper, we explore leakage and containment as ontological conditions. We begin by situating their presence, spanning from the life-giving vessel of the womb to the numerous technologies that render human and more-than-human worlds thinkable, knowable and liveable. Inspired by our ongoing work on waste, and drawing from science and technology studies, Sloterdijk's spherology, Simondon's philosophy of individuation, and Deleuzoguattarian geophilosophy, we suggest that leakage and containment need to be examined beyond dichotomies such as active/passive and stasis/flow. We do so in three successive analytical moves: first, we examine containment as a crucial spatial technology of ordering; containers hold together in place what would otherwise fluctuate in perpetuity. Second, and since the occasion of leakage disrupts – with the potential to upend – any established order, we argue that containment and leakage, closely linked to the ideas of interiority, exteriority, and metastability, need to be understood as occasions of reciprocal presupposition. Thereby, we wish to depart from the line of thinking where containment becomes merely a teleological principle for the control of leakage. Finally, and moving beyond prevailing negative conceptualisations and practices that render leakage a synonym of mismanagement and disorder, we outline an affirmative take that equally considers leakage and containment as affective, expressive, and intimate processes of sympoiesis whereby interiorities and exteriorities have a provisional character instead of a permanent, fixed, and oppositional one.

The Tragedy of Disruption

Freya Häberlein (Leuphana University Lüneburg, Germany)
in person

Disruption as thought of by Bernard Stiegler describes a cathartic state of an organological system of relations, preceded by a loss of individuation for the dynamic relationship between psychological, technical and social becoming. As such, disruption signals a nihilism co-emerging with and through algorithmic technology: Due to an extreme account of rationalization, there is a loss of reason. According to Stiegler, especially young people experience such disruption through a degradation of values; he refers to a generation that understands itself “to be the last, or one of the last”, ultimately giving up on the pursuit of values such as family, work or politics. The absence of such collective

protentions is in Stiegler's analysis accelerated by 'algorithmic exosomatization' (algorithms as mnemotechnics): the anticipation of affective motifs by digital technologies, annihilating the possibility of normativity. While Stiegler principally focuses on the fatal force of disruption in his writings, he also hints towards a pharmacological understanding of it, undermining just those values whose decreasing significance is often pointed out by him: "As a pharmakon [...] (algorithmic exosomatization) makes it possible to short-circuit these social organizations, setting up the tragedy of disruption [...]. But this would then amount only to the first stage of a doubly epokhal redoubling, which could and should lead to a second stage: not just to a new epoch, but to a new era." The following question should then be pursued by departing from Stiegler's last texts: How can algorithmic exosomatization be perceived otherwise than in terms of a techno-social nihilism? This question is developed by tracing an axiology of excess alongside that of disruption in Stiegler's work, and by catching up on Nietzsche's influence in the former's account of algorithmic technologies.

Panel #50 (Techno)Biopolitics in Healthcare and Welfare Institutions

Stream: Bodies & Governance
ABS/213
Chair: Timothy E. Stroud

Aggression Leaks: On Bodies and Situations in De-escalation Practices

Maja Sisnowski (University of Amsterdam, Netherlands)
in person

In this contribution, I examine aggression as leaky matter. To do so, I draw on ethnographic engagement with de-escalation practices in the German health and social welfare sector. A professional approach towards aggressive and potentially violent behavior, de-escalation in health and welfare services can be conceptualized as a traveling technology. While it is mandated as a standard of care in some fields, especially in psychiatry, de-escalation trainings are primarily a workplace safety measure, thus situated at a conflux of care and security concerns. Engaging with de-escalation practices in Berlin homeless shelters and with de-escalation trainings, I query how aggression figures in these sites. Fraught with the potential of physical harm, de-escalation is profoundly embodied, with aggression, tension and fear locatable in and expressed through individual bodies. Yet, practitioners also attend to aggression as leaky matter, not located solely in individual persons, but rather leaking out of institutional routines, transpiring between bodies and in situations, and palpable in charged atmospheres. Thinking bodies as material-semiotic nodes (Haraway 1991), I trace how bodies in (de-)escalation come to relate to other bodies, to identity categories, and to different forms and distributions of violence and aggression along axis of inequality. I examine how leaking and thus translation across bodies and experiences depends on notions of sameness and difference, and follow embodied, situated modes of knowing. Such modes of knowing animate both everyday practices and de-escalation trainings, where they are partially articulated through methods such as stories, models or role-play. Leakage as metaphor and aggression as leaky matter enable an exploration of bodies as interdependent and variously (in)capacitated in escalating situations. What relations between bodies render aggression knowable, senseable and relatable? What does it take to understand aggression as leak, as indexical of pressure, corrosion or material fatigue?

Spines in Motion

Alina Lambrecht (Goethe University Frankfurt, Germany)
in person

Our spine, literally our backbone, is thought to give our bodies stability. However, it is not a single bone, but a complex system of discs, nerves, muscles and tendons that work together to provide support and movement. What happens when this system does not function as supposed and needs attention? In this conference contribution I address the medical diagnosis of *scoliosis* from the perspective of affected youth with a textual ontological analysis. Following Annemarie Mol's notion of 'doing health' (2002), I explore how the care for, and treatment of scoliosis is done in practice. To do so I draw from medical STS, care studies and biopolitics. Scoliosis is a medical diagnosis often given to still growing young adults. It is characterized by an atypical curvature of the spine that may progress and is associated with symptoms such as back pain or postural changes. The treatment of scoliosis aims to (re)shape the human spine through exercises, temporary fixation, or permanent spinal stabilization to relieve symptoms. Some of these approaches are highly dependent on patient compliance, are time-consuming and painful. In order to gain a more than medical understanding and to focus on the perspective of the patients, I use biographical accounts taken from a German self-help group website as the basis for my analysis. Taking text seriously as a means of organizing realities. Contextualized by medical literature, I identify the difficulties of scoliosis care. With my analysis I argue how spines in motion are shaped by – and shape – social circumstances such as technologies, family structure and the medical system. The spines in motion acknowledge cracks in the classical understandings of health/disease, patient/treatment, body/technology and offer a non-dualistic perspective that leaks beyond these categories.

The Leakage Between Actual and Virtual Realities in the World of Simulated Nursing Education

Jamie B. Smith (Charité - Universitätsmedizin Berlin, Germany)
Eva Willis (University of Siegen, Germany)
in person

We explore the leakage between actual and virtual realities in the world of simulated nursing education. We currently observe a worldwide shift in nursing education from in practice clinical placements towards simulation training (WHO, 2018; Bridge, 2022). Recommendations in several western countries suggest replacing up to 50% of nurse clinical placements with simulation training (Alexander et al., 2015; CASN, 2021; 2015; NMC, 2021). We are worried by the mostly enthusiastic acceptance in academia and lack of addressing the omission of knowledge by reducing clinical placement time by half. Material encounters leak into the forming of professional identities. Students learn to cooperate with and care for people with vastly different and sometimes contradicting ethical coordinates than the people they already know. Clinical placements offer encounters with diverse patient groups that confront nursing students with repeating social friction and dilemmas over time. We believe these are crucial to become nurses that deliver equitable care. In the first half, we look at whose reality is produced in simulation. We ask what complexities are reduced through simulation and what forms of knowledge might be excluded from leaking into professional knowledge formation (Benjamin, 2019)? We interrogate how simulation might increase biased thinking (Smith, Willis & Hopkins-Walsh, 2022) and disenfranchisement of patients and neurodivergent nursing students (Epstein, 2021a, 2021b). We do not stand in opposition to simulation training. In the second half, we advocate for simulation that extends the replication of reality in simulation and encourage the leaking of speculative utopian and dystopian scenarios to inform the ethical positionality of healthcare workers (Haraway, 2016). We consider the potential of simulation-training from a posthuman perspective with affirmative ethics (Braidotti, 2013, 2022) to provide novel ethical, empirical and practical insights. Through this approach, simulation could create situations that expand from individual cases to how nursing teams create and influence shared realities.

Panel #51 Up to Date in STS: Early Career Reading Group

Stream: stsing
ABS/0017
Chairs:
Ruzana Liburkina (Goethe University Frankfurt, Germany)
Patrick Bieler (Technical University of Munich, Germany)

The working group "Up do Date in STS" is a forum for early career researchers (postdocs and docs) who are interested in reading recently published or yet unpublished cutting-edge STS books/book manuscripts and discussing them with the respective authors on a regular basis. Currently, we are a small group of (post)docs working in different cities and fields. As such, we explicitly aim to keep track of monographs and edited volumes we consider key or even path-breaking contributions to STS beyond specific research communities. To do this, we collectively gather and decide upon literature suggestions in preparatory meetings ahead of the actual 'book club' sessions.
In this meeting, some of the members of the working group will briefly present their recently and soon to be published monographs and edited volumes. Moreover, we will evaluate and fine-tune the working of the reading group and plan ahead for the upcoming summer semester. Interested newcomers are very welcome!

Panel #52: Leaking Infrastructures of Decarbonisation

Stream: Energy Infrastructures & Contamination
ABS/214
Chair: Livia Cahn

Hydrogen Markets in the Making

Kristiane Fehrs (TU Dresden, Germany)
in person

Hydrogen Backbone, Hydrogen Valley, durch^{2}atmen (engl.: take a breath^2), North^2, HZwo...– a list of current hydrogen projects (and their title's puns and word creations) could be continued to fill a whole paragraph. The sheer mass of projects gives an impression of current efforts to build up a hydrogen market, in which hydrogen is presented as an important pillar for an energy transition in Europe and Germany. In my contribution I will share insights into an ethnographic research project on hydrogen in the context of energy transition in Saxony. Hydrogen technologies have had its ups and downs long before the element took its role in current ambitions towards decarbonization. The uncertainty of its success continues to be prevalent in negotiations around a 'hydrogen market acceleration' in Germany. Along the lines of Michel Callon's 'Markets in the Making' I will take a look at hydrogen market agencements and outline hopes and doubts that hydrogen evokes.

Chemical Life of Hydrogen Infrastructures: Intensive and Extensive Gas Leakage

Johannes Hollenhorst (The London School of Economics and Political Science, UK)
in person

The global climate movement has forced many nation states and global companies to commit to net zero. To realize these commitments, the materialization of hydrogen energy and infrastructures has become a new priority. While hydrogen can indeed act as a carbon-neutral energy material, the

infrastructures for its production, transport, and use entail at least two key difficult engineering challenges which I am analyzing through the socio-material concept of chemical life: One intensive and one extensive leakage problem. The intensive leakage problem is caused by the small size of hydrogen molecules, making hydrogen leak through and crack open materials like steel pipelines and pressurized storage systems which were designed for methane gas. Despite this problem, fossil fuel companies have identified hydrogen as the means to extend the lifetime of their carbon assets into the net zero future. But even if this intensive problem is overcome in the interest of company profits, the extensive leakage problem remains unaddressed: The production of hydrogen at a scale comparable to methane to replace its use. As electrolysis capacity to extract hydrogen from water remains difficult to scale for various reasons, fossil fuel companies envision the use of Carbon Capture and Storage (CCS) to extract hydrogen from methane in a climate neutral way. The production of this so-called blue hydrogen however perpetuates the extraction of natural gas and the respective problem of methane leakages. In addition, it also creates new leakage risks in relation to the carbon dioxide stored underground. To address these leakage problems in a meaningful way, the use of hydrogen itself accordingly has to be carefully reviewed.

Leaking In Or Leaking Out – Conceptual Thoughts on Underground Leakage

Danny Otto (Helmholtz Centre for Environmental Research GmbH – UFZ, Germany), Alena Bleicher (Harz University of Applied Sciences, Germany), Matthias Groß (Helmholtz Centre for Environmental Research GmbH – UFZ, Germany, remote), Marco Sonnberger (Friedrich Schiller University Jena, Germany, remote) & Maria Pfeiffer (Friedrich Schiller University Jena, Germany)
in person

With this talk we propose a theoretical approach to the understanding of leakage through the lens of underground uses such as gas storage and geothermal energy extraction. The argument we unfold addresses two ambivalent and intermingled dynamics related to leakage: 1) physical processes of things leaving a certain demarcated space, 2) symbolic processes of (not) constructing leakage (e.g. something leaks out of a partially stabilized normality. Furthermore, we establish the figure of leaking in and leaking out as a conceptual vantage point. We draw on STS and environmental sociological research on carbon capture and storage (CCS) and geothermal energy use to develop this conceptualization of leakage and to exemplify both the physical and symbolical dimensions of leakage. These cases provide a good basis for the investigation of actors and elements involved in the establishment and upholding of (imagined) barriers, control and retention as well as the temporalities of constant micro-leakage and observed loss of containment security. This approach allows us to unfold the networks of underground containment and leakage and their dialectic relation. It provides an opportunity to study tipping points in societal interpretations of what it means to be sealed and to be leaking and to grasp the continuum between these ideal typical states.

Panel #53 Technoscientific Worldmaking, Environments, and Futurity

Stream: Tech & Data
ABS/105
Chairs: Estrid Sørensen

Challenging Environments: The Outside of Technoscience

Lisa Anders, Hannah Link (Johannes Gutenberg-Universität Mainz, Germany)
in person

Recent discourses of science and technology studies increasingly acknowledge the social relevance of non-human entities: either as stabilizing actors (Latour 1995), social partners (Knorr-Cetina 1998), or mediators of chiasmic relations (Verbeek, de Boer 2022). In particular, neo-materialism is contesting the classical dichotomy of nature and culture while proposing their constitutive entanglement. Nature is re-imagined, not as a passive and malleable resource for societal and economic processes, but as a lively opponent (Lovelock 2021), as a self-organizing being (Braidotti 2014), and as an unpredictable trickster (Haraway 1988).

Behind this backdrop, we aim at destabilizing a preconceived notion of the environment as a natural resource. Based on ethnographic observations of military aeronautics and robotics, we examine technoscientific habits of enacting environments, as spheres that exist beyond situational-drawn ontological boundaries. Based on empirical data, we propose an analytical systematization of the various ways of technoscientific practices of un/relating: Practices that oscillate and thereby constitute exclusion and inclusion, that we understand as reflexive interpenetration of the inner and outer spheres. What and who is perceived and addressed as 'the environment' is considered a product of socio-material practices. Therefore, the notion of 'the environment' exceeds an understanding of nature and encompasses both human and technological entities as equally influential. In this context, nature appears as a dynamic, influential force that offers resources, introduces disruptions, and exhibits inherent unpredictability. Therefore, our perspective promotes a comprehensive and procedural understanding of the environment, acknowledging the intricate interactions among nature, humans, and technology while recognizing their cooperative and conflicting dynamics.

The Technosphere: Technology as a natural phenomenon

Justus Pötzsch (Johannes Gutenberg-Universität Mainz, Germany)
in person

The unused and accumulating piles of human waste are probably one of the main threats for current planetary life. As geologist Peter Haff points out: "In a closed environment like the Earth [...], every metabolizing system must eventually recycle its own waste products [...], otherwise accumulation of spent material (i.e. pollutants) will impair system function." But piling up of waste is apparently one of the main anthropogenic influences on Earth and ecosystems. Especially since the 20th century, carbon dioxide and other greenhouse gasses are being built up in the atmosphere and oceans, thus dangerously altering the chemical composition of habitats on land and water. Additionally, plastic and other synthetic materials aggregate in gigantic landfills, circulate through global waterways and infiltrate ecosystems as well as organisms. The remnants of this wasteful lifestyle of (some) humans are even most likely becoming the chrono-stratigraphic marker that define the dawn of a new geological epoch – the Anthropocene. Apparently, we (or actually just a few of us) have literally wasted our future.

But instead of giving in to the lament of an annihilation of mankind or the invocation of heroic human geoengineering, I would like to propose an alternative perspective on the problem of global waste and pollution. Even though mitigating the destructive effects of human influence on planetary

systems should be of primarily concern, radically changing environmental conditions are not without precedence in planetary history. In fact, the deep time of Earth's past reveals several drastic geosystemic transformations, when the accumulated unused or waste matter became the stimulus for evolution and subsequently the 'fuel' for new forms of life. In this way oxygen (as a by-product of early oxygenic photosynthesis) and coal (as remnants of ancient rotten carboniferous forests) have once undergone a transformation from a by-product or dead matter to new resources for natural as well as cultural forms of evolution.

Mond im Vakuum. Leere aufbewahren und Zukunft produzieren.

Rachele Delucchi (Eidgenössische Technische Hochschule Zürich, Switzerland)
in person

Die Astronauten der Apollo 17 Mission brachten im Dezember 1972 einhundert Kilo Mondgestein zurück zur Erde mit. Darunter war ein ca. 35 cm langer Bohrkern, der bereits auf der Mondoberfläche in einem vakuumversiegelten Behälter aus Edelstahl eingeschlossen worden war, den sogenannten Core Sample Vacuum Container (CSVC). Diese Mondprobe war als "posterity sample" bestimmt: eine Mondprobe in lunarem Vakuum, die für viele Jahre so intakt und unangetastet wie möglich in der Lunar Sample Collection der NASA konserviert werden sollte.

Am 23. Februar 2022 griff ein internationales Team von Wissenschaftler*innen nach der Mondprobe 73001. Der Boden des CSVC wurde durchstochen. «A satisfying 'pop' marked successful piercing» (McDonald 2022). Die Geschichte der Mondprobe 73001 kann als Erfolgsgeschichte von technischen, organisatorischen und institutionellen Dispositiven der kontrollierten Schliessung und Öffnung erzählt werden. Oder als die Geschichte einer 50 Jahre langen, sich um ein Leck konfigurierenden Erdlandung. Dieses Leck war eine einzukalkulierende Eventualität und eine auszuschließende Option, ein virtuelles Ereignis und ein messbarer Effekt. Vakuumtechnik sollte eine fortwährende Gegenwart und eine offene Zukunft für die Mondprobe produzieren. Wie wurde das Leck in den räumlichen und zeitlichen Verflechtungen dieser Erdlandung verhandelt? Wie beeinflusste das möglichst hermetische Verschliessen der Mondprobe die Möglichkeit einer späteren kontrollierten Öffnung?

Panel #54 Queer-feminist Transformations (German Panel)

Stream: Ecological Leakage and the More-than-human
ABS/114
Chair: Lisa Krall

Just Transition meets Muttermilch: Ansätze für eine feministische Transformationsforschung

Dorothea Hamilton, Sören Becker (Philipps-Universität Marburg, Germany)
in person

Während der Kohlekompromiss ermöglicht, dass einige Großkraftwerke bis 2038 emittieren, wurden in den deutschen Kohlerevieren jeweils Strategien für einen nachhaltigen Strukturwandel entwickelt. In der Forschung und in Programmen der Europäischen Union werden unter dem Begriff just transition regionale Entwicklungsprozesse untersucht, in denen möglichst unterschiedliche Akteursgruppen eingebunden werden sollen. In den resultierenden Debatten stehen Arbeitsplätze, Standortfaktoren und Technologieförderung im Vordergrund – während Frauen häufig unsichtbar bleiben. In einem Zwiegespräch zwischen einem Vertreter der akademischen Transformationsforschung und einer stillenden, prekär beschäftigten Geographin werden wir darüber diskutieren, wie die Agenda einer feministischen Transformationsforschung aussehen könnte.

Ausgangspunkt sind dabei die Impulse, die aus der Brust auslaufende (leaking) Muttermilch in Verbindung mit feministischen Theorien aus dem globalen Norden und Süden bieten, um die kritische Transformationsforschung um eine feministische Dimension zu erweitern. Wir diskutieren anhand der durch Muttermilchflecken sichtbar gewordenen Leerstellen der aktuell dominanten Transformationsforschung, und wie diese für eine Forschungsagenda umsetzbar gemacht werden können. Basierend auf der Erfahrung der unwillentlich austretenden Muttermilch gehören dazu die Betrachtung von Körperlichkeit, Alltagshandlungen, „Care“ und von mehr-als-menschlichen als Akteuren.
Wir benutzen die Erfahrung, um die Fragen zu diskutieren, was Transformation ist, was Transformationsforschung untersuchen soll und wer aus den verschiedenen Perspektiven agiert. Im Format des Zwiegesprächs werden der Positionalität der Forschenden Rechnung getragen und Anwendungsmöglichkeiten, Grenzen und offene Fragen entwickelt, z.B. wer über etwas sprechen, schreiben oder forschen darf.

Queere landwirtschaftliche Mensch-Natur-Beziehungen

Sophie von Redecker (University of Kassel, Germany)
in person

Dieser Vortrag fokussiert auf Mensch-Natur-Beziehungen in landwirtschaftlichen Praktiken. Wie können dichotome Konstruktionen von Mensch-Natur-Trennungen aufrecht erhalten werden, wenn die Hand der landwirtschaftlich wirtschaftenden Person auf Lebzeiten Erde in sich trägt? Inwiefern werden anthropozentrische Ontologien durch die Liebe von Gärtner*innen zu ihrem Garten irritiert? Und inwiefern zeigen landwirtschaftliche Praktiken und Akteur*innen auf, dass „die Natur“ nicht gemacht ist, um heteronormative Konstruktionen zu untermauern? Mit Bezugnahme auf queer-feministische Theorieansätze, Neue Materialismen, STS und more-than-human Zugänge untersucht dieser Votrag sowohl anhand autobiographischer Erzählungen als auch auf Grundlage von arts-based-research Methoden counterhegemoniale Mensch-Natur-Verhältnisse und lässt leakages an den Rändern der Disziplin der Agrarwissenschaften aber auch den Auffassungen dessen was Mensch, was Natur, was Gärtnern und Ackern ist, entstehen. Die zwei Dokumentarfilme „Out Here“ (Jonah Mossberg, 2010) und „Queer Gardening“ (Ella von der Haide, 2022) werden dabei immer wieder als Beispiel herangezogen und vor dem Hintergrund der oben genannten Fragestellungen und Forschungsansätze untersucht. Um andere Formen des Wahrnehmens zu wecken, wird das starre Modell wissenschaftlichen Vortragens durchbrochen, indem die Zuhörenden eingeladen werden, “reading with our senses attuned to stories told in otherwise muted registers.”, wie Carla Hustak and Natasha Myers (2012,77) es beschreiben, zu erproben. Mit Blick auf dekoloniale und indigene Forschungsansätze werden all jene Zusammenhänge auch wissenschaftsgeschichtlich eingeordnet und somit auch epistemische Machtverhältnisse aufgedeckt.

Panel #55 States of Emergency and Technological Sensing

Stream: Tech & Data
ABS/213
Chair: Jannis Steinke

COVID-19 Simulationsmodelle als Experimentalsysteme im Notstand

Finn Langbein (Philipps-Universität Marburg, Germany)
in person

Due to its high level of uncertainty, unpredictable dynamics, and potentially devastating consequences, the COVID-19 pandemic has not only challenged political decision-making but has also impacted scientific research. The public health emergency profoundly changed the practice of pandemic simulations situated at the intersection of these fields. On the one hand their conventional methods of mathematical data analysis and projection of future scenarios were used, but on the other hand the characteristic uncertainties within the pandemic event-space were different and the temporal and spatial horizons of knowledge practices were transformed by the pandemic as well. Some modelers even entered the public sphere and actively advocated their public health positions. Drawing on Rheinberger's notion of “experimental systems,” I investigate how the state of emergency was inscribed into the experimental practice of pandemic simulations during the COVID-19 pandemic. By demonstrating how scientific practices were infused with their constitutive other – the political – which leaked into them in the form of the state of emergency, my work contributes to the debate on “governing emergencies,” shifting the focus from the executive to the technoscientific level. While, in my dissertation, I aim for a comprehensive analysis of this process, in my talk, I use the shift in the temporality of work horizons of simulation practice as an example. To illustrate this, I use excerpts from interviews I conducted with German modelers. The material highlights that pandemic simulations under emergency conditions were characterized by external pressure toward significantly shorter work cycles and decision timeframes, enabling the short-term production of scientific evidence in the face of a precarious data situation, which could, however, rapidly lose relevance.

When Uncertainties Seep Into Trading Zones: Tsunami Warning System Technologies

Irina Rafliana (German Institute of Development and Sustainability, Germany)
in person

"When it comes to recognizing and forecasting tsunami events, tsunami modeling is a fascinating knowledge practice. Diverse scientific disciplines, (geo)political interests often entangled, hence establishing ‘trading zones’ to exchange and negotiate trade-offs of what modeling could and could not offer as an effort to reduce tsunami risks. The trading zones become the place to exchange understandings, techniques on observation and analysis. Oftentimes, these trading zones create uneven spaces of exchanges and epistemic hierarchies that occur due to the asymmetries of knowledge, infrastructures and funding constraints at local and more than local level. What makes the exchanges more intriguing are when epistemological and ontological uncertainties factors in. The presentation aims at demonstrating vignettes where uncertainties seeps into the manifolds of ‘trading zones’ observable in tsunami modeling, that shapes warning system technologies as social practices. While it sheds light on the impossibility of warning system to completely serve its purposes (with reference to 2011 East Japan tsunami, Palu tsunami and Krakatau volcano flank collapse tsunami in Indonesia, among others), such uncertainties are arguably generative in forcing knowledge and modeling improvements to continuously understand the earthly dynamics, and transform warning system technologies from merely a ‘fixed and done’ infrastructure projects, into ‘continuously becoming’.

Geo-Aesthetic Witnesses: A Case Study Of Joshimath

Ayush Biswas (Goethe University Frankfurt, Germany)
Shrestya Saraswati (University of Bonn, Germany)
in person

Joshimath is located on the downslope in the Chamoli District in Uttarakhand, India. The first crack in a residential building was identified in the month of October 2021 in Gandhi Nagar village. By January 2022, it was noted that around 800 residential buildings, including some commercial ones, had developed and were showing visible structural damage. The cause of the said damages was attributed to land subsidence owing to heavy anthropogenic activities. The Bureau of Indian Standards has categorized the seismic zones of India into four categories, with 'II' being the lowest and 'V' being the highest. Joshimath is categorized as Indian Seismic Zone V. According to a report by the National Remote Sensing Centre (NRSC)and the Indian Institute for Remote Sensing (IIRS), relying solely on satellite images and GIS systems, they concluded that Joshimath has been sinking at 2.5 inches per year between 2020 and 2022. This proposal initially will look, through Susan Schuppli's use of 'Material Witness', at the historical materials that foresee the coming of this traumatic event. Specifically for this instance, the work will look into the Misra Committee Report of 1976; a recommendation report which performs its leaky spectrality of State-Science, foreseeing the coming geological vulnerability of this site. Along with this review, the proposal will also look into the assemblage of geo-aesthetics- that is performed and produced by micro-nodal infrastructures of State-Science. To locate this material-semiotic mesh of evidence, we will primarily look into the satellite images produced through remote sensing and GIS technologies by the previously mentioned -NRSC and the IIRS. This mesh would also attempt to draw out the power architecture of state science mainly in the suppression of the very scientific (non) evidence produced by its nodal points i.e., bodies like the IIRS and NRSC.

#56 Conversation Café Leaking Ethnographies: A Conversation Café on Methodography

Stream: Methods & Epistemology
ABS/214
Chair: Hannah Grün & Aurora A. Sauter

Lisa Wiedemann (Helmut Schmidt University, Germany), remote
Aurora A. Sauter (Johannes Gutenberg-Universität Mainz, Germany)
Hannah Grün (Helmut Schmidt University, Germany)
Milena Bister (Humboldt-Universität zu Berlin, Germany)
Stefan Laser (Ruhr University Bochum, Germany)
Maxime Le Calvé (Humboldt-Universität zu Berlin)
Ingmar Lippert (Goethe University Frankfurt, Germany)
Britta Acksel (Ruhr University Bochum, Germany)
Julie Sascia Mewes (Museum für Naturkunde – Leibniz Institute for Evolution and Biodiversity Science, Germany)

STS ethnographers have been concerned with methods for a long time. On the one hand, we have approached the method as an object of research when studying how facts are produced in scientific laboratories. On the other hand, our concern with method is a matter of reflexivity about how methodological choices shape our ethnographic fieldwork, analysis, and interpretations. Within this context, John Law (2004), for example, showed in his book After Method that methods are always political. In a recent Special Issue on "methodography," Julie Mewes and Ingmar Lippert (2021) seek to advance discussions on reflexivity by developing a genre of writing that challenges us to "[understand] our method of ethnography ethnographically." They invite STS ethnographers to

examine the worlds their methods are performative of, how they negotiate epistemic authority about the kinds of worlds presented in their stories, and the implicit standards, affective experiences, as well as practical considerations that guide their research. Methodography is a response to the diagnosis that hegemonic formats of publishing and conferencing do not leave enough room for engaging with our own situated knowledge- making practices. In our view, leakage is an applicable "sensitive term" (Mol 2010), which captures the complexities, messiness, and vulnerabilities of doing ethnography. Leakage is that which is not contained in the polished presentations of our research results, which draws straight lines through an oTen intuitive and contingent research process marked by uncertainty. That which, for the most part, only drips into (unpublished) conversations in the form of affects, feelings, or insecurities (Smolka et al. 2022). Sometimes, it dissolves into our research practices. Hence, engaging with leakage requires us to attend to epistemic instabilities and unintended consequences. The concept allows us to describe what we usually consider as 'unnecessary' data, troublesome ethical considerations, implicit decision-making processes, and those experiences or activities that are difficult to put into words. In this panel, we invite contributors to think and discuss methodography through the concept of leakage. The contributors are STS researchers who are experienced in the genre of methodography. Each of them will host a thematic 'coffee table.' Examples of themes are 'overlooked', 'hidden,' and 'unnecessary' data, situated ethics, choices in analysis, the art and craT of writing, epistemic agency, and serendipity in field construction. Contributors will give a short input on their theme to create a focused space for discussions and exchange. Panel participants are invited to bring in their own ideas, experiences and questions to draw attention to and collect 'methodographic leakages.' The tables will be hosted by Britta Acksel, Milena Bister, Juliane Jarke, Jonna Josties, Maxime Le Calvé, Julie Mewes, Ingmar Lippert, and Stefan Laser. The panel will be organized in English, but discussions at the coffee tables may be in German if preferred. The aim is to create a conversation café where we can collaboratively develop a sensitivity for attending to our own ethnographies methodographically and create room to reflect collectively on our situated knowledge-making practices.